

Video Info

File name: P-CDL-I-M2-1-Exploring Data Transformation with Google Cloud

Presenter:

02

Exploring Data Transformation with Google Cloud

Module 2

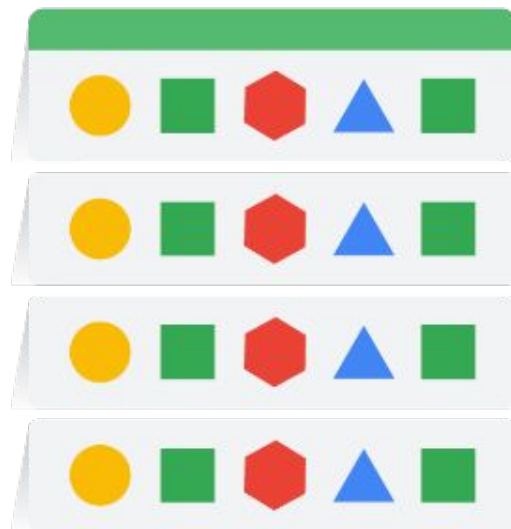
Exploring Data Transformation with Google Cloud



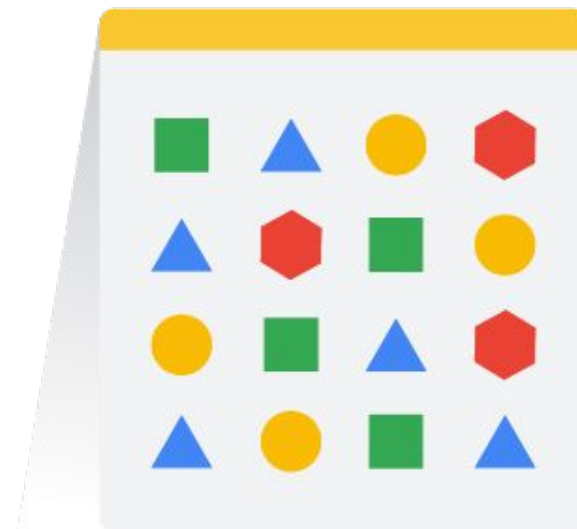
Lessons

- 01 The value of data
- 02 Google Cloud data management solutions
- 03 Making data useful and accessible

Three main types of data



Structured data

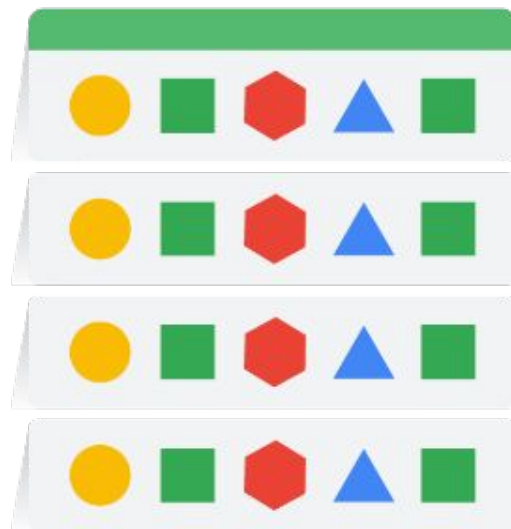


Semi-structured data



Unstructured data

Structured data



Structured data

- ✓ Is highly organized and well-defined.
- ✓ Is typically stored in a table.
- ✓ Includes spreadsheets and databases.
- ✓ Is easy to analyze.

Unstructured data



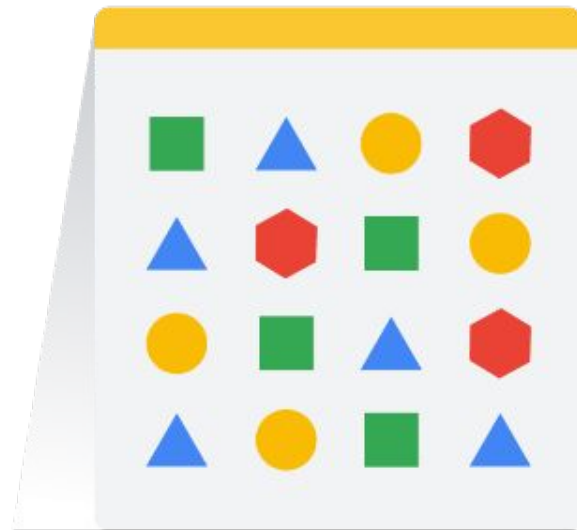
Unstructured data

- ✓ Doesn't have a predefined data model.
- ✓ Isn't organized in a predefined manner.

Categories

- Text, like documents and presentations
- Data files, like images, audio, and video
- Infrastructure activity and performance data

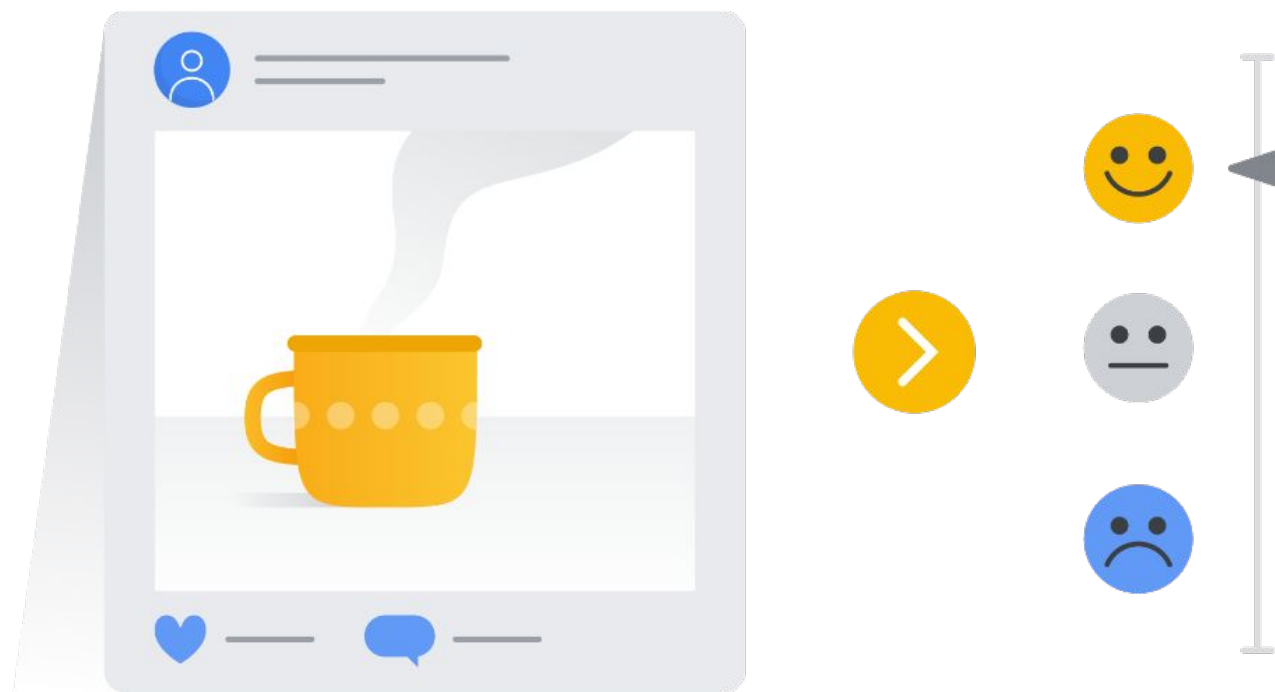
Semi-structured data



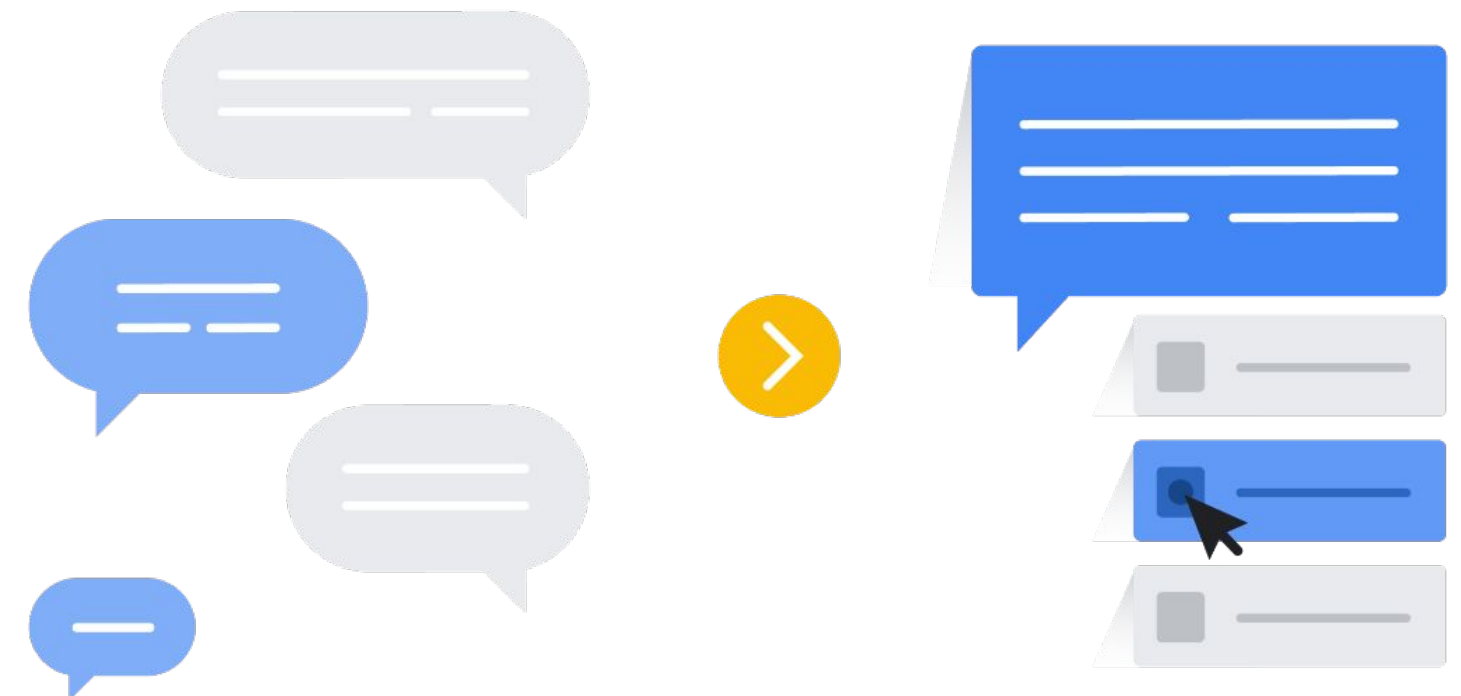
Semi-structured data

- ✓ Is organized into a hierarchy.
- ✓ Lacks full differentiation or order.
- ✓ Includes examples like emails, HTML, JSON, XML.
- ✓ Doesn't have a formal structure.
- ✓ Contains tags for easier analysis.

Organizations can use unstructured data in many ways

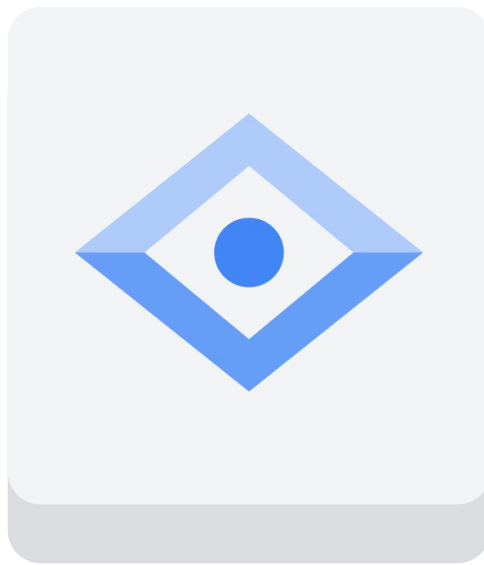


Analyze social media posts to identify sentiment toward a brand.



Analyze customer communications to provide responses for chatbots.

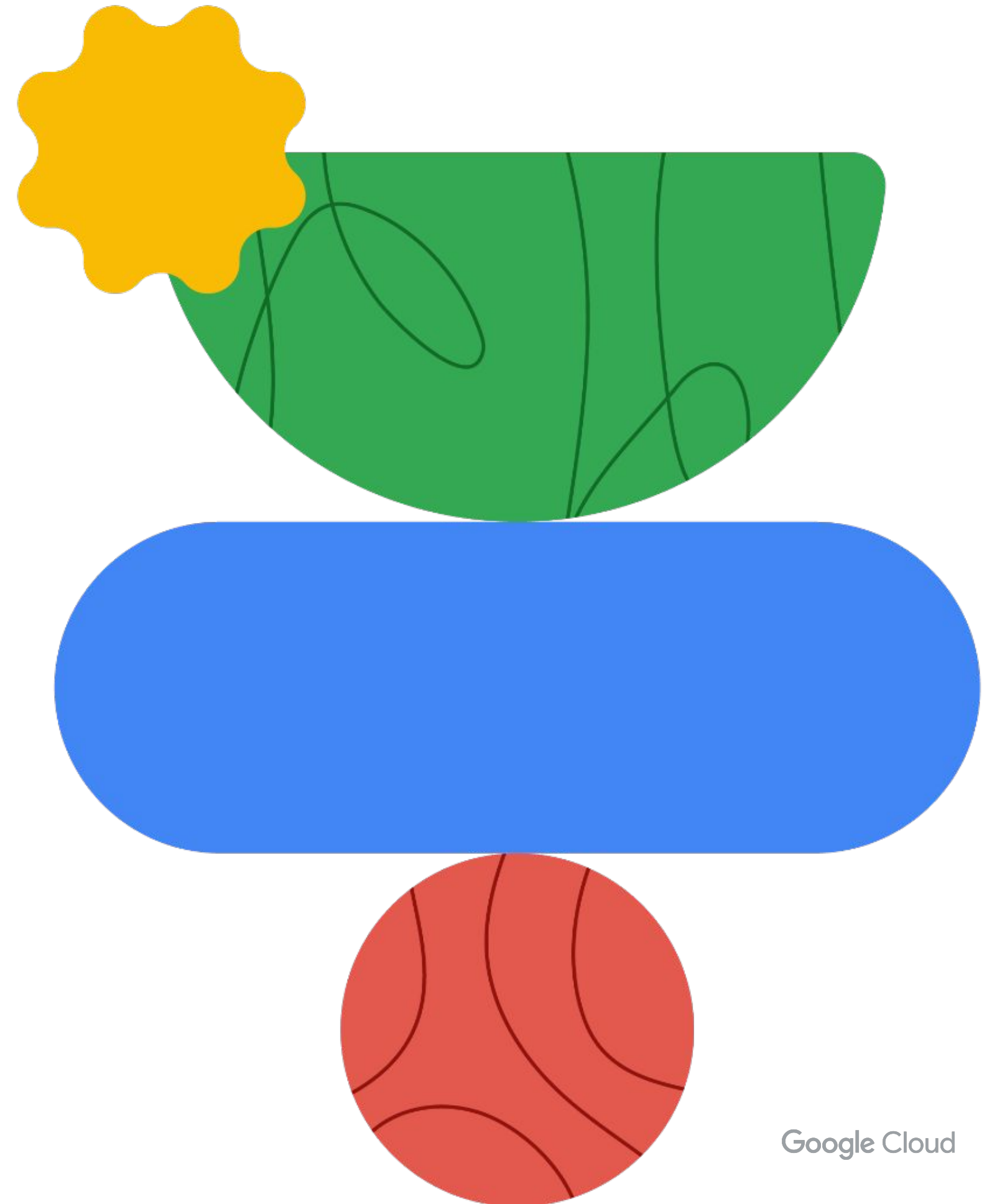
APIs can help extract value from unstructured data



Cloud
Vision API

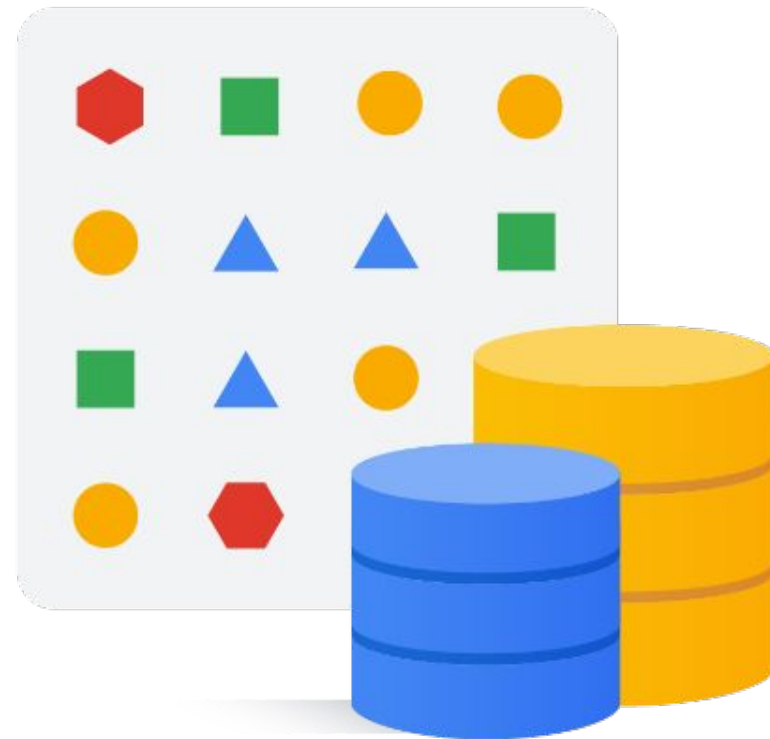
- ✓ Uses machine learning.
- ✓ Detects products within a picture.
- ✓ Labels the picture to describe its contents.

Database, data warehouses, and data lakes

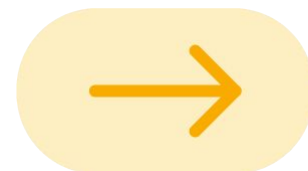


Database

An organized collection of data stored in tables and accessed electronically from a computer system.



Databases



Relational databases



Non-relational databases

Relational database

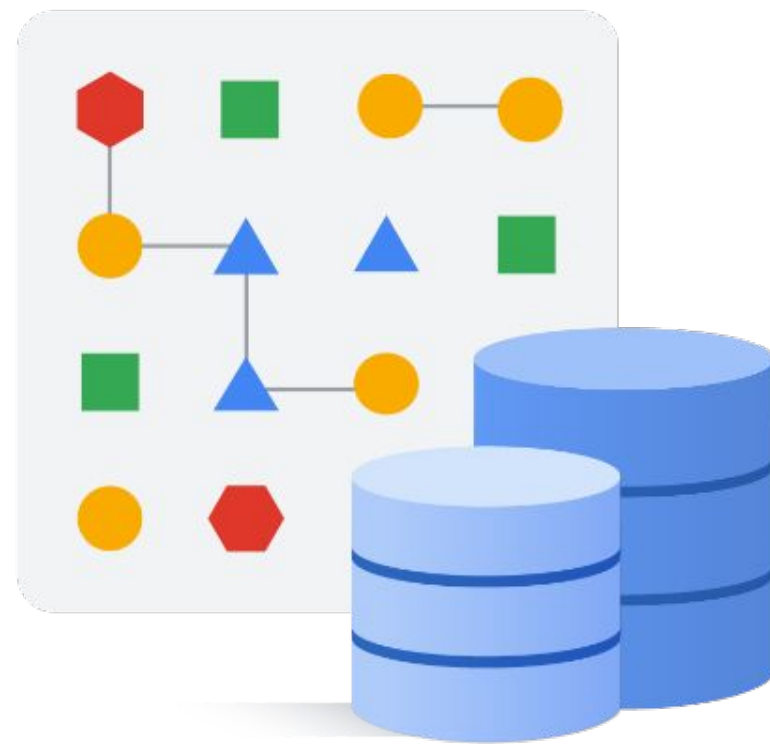


Relational databases

Stores data points in tables, rows, and columns that have a clearly defined schema.

- ✓ Is highly consistent and reliable
- ✓ Suited for large amounts of structured data.
- ✓ Is designed for business data processing.
- ✓ Is designed for storing online transactional data.

Non-relational database

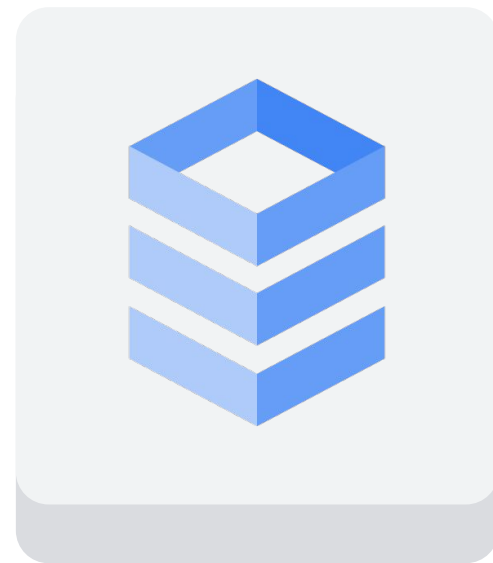


Non-relational databases
(NoSQL)

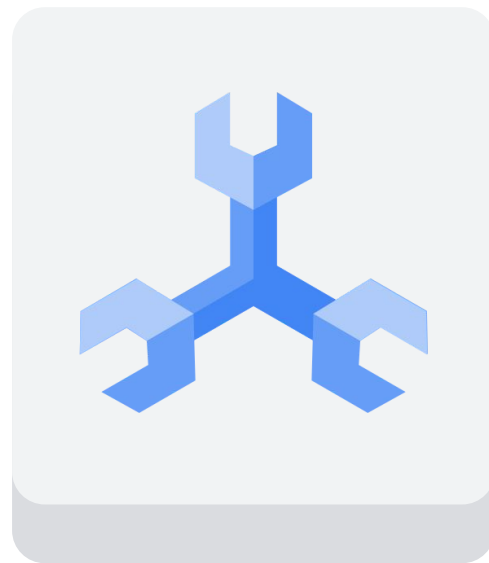
- ✓ Doesn't use a tabular format.
- ✓ Follows a flexible data model.
- ✓ Ideal for data with changing organization.
- ✓ Ideal for applications with diverse data types.

Google Cloud database products

Relational databases

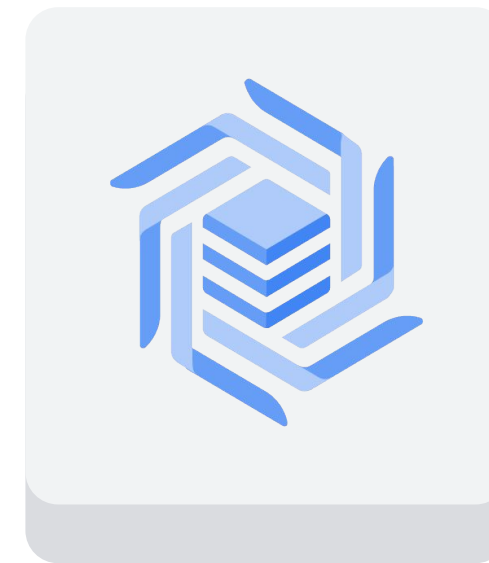


Cloud SQL



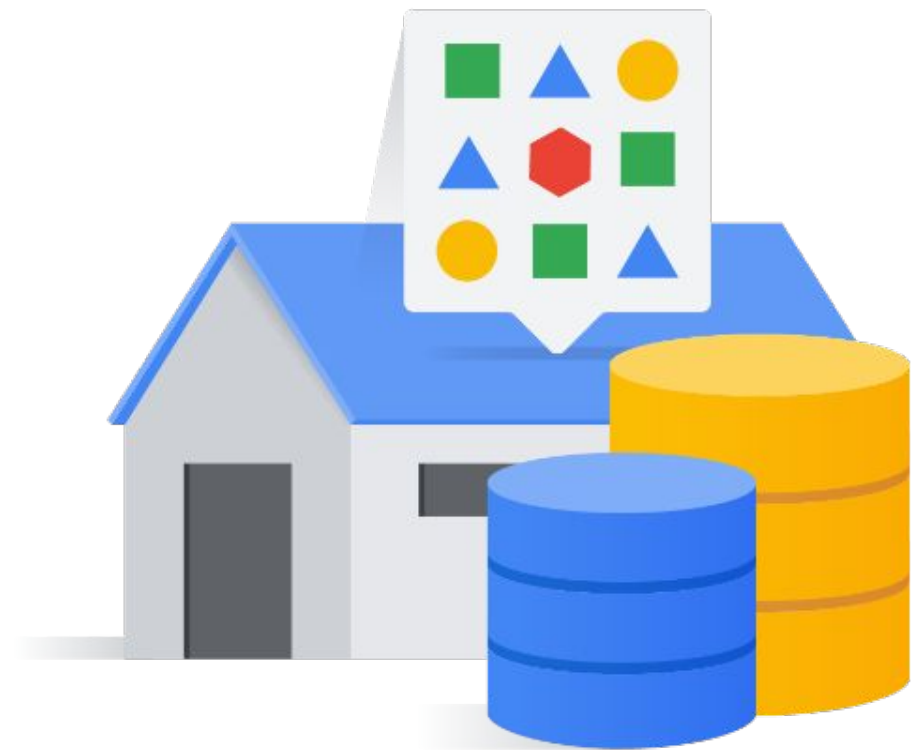
Cloud
Spanner

Non-relational databases



Bigtable

Data warehouse



Data warehouse

An enterprise system used for the analysis and reporting of structured and semi-structured data from multiple sources.

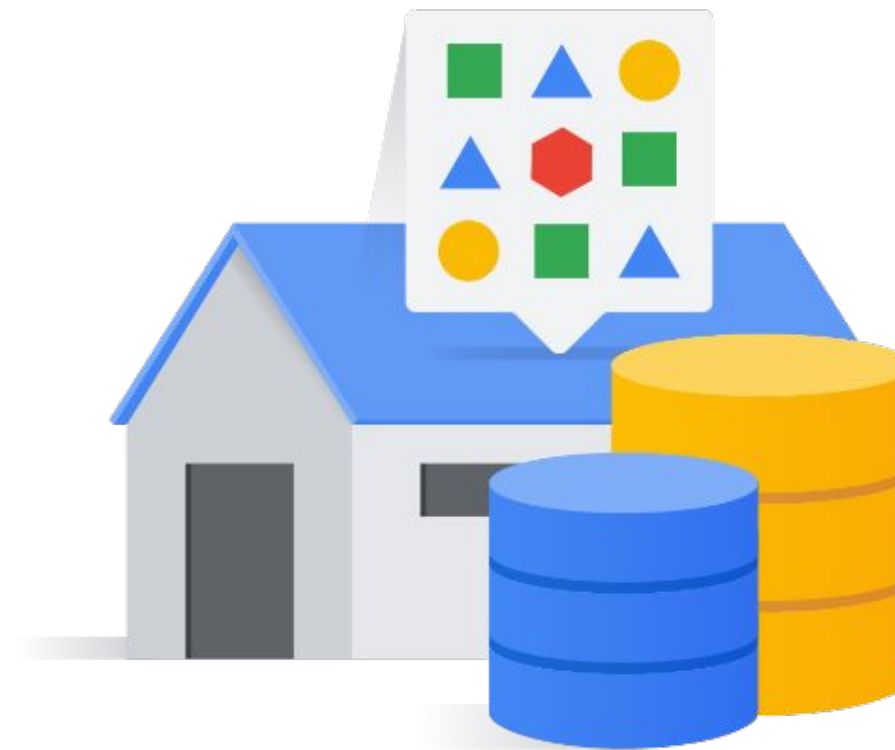
Business data:

- Point of sale
- Marketing automation
- CRM data

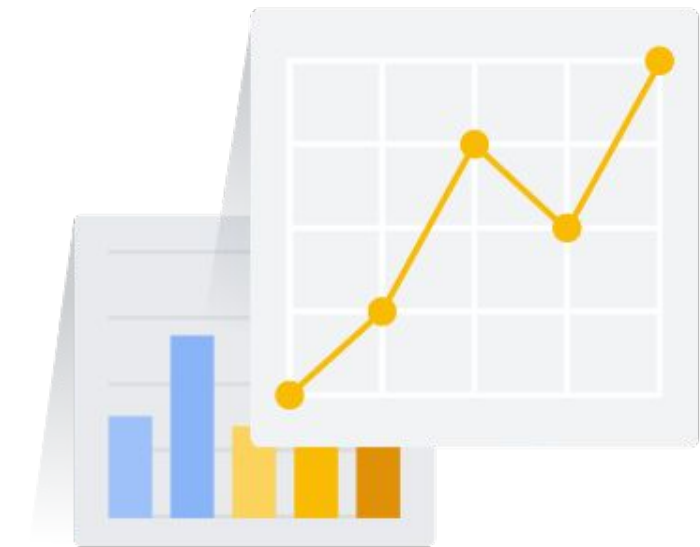
A data warehouse provides a long-range view of data over time



Current
and historical data



Data warehouse

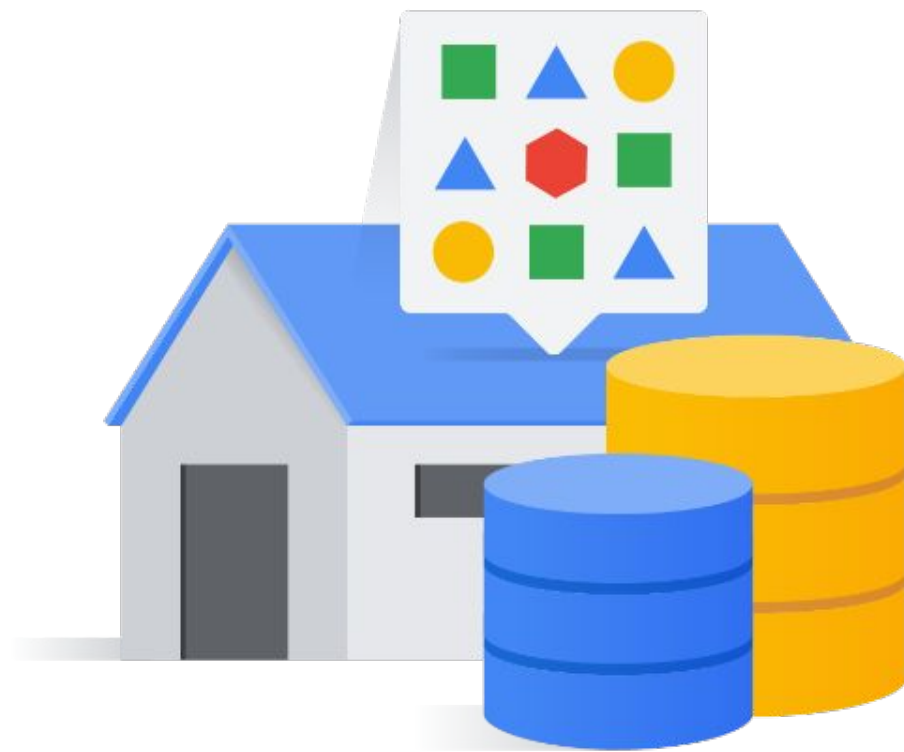


Analyze sales

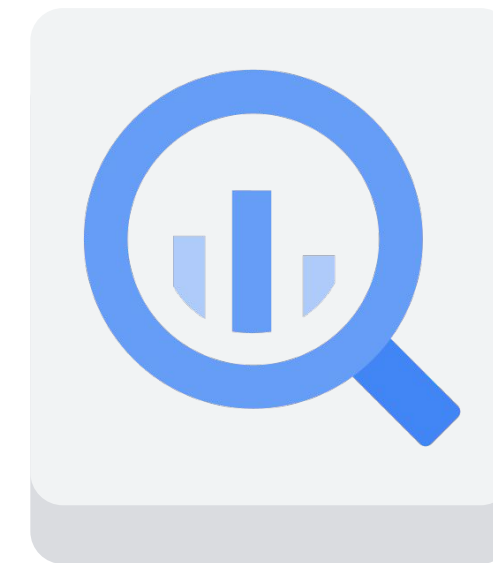


Identify trends

BigQuery is Google Cloud's data warehouse offering

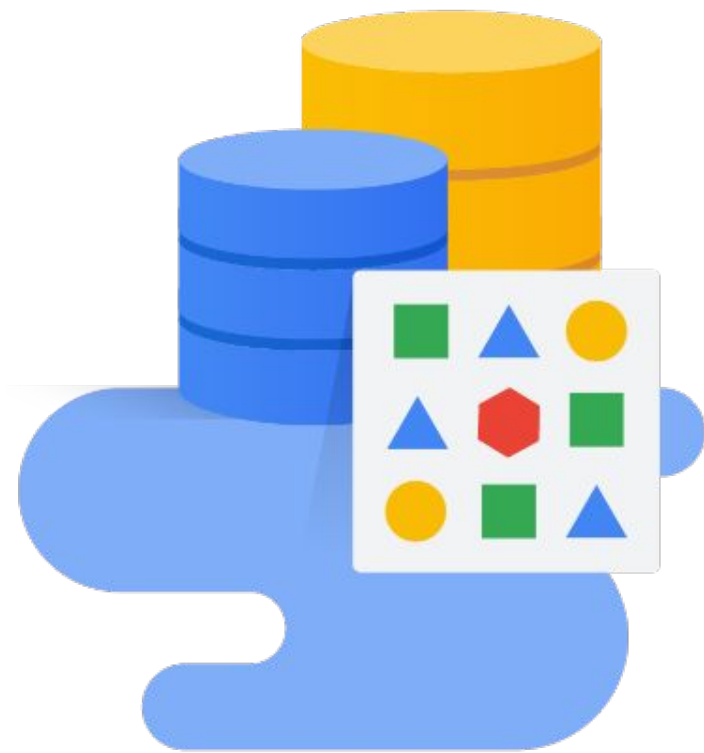


Data warehouse



BigQuery

Data lake



Data lake

A repository designed to ingest, store, explore, process, and analyze any type or volume of raw data

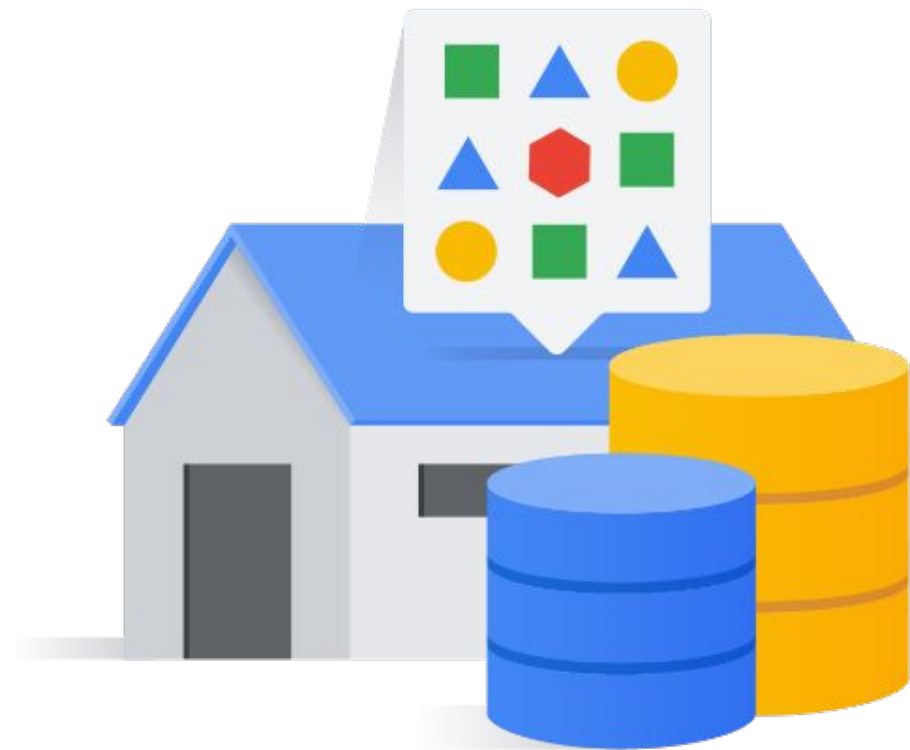
- Operational systems
- Web sources
- Social media
- IoT

It can store different types of data:

- In its original format
- By ignoring size limits
- Without much preprocessing
- Without adding structure

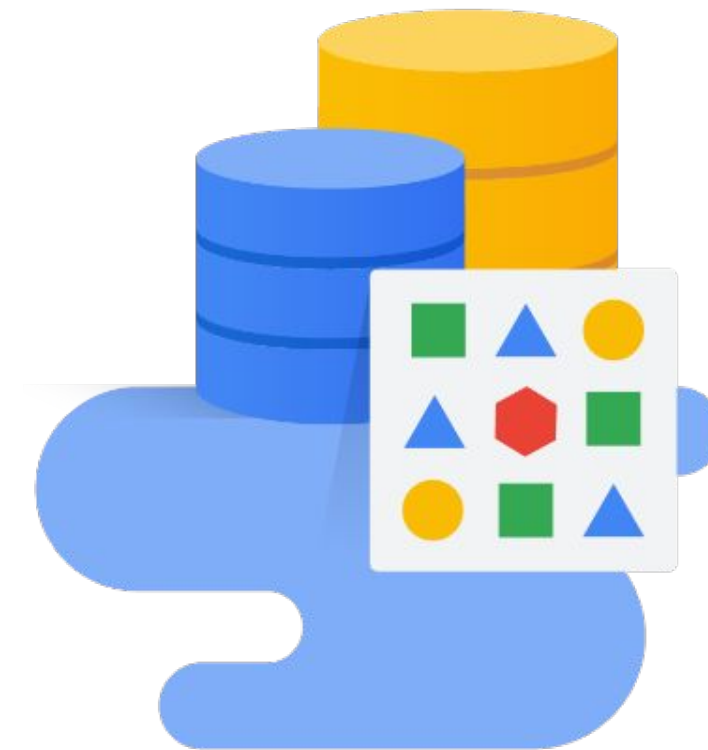
Data warehouses and data lakes are complementary tools

Data warehouse



Users: Business intelligence analysts
Goal: Use the data to answer questions.

Data lake



Users: Business intelligence analysts, data engineers, and data scientists
Goal: Explore, mine, and experiment with raw data.

Types of business data

First-party data

The proprietary customer datasets that a business collects from customer or audience transactions and interactions

Second-party data

First-party data from another organization that can be easily deployed to augment a company's internal datasets

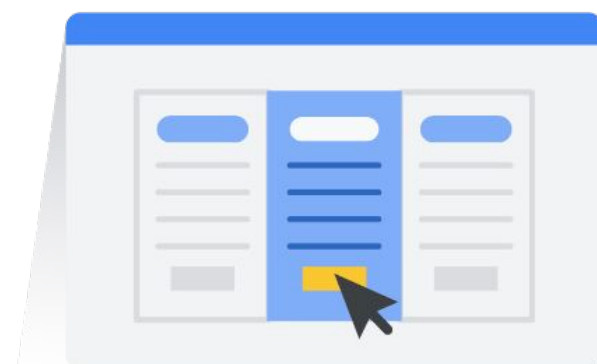
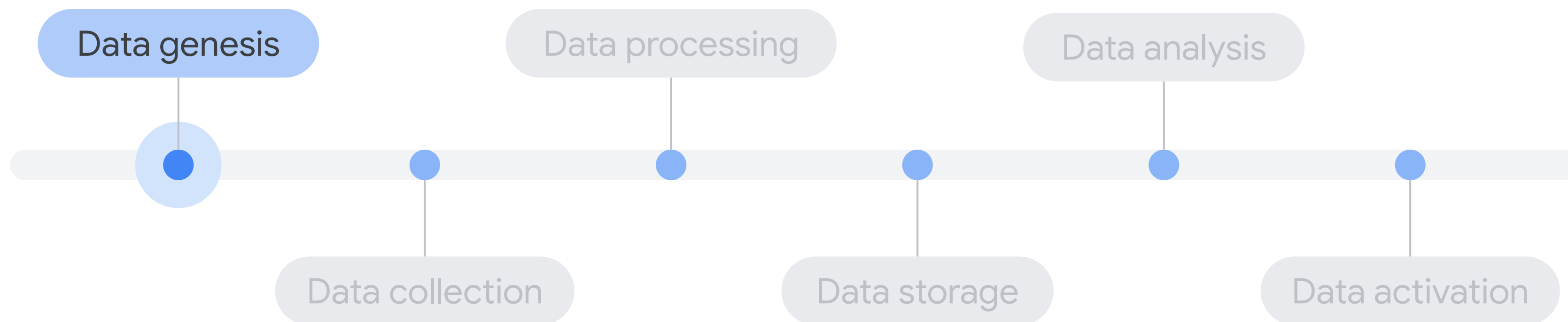
Third-party data

Datasets collected and managed by organizations that don't directly interact with an organization's customers or business

Data value chain



Step 1: Data genesis



Website click

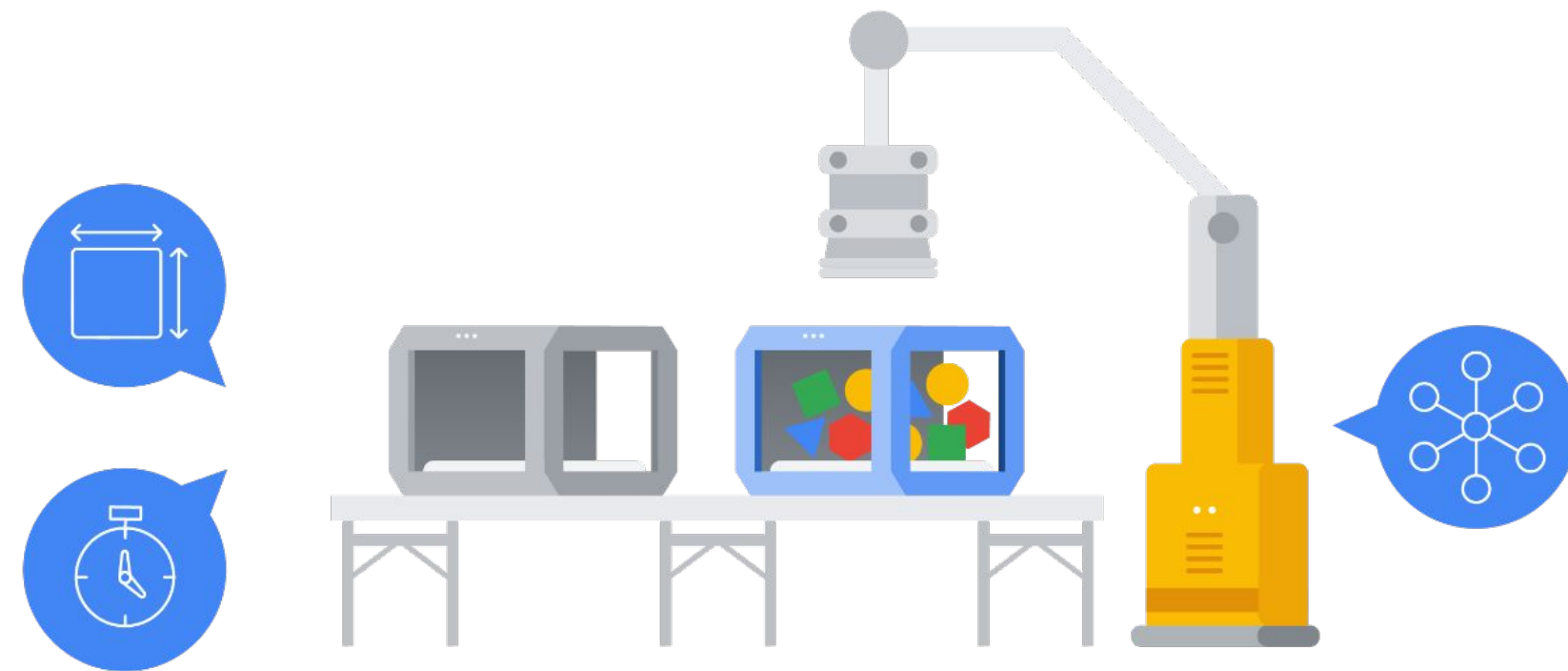
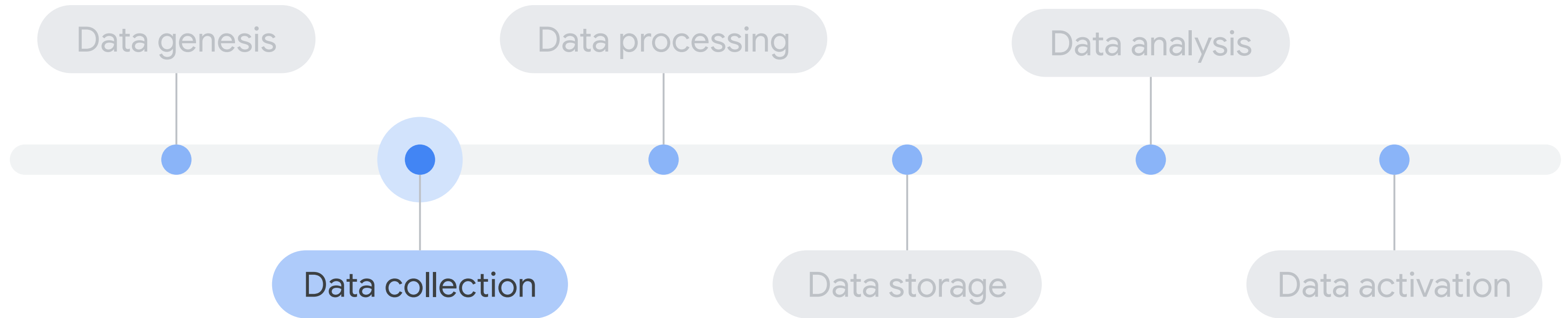


Credit card swipe

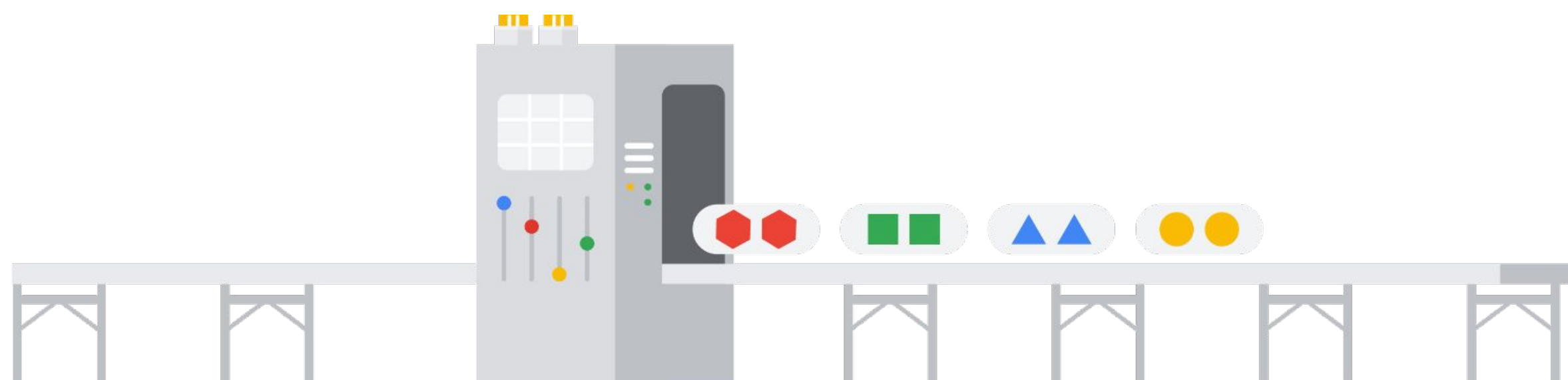
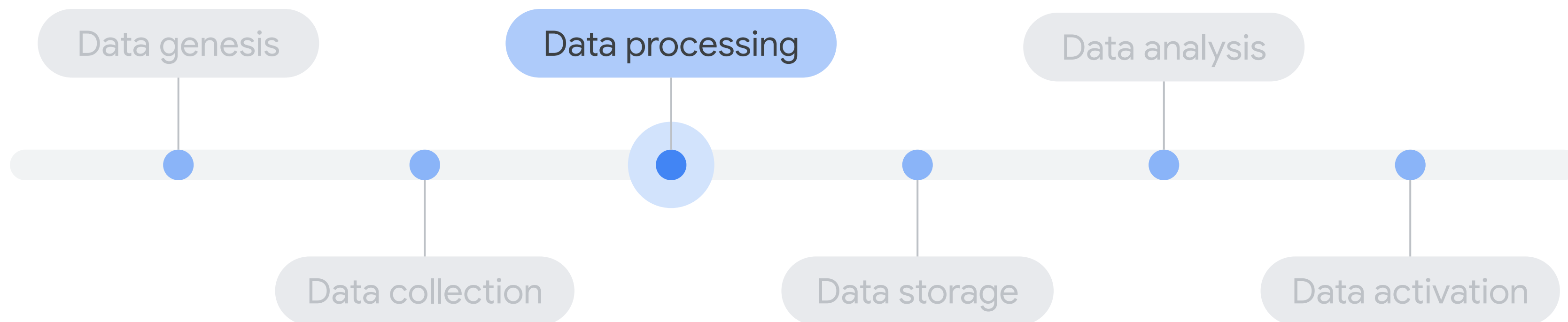


IoT sensor recording

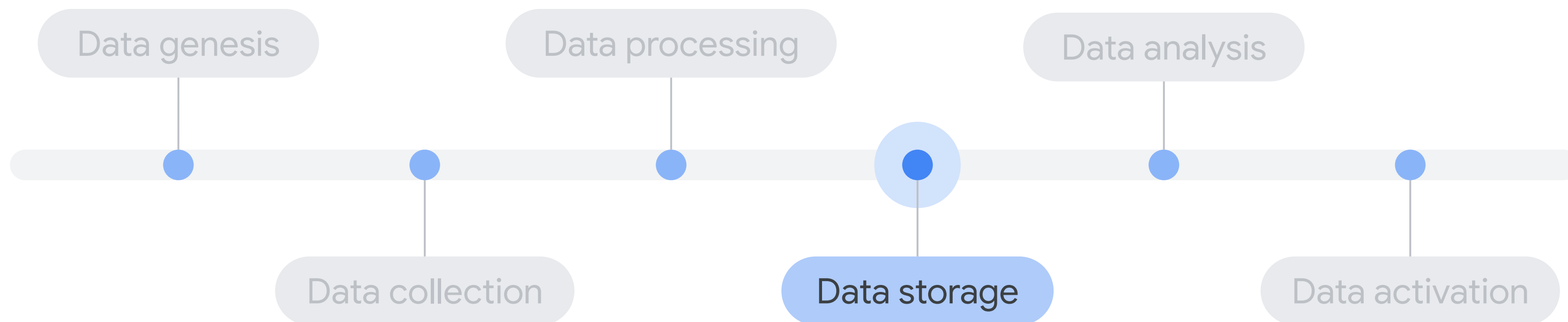
Step 2: Data collection



Step 3: Data processing



Step 4: Data storage



NoSQL

Fast reads and writes



Data warehousing

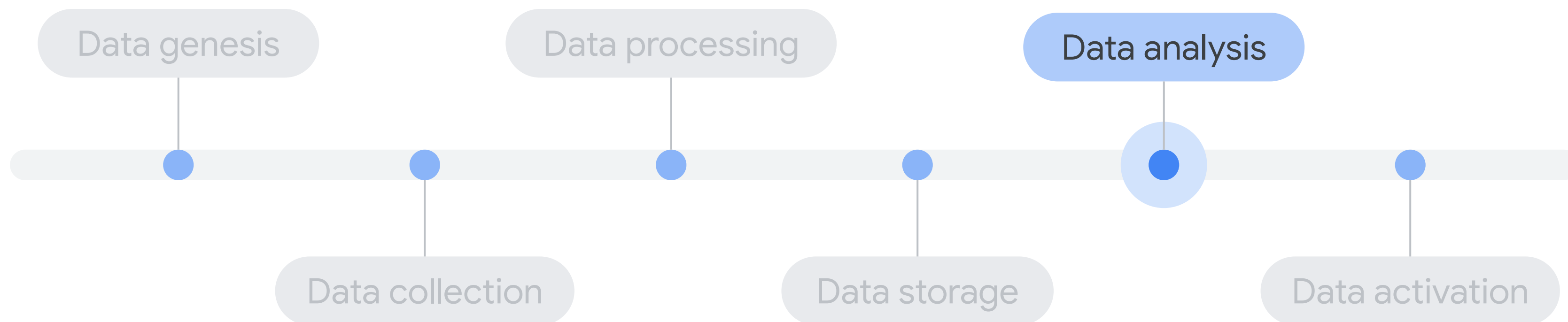
Fast access to analysis



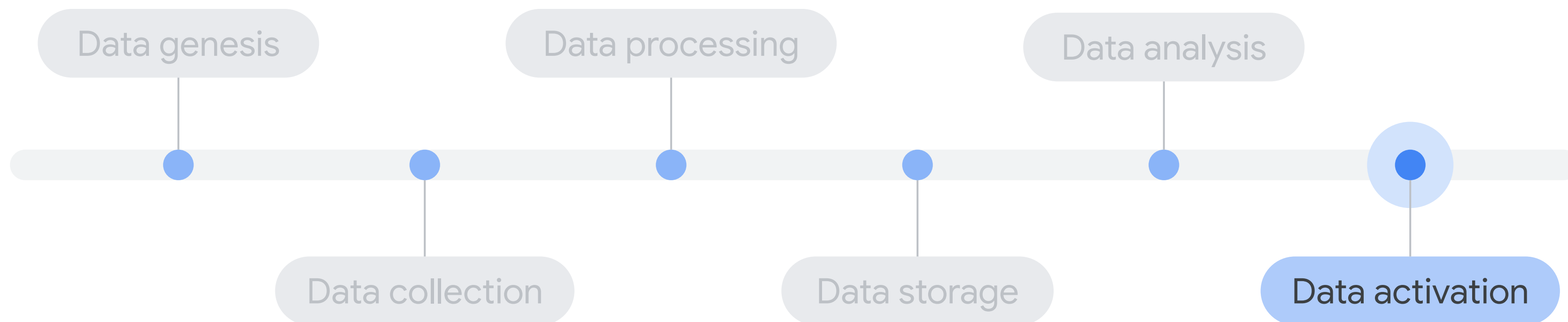
Object storage

Unstructured data

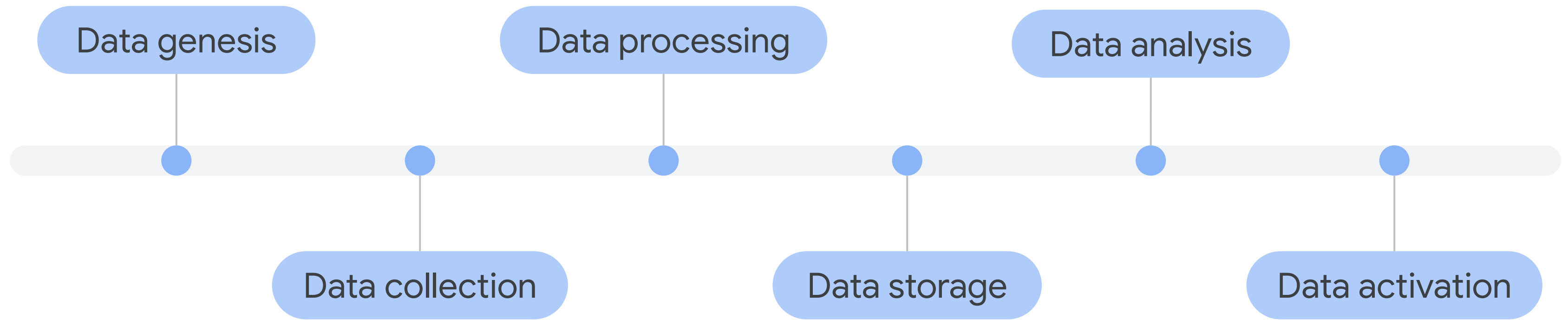
Step 5: Data analysis



Step 6: Data activation



There is no one way to assemble a data value chain



Quiz

Question

Which of these best defines the **data value chain**?

- A. The steps involved in collecting, storing, and processing data.
- B. A framework for evaluating the efficiency of data processing systems.
- C. A set of best practices for data security and compliance.
- D. The entire process of creating and using data to generate business value.

Quiz

Answer

Which of these best defines the **data value chain**?

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Quiz

Question

Which step in the data value chain describes when collected raw data is transformed into a form that's ready to derive insights from?

- A. Data genesis
- B. Data processing
- C. Data storage
- D. Data analysis

Quiz

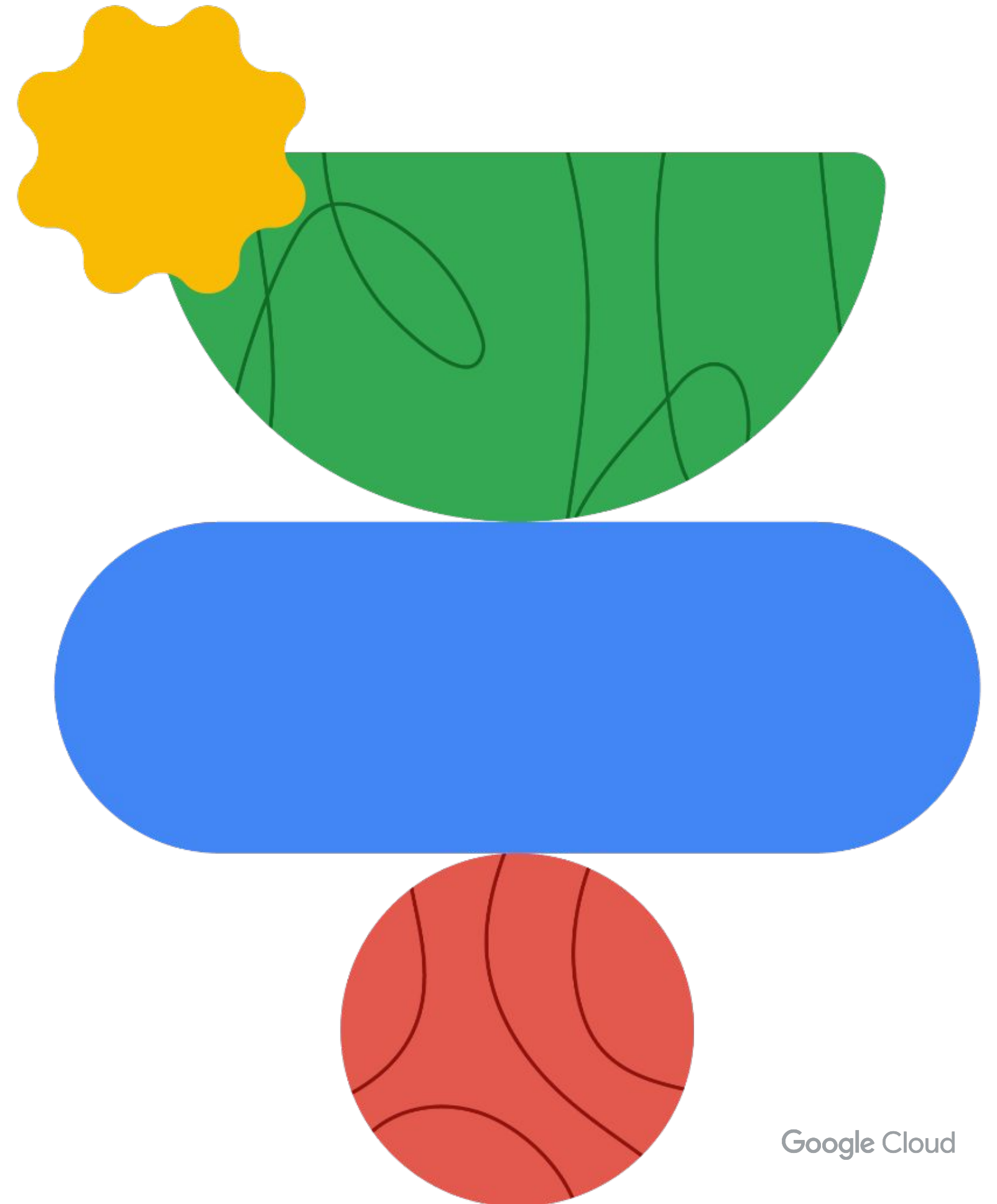
Answer

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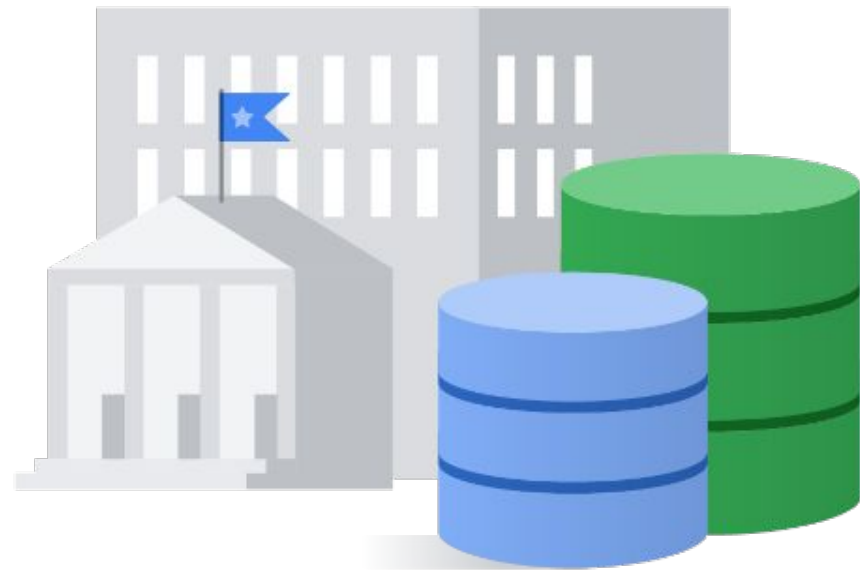
- A. Data genesis
- B. Data processing
- C. Data storage
- D. Data analysis



Data governance



Data governance



- ✓ Setting internal standards for data
- ✓ Granting access permissions
- ✓ Complying with external standards

Data governance benefits

It makes data more valuable.

It helps users make better, more timely decisions.

It improves cost controls.

It enhances regulatory compliance.

It helps earn greater trust from customers and suppliers.

It helps manage risk.

It allows more personnel access to more data.

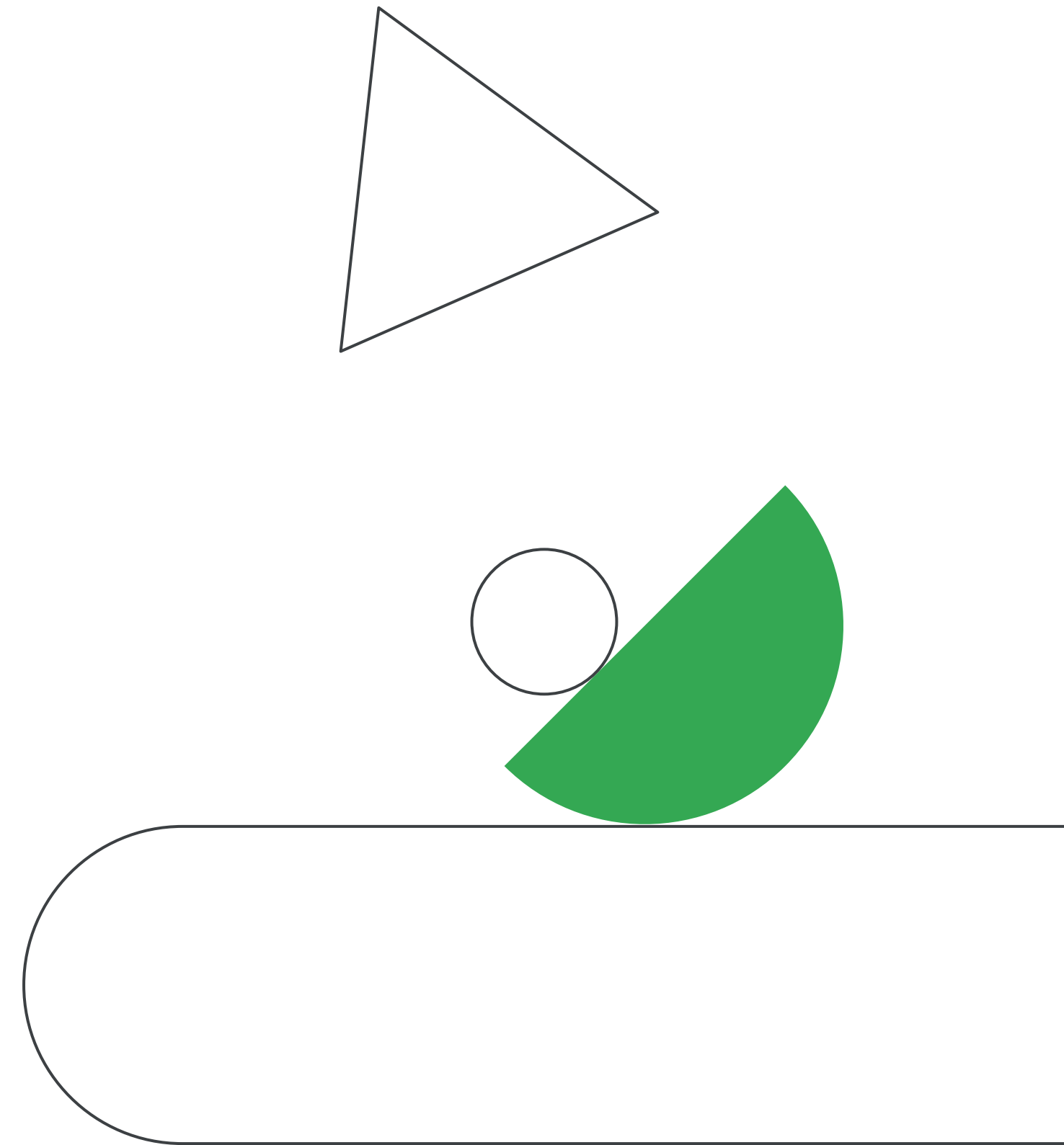


Data governance

Example 1

A coworking office rental business uses an **online tool to record** daily desk, room, and meeting **bookings**. If a client books a desk for the day, that data is captured and desk availability is **updated in real time** on all customer channels. The rental business now wants to do even more with their data. They **want to use multiple types and sources of data to gain insights about facility quality** and, ultimately, to **improve their service** to customers.

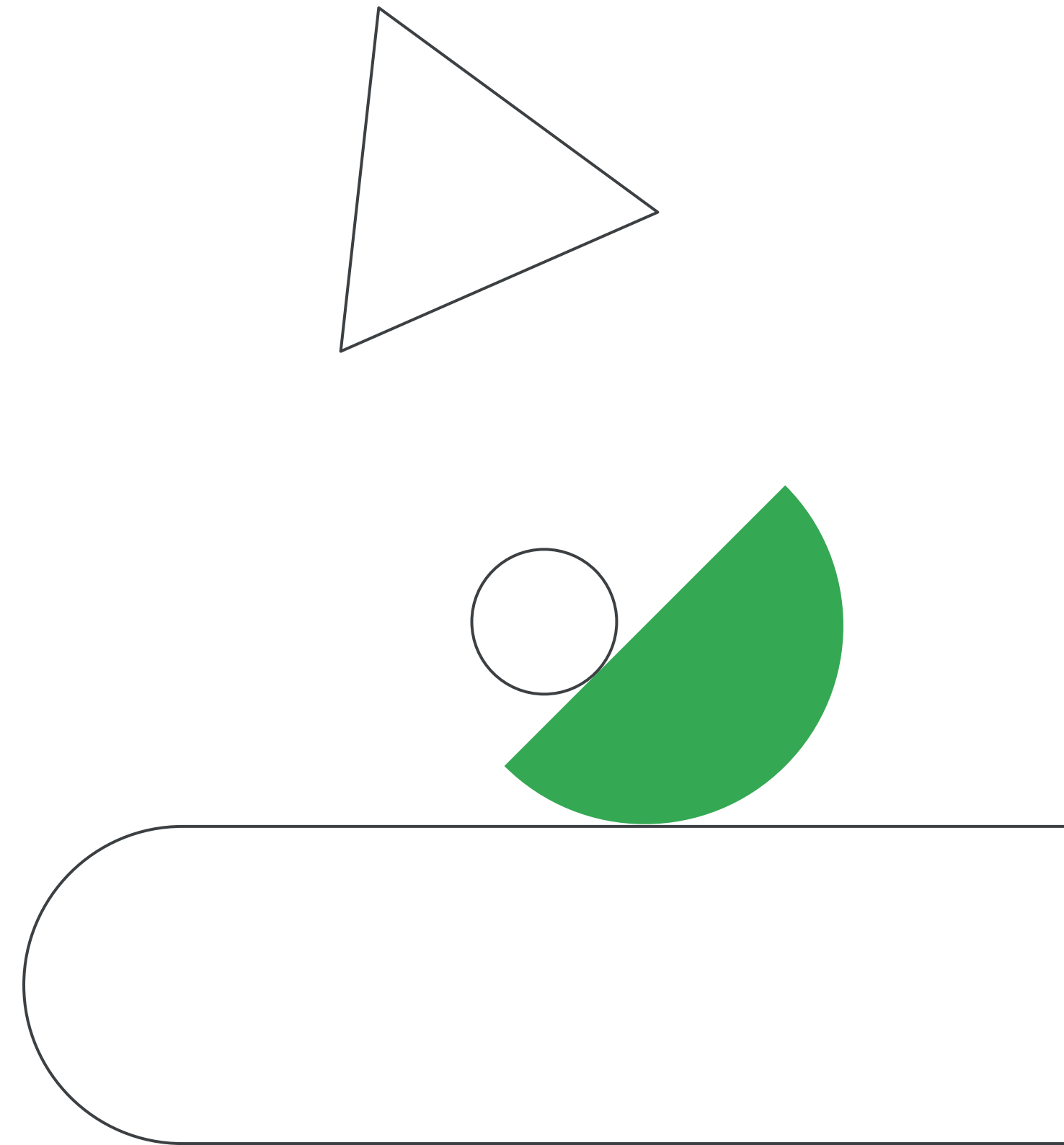
Which data management solution best fits the coworking business' needs?



Example 1

A coworking office rental business uses an **online tool to record** daily desk, room, and meeting **bookings**. If a client books a desk for the day, that data is captured and desk availability is **updated in real time** on all customer channels. The rental business now wants to do even more with their data. They **want to use multiple types and sources of data to gain insights about facility quality** and, ultimately, to **improve their service** to customers.

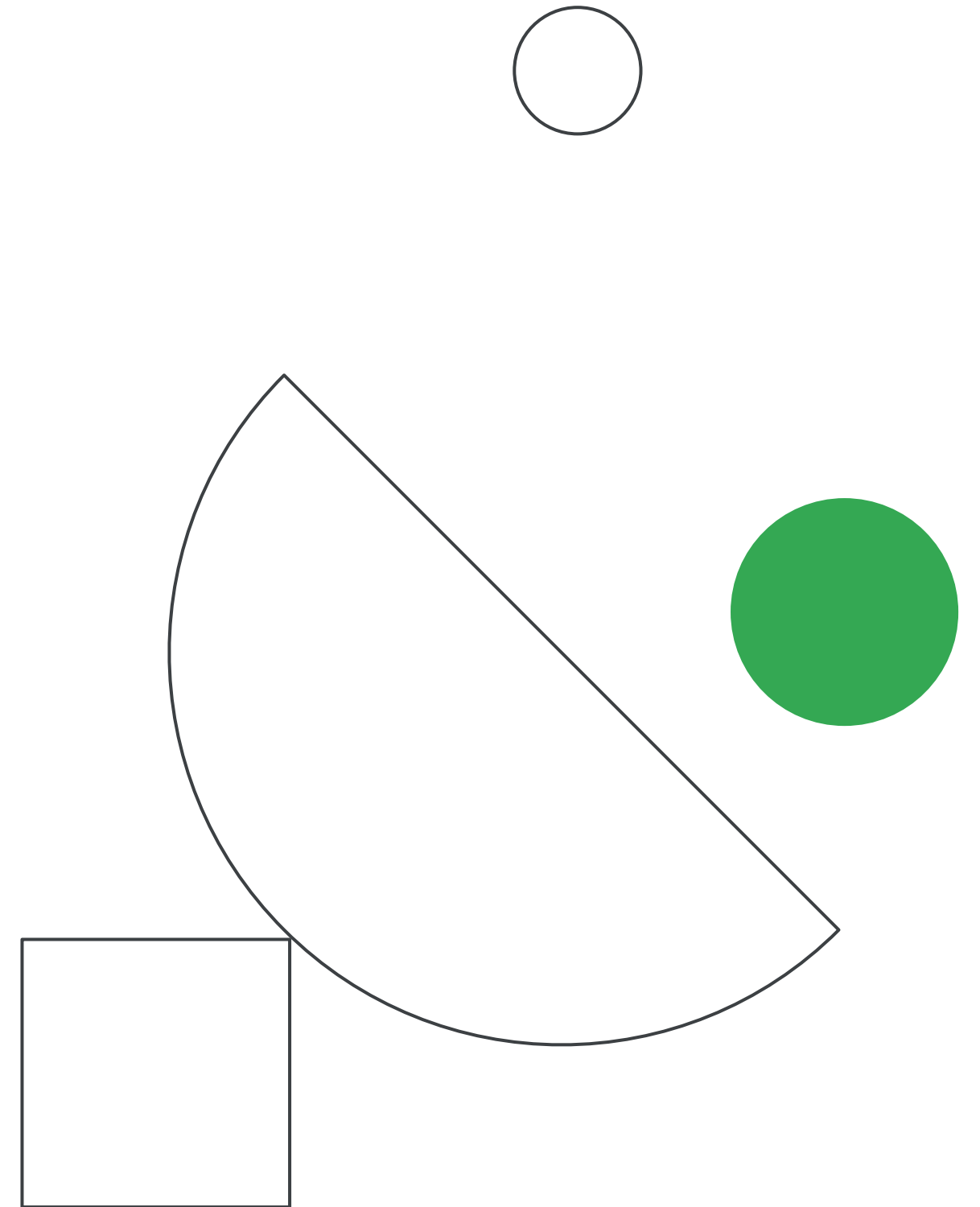
Answer: **Data warehouse**



Example 2

A bank is launching a **mobile banking app**, and wants to **track money transfers** from one account to another. They want to ensure that the transferred figure is **updated** in the bank's records in **real time** and the user is able to see the **most up-to-date account balance**.

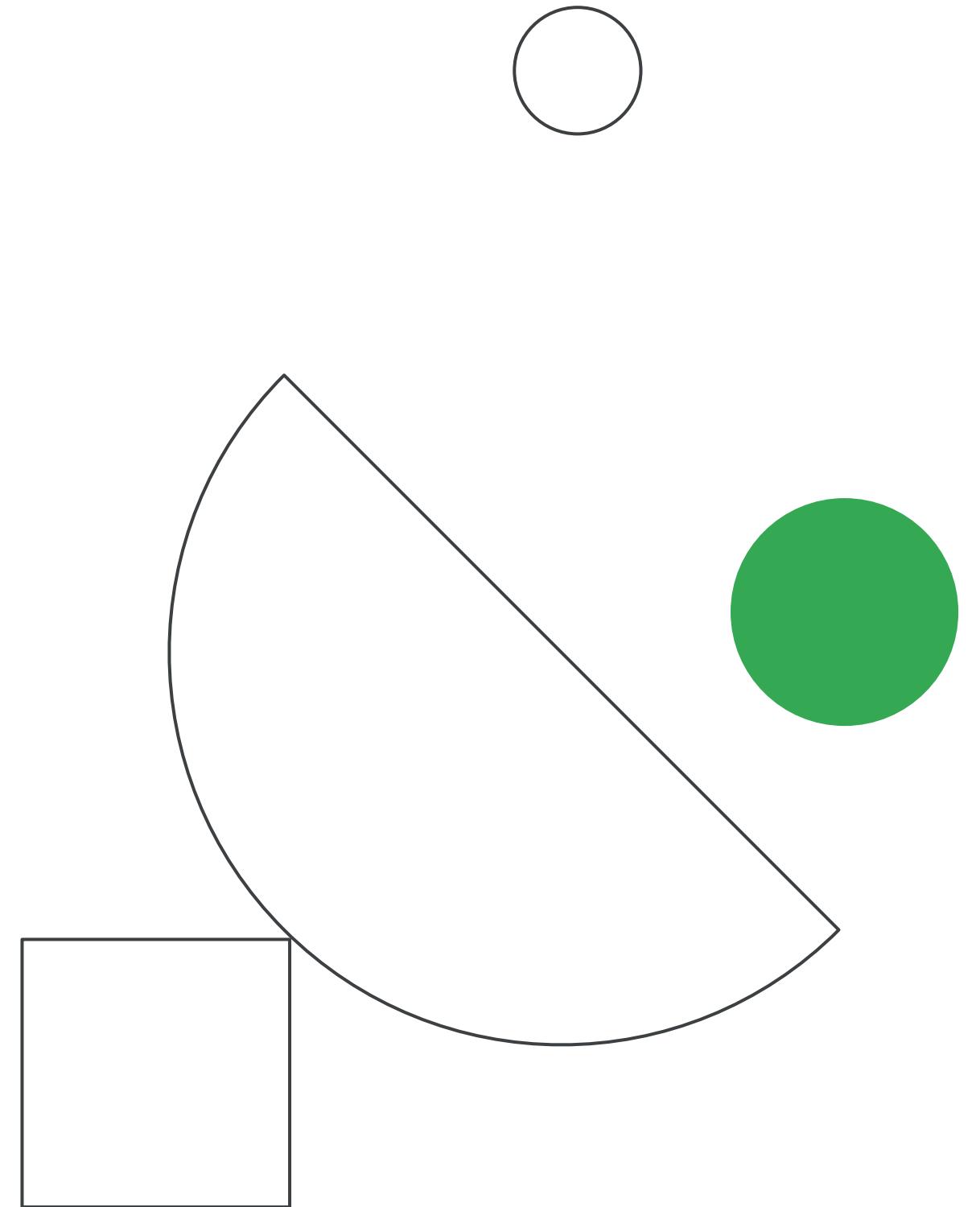
Which data management solution best fits this bank's needs?



Example 2

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Answer: **Database**



Module 2

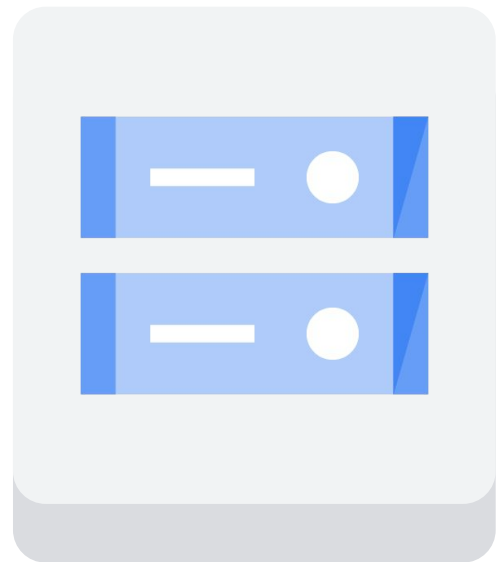
Exploring Data Transformation with Google Cloud



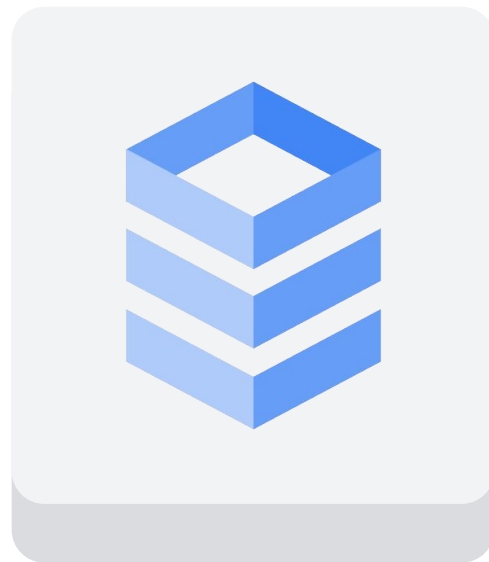
Lessons

- 01 The value of data
- 02 Google Cloud data management solutions
- 03 Making data useful and accessible

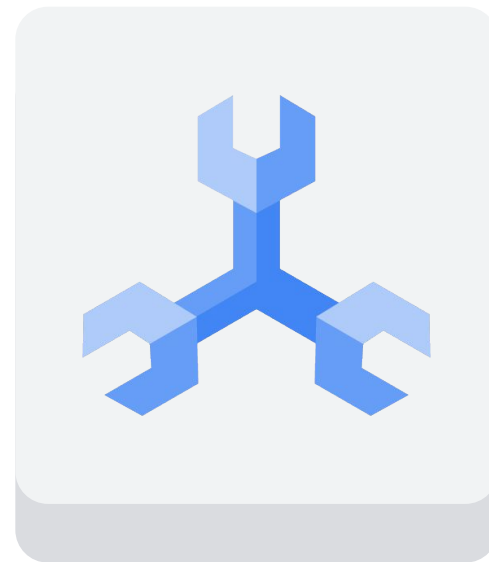
Google Cloud's core storage products



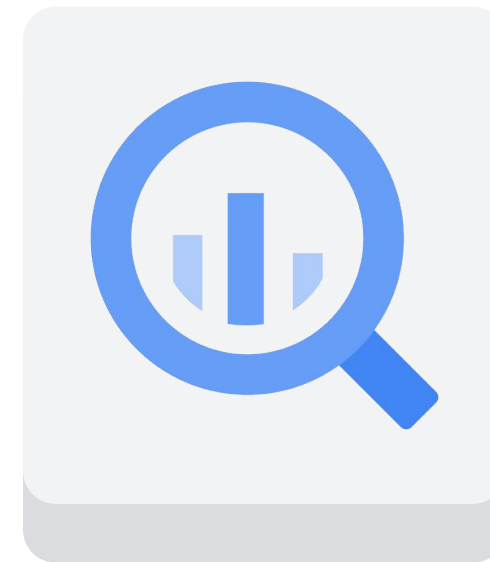
Cloud Storage



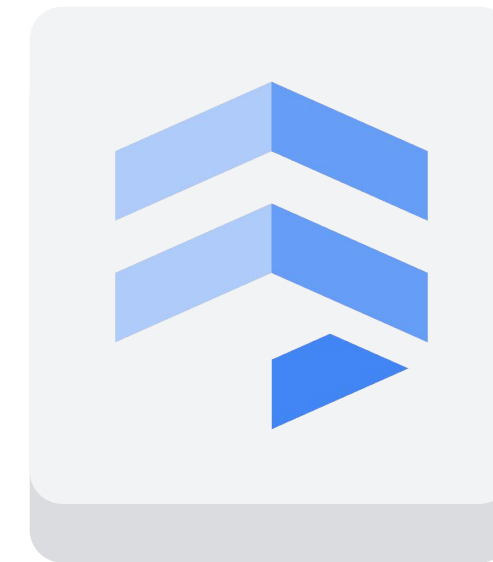
Cloud SQL



Cloud Spanner



BigQuery

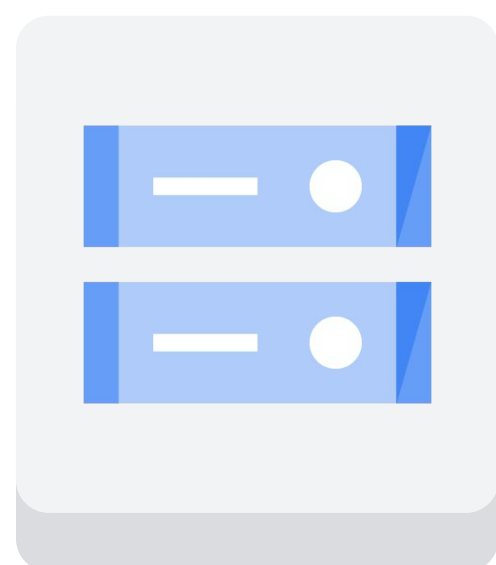


Firestore

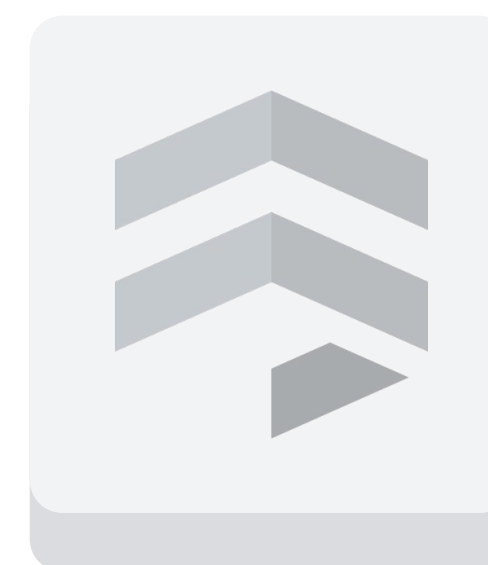
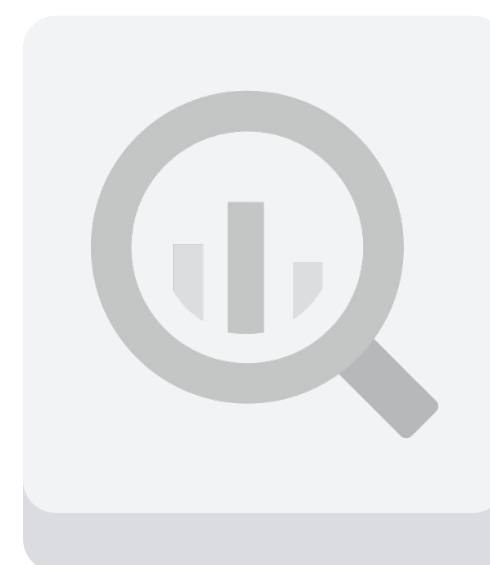
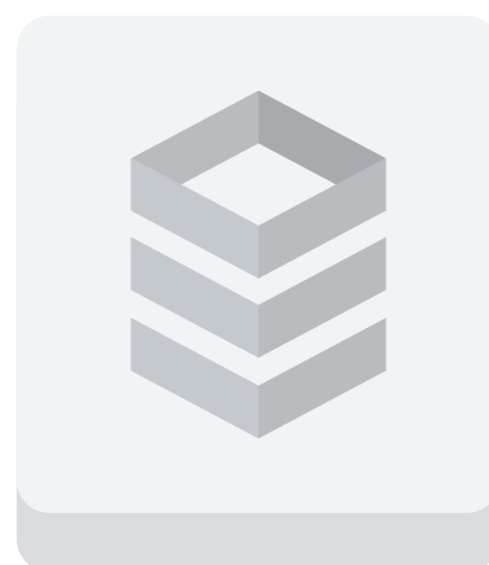


Bigtable

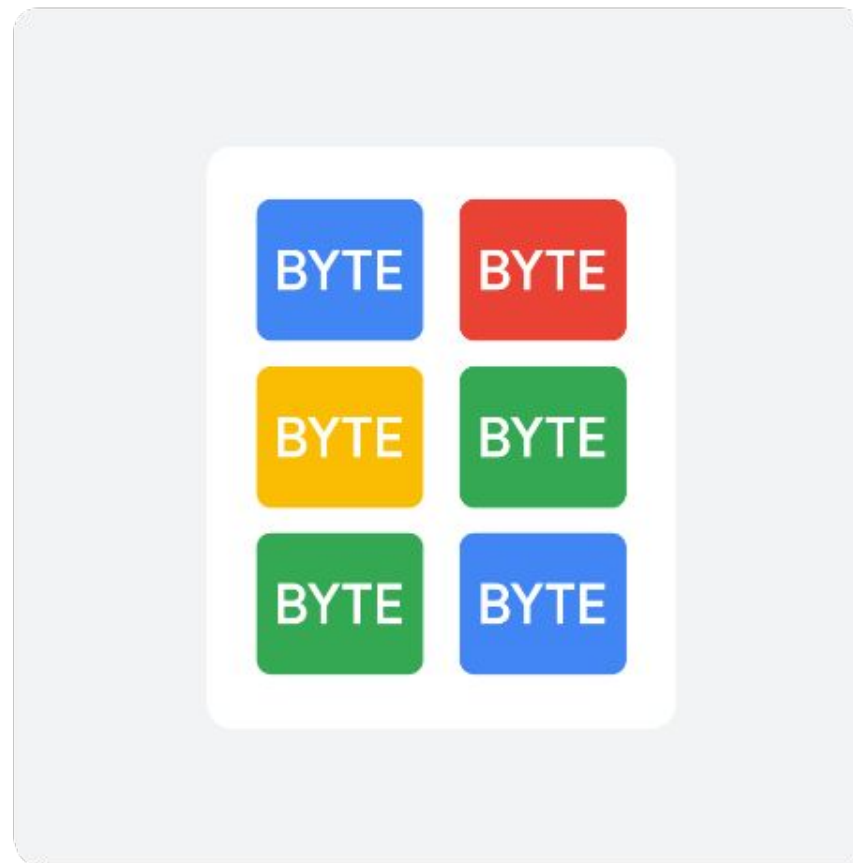
Google Cloud's core storage products



Cloud Storage



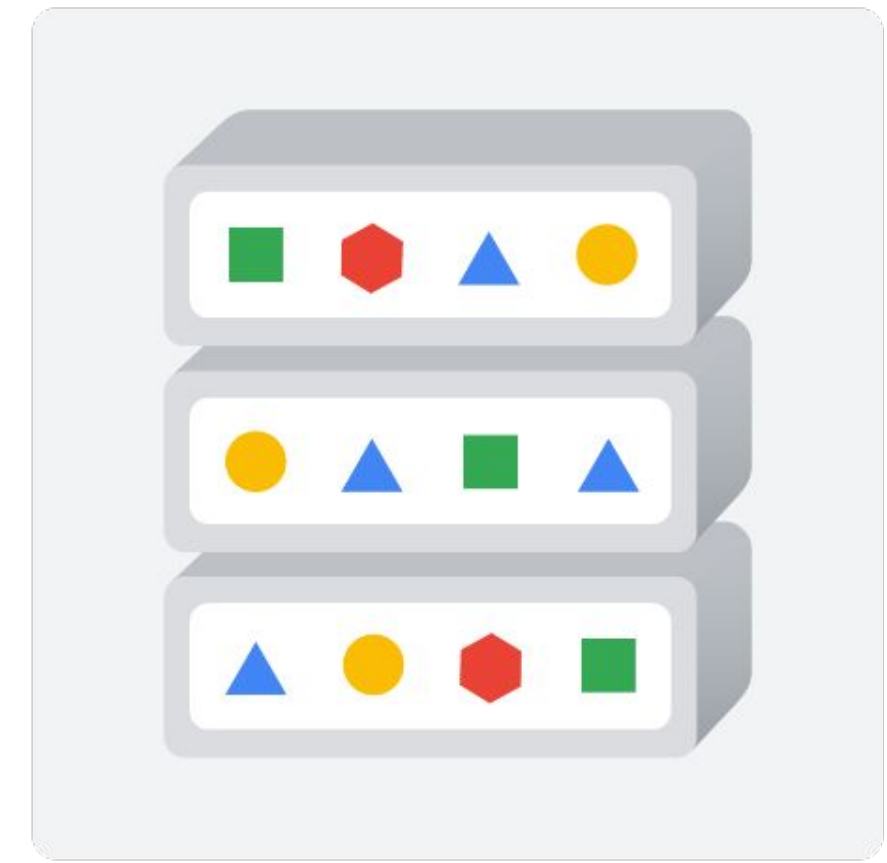
What is object storage?



Object storage
is a computer data storage
architecture that manages
data as *objects*.

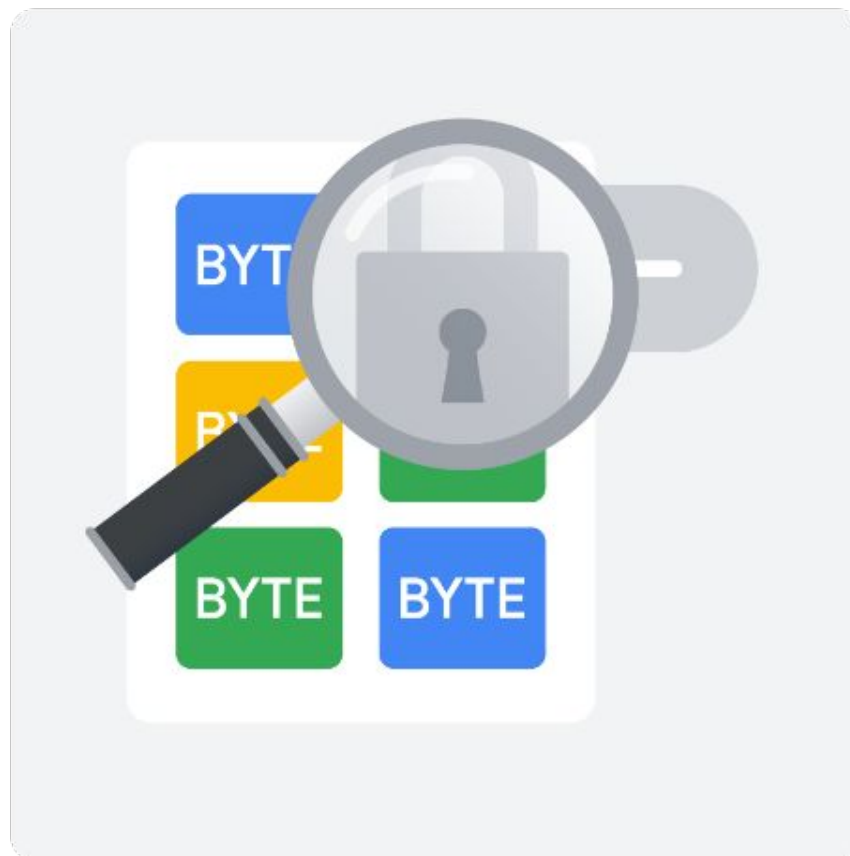


File storage
stores data in a file and
folder hierarchy.



Block storage
stores data as fixed-sized
chunks on a disk.

Object storage



Object packaged format contains:

- ✓ Binary form of the actual data itself.
- ✓ Relevant associated metadata.
- ✓ Globally unique identifier.

Cloud Storage's four primary storage classes

Standard storage



Hot data

Nearline storage



Once per month

Coldline storage



Once every 90 days

Archive storage

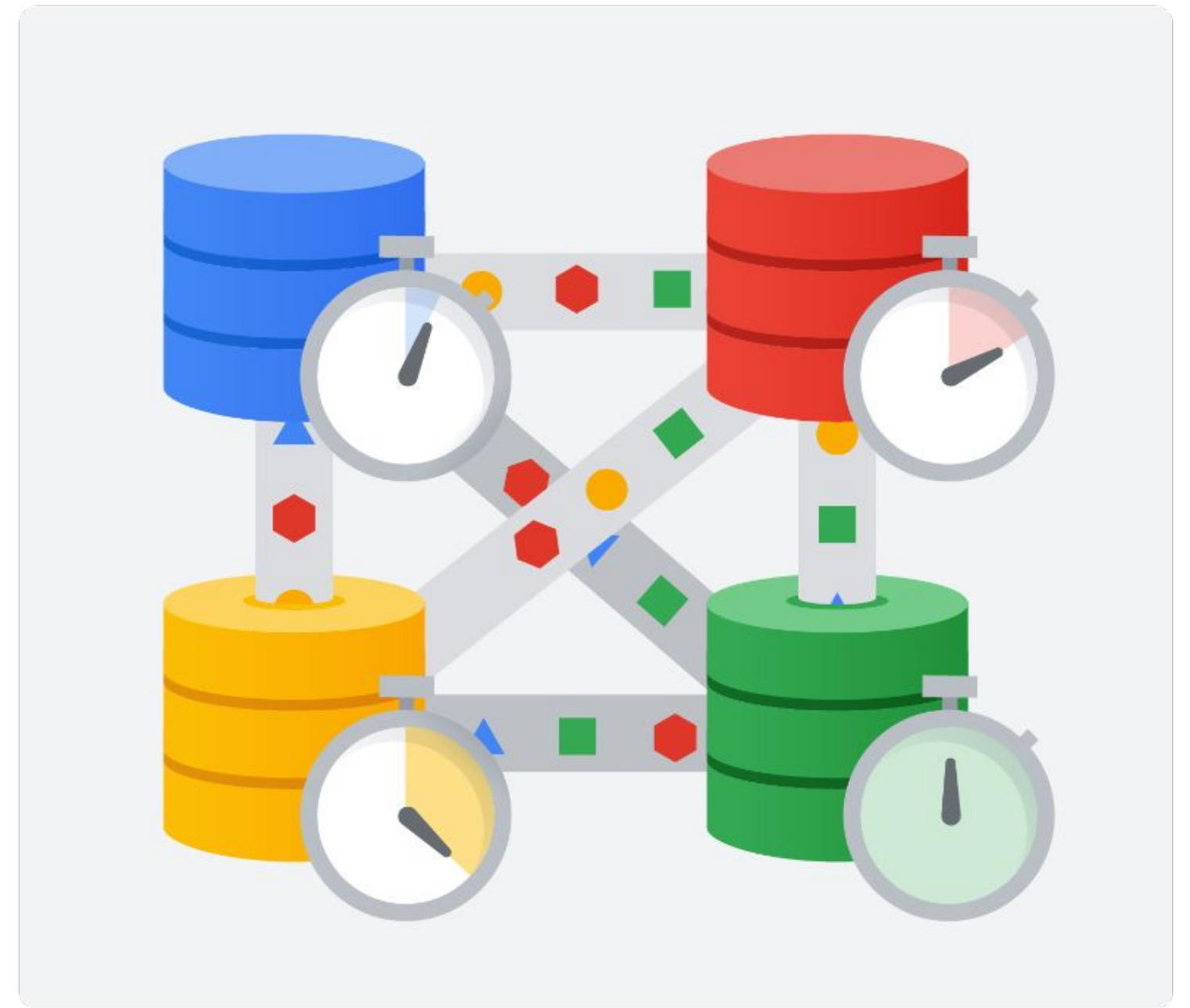


Once a year

Autoclass automatically transitions objects to appropriate storage classes

Autoclass

- ✓ Moves data that is not accessed to colder storage classes to reduce storage cost.
- ✓ Moves data that is accessed to standard storage to optimize future accesses.



Quiz

Question

A data analyst for an online retailer must produce a sales report at the end of each quarter. Which Cloud Storage class should the retailer use for data accessed every 90 days?

- A. Coldline
- B. Standard
- C. Nearline
- D. Archive

Quiz

Answer

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- A. Coldline
- B. Standard
- C. Nearline
- D. Archive



Quiz

Question

Which characteristic is true for all Cloud Storage classes?

- A. Maximum storage limits
- B. Accessibility only within one region
- C. Geo-redundancy if data is stored in a multi-region or dual-region
- D. High latency and low durability

Quiz

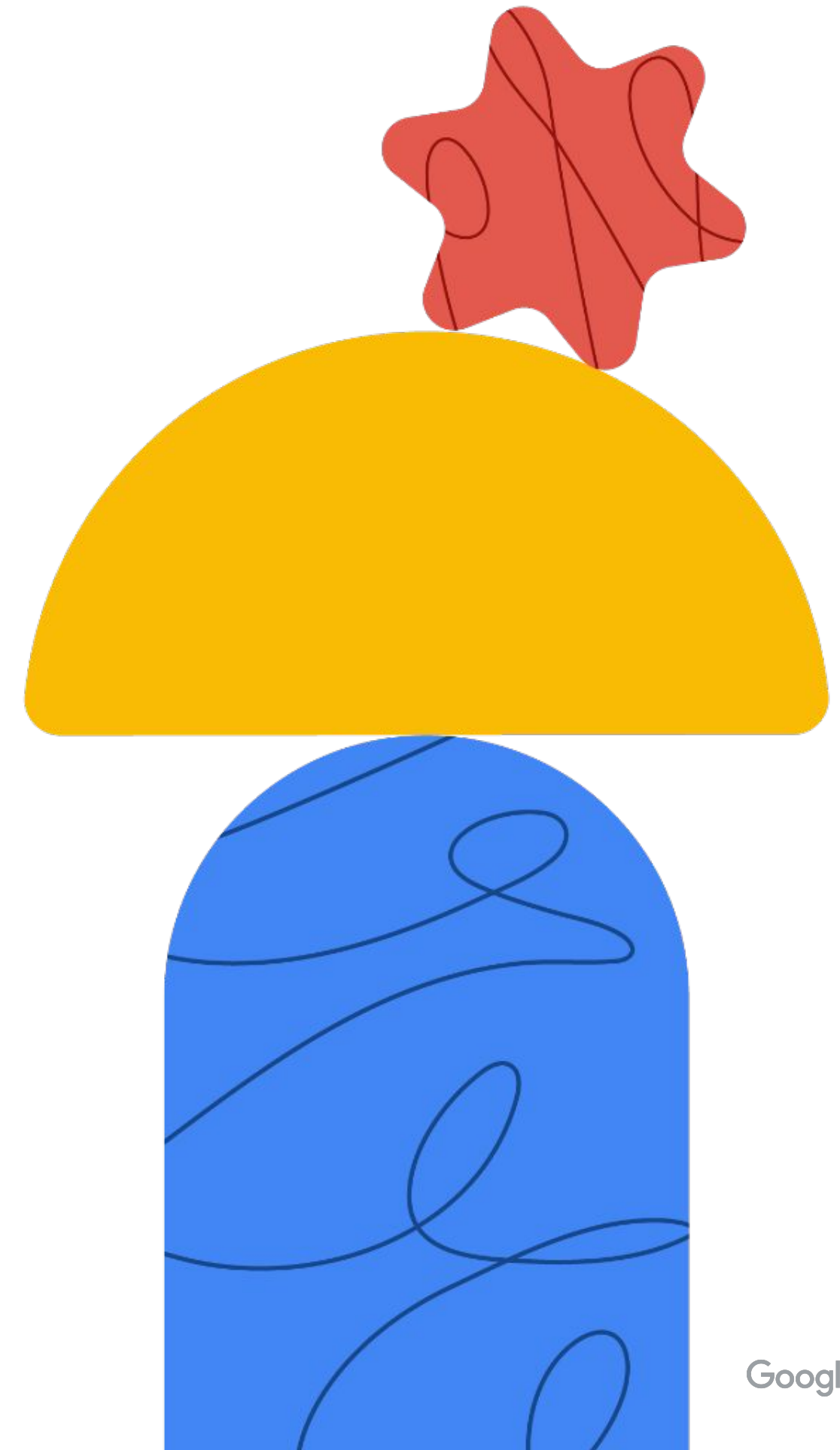
Answer

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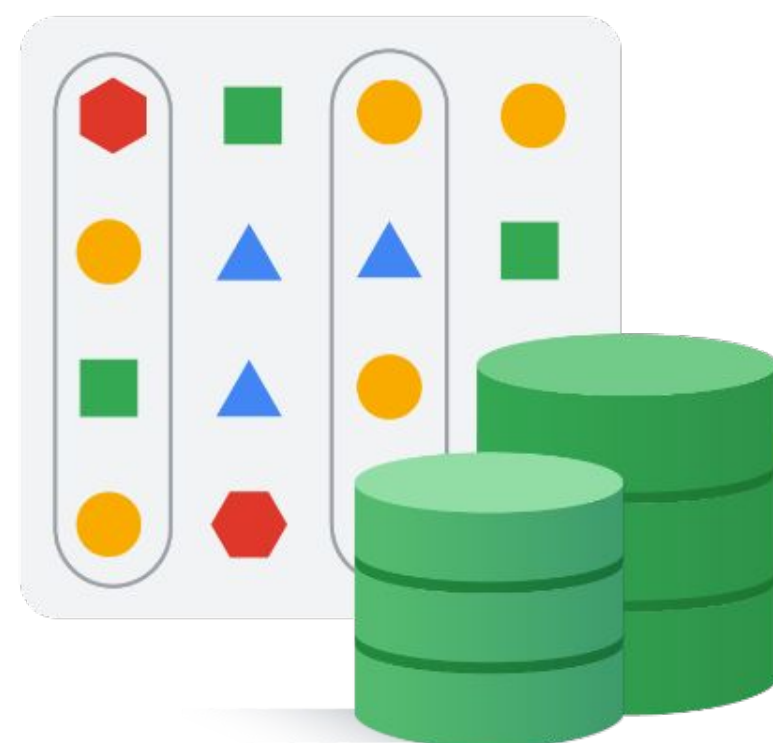
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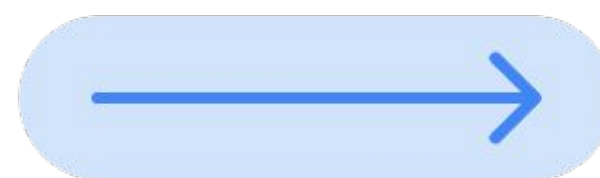
Structured data storage



Structured data



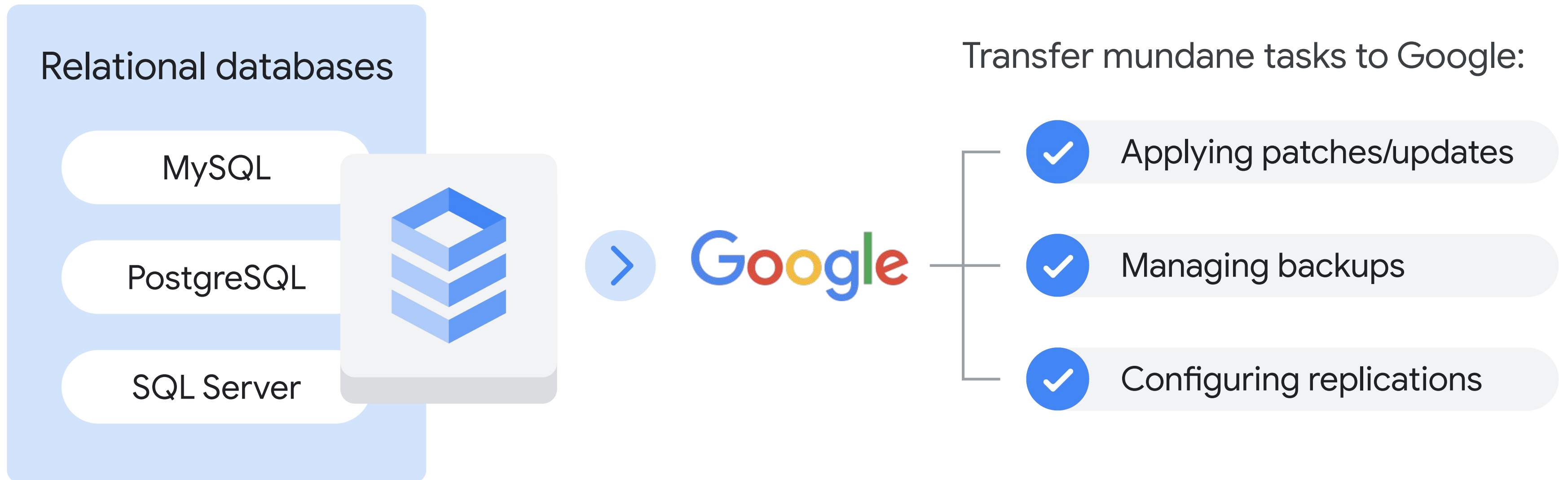
Relational databases



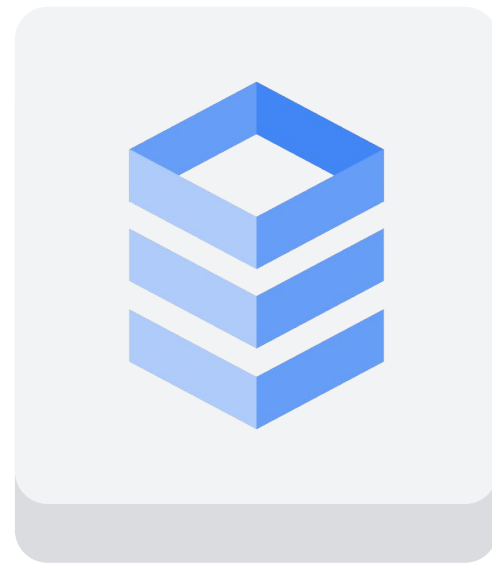
green square	blue triangle	yellow circle	red hexagon	yellow circle
yellow circle	red hexagon	yellow circle	blue triangle	green square
blue triangle	yellow circle	red hexagon	green square	blue triangle
red hexagon	green square	blue triangle	yellow circle	green square
green square	blue triangle	yellow circle	red hexagon	blue triangle

Stores information in **tables**, **rows**, and **columns** that have a clearly defined schema that represents the structure or logical configuration of the database.

Cloud SQL



Cloud SQL benefits



Cloud SQL

Is trusted by thousands of the largest enterprises around the world.

Doesn't require software installation or maintenance.

Supports managed backups.

Encrypts customer data.

Includes a network firewall.

Cloud Spanner



A fully managed, mission-critical, relational database service that scales horizontally to handle unexpected business spikes



Differences between Cloud SQL and Cloud Spanner

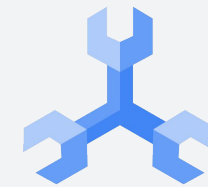


Cloud SQL

It's a fully managed relational database service for MySQL, PostgreSQL, and SQL Server.

Its availability is greater than 99.95%.

DMS makes it easy to migrate your production databases to Cloud SQL with minimal downtime.



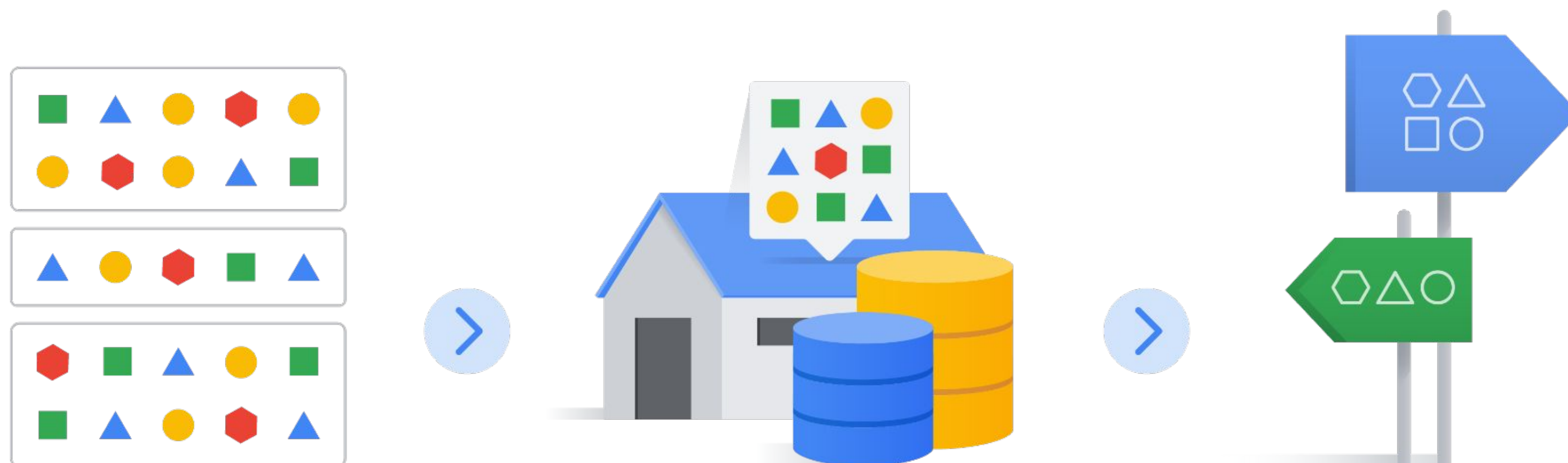
Cloud Spanner

It's a fully managed relational database with unlimited scale, strong consistency.

It's availability reaches 99.999%.

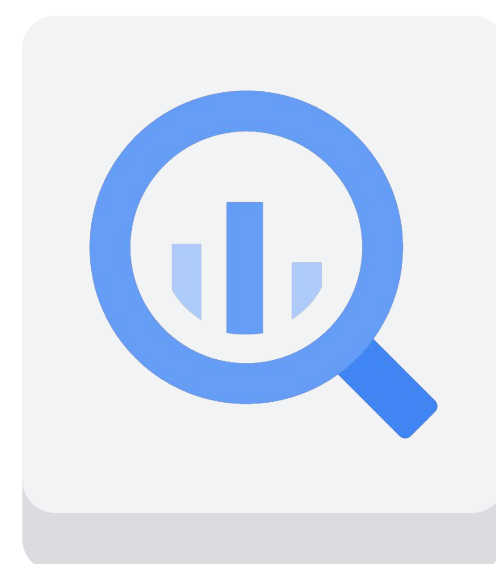
It handles replicas, sharding, and transaction processing, so you can quickly scale to meet any usage pattern and ensure success of products.

BigQuery

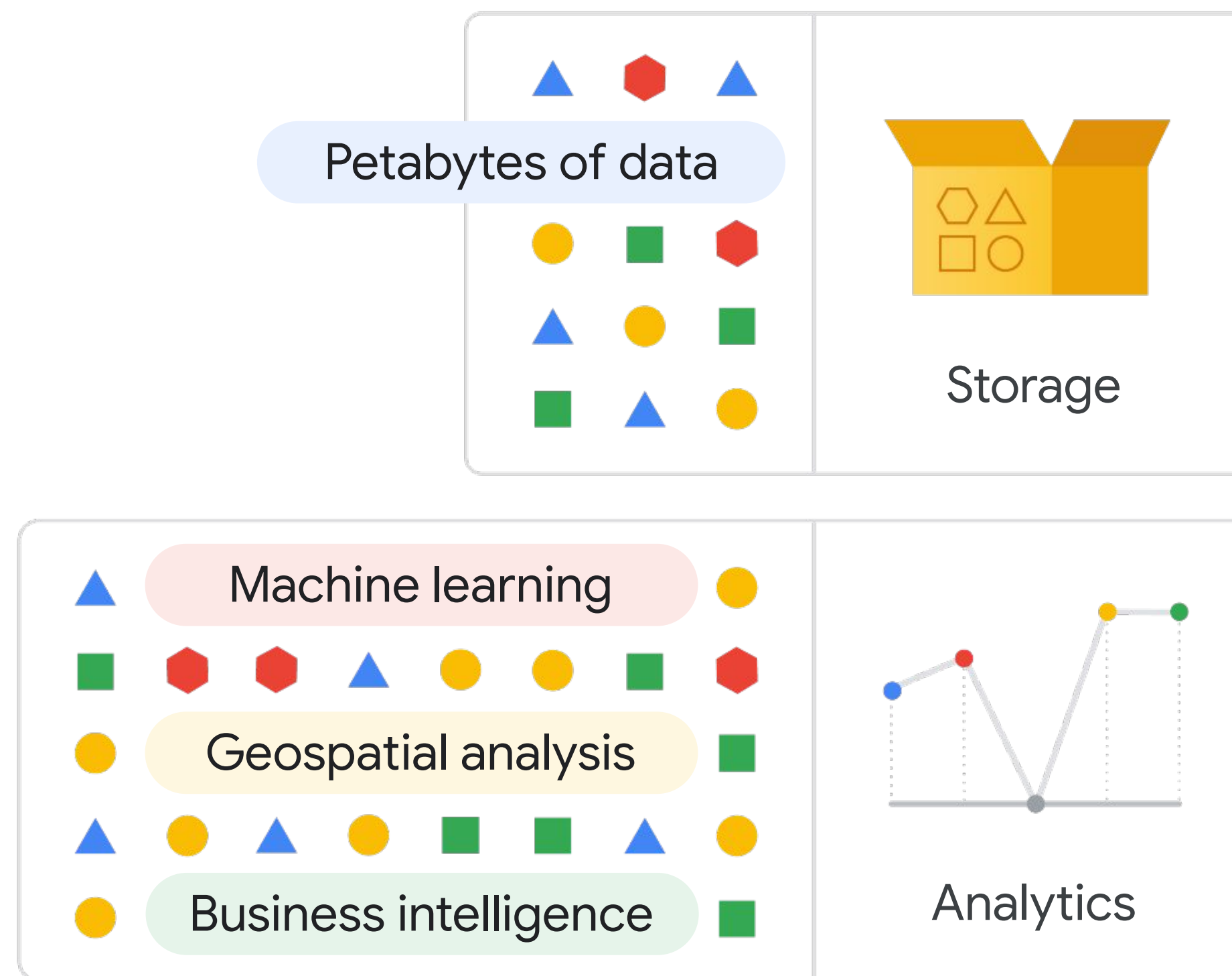


BigQuery is a fully-managed data warehouse.

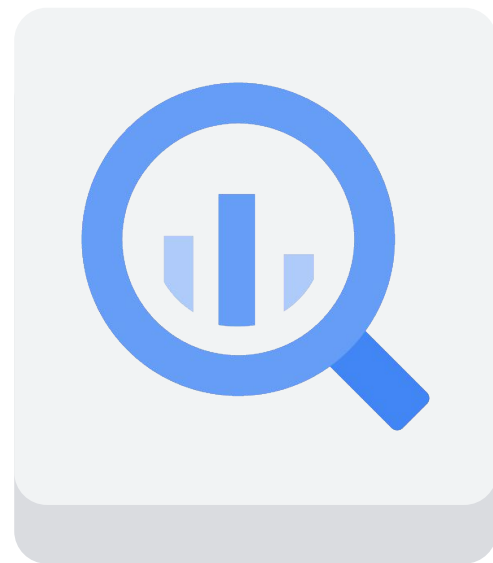
BigQuery provides two services in one



BigQuery



BigQuery features



BigQuery

Data in BigQuery is encrypted at rest by default.

BigQuery seamlessly integrates with the partner ecosystem.

BigQuery works in a multicloud environment.

BigQuery has built-in machine learning features.

Quiz

Question

Which is the best SQL-based storage option for a transactional workload that requires local or regional scalability?

- A. Cloud Storage
- B. Cloud Spanner
- C. BigQuery
- D. Cloud SQL

Quiz

Answer

Which is the best SQL-based storage option for a transactional workload that requires local or regional scalability?

- A. Cloud Storage
- B. Cloud Spanner
- C. BigQuery
- D. Cloud SQL



Quiz

Question

Which would be the best SQL-based storage option for a transactional workload that requires global scalability?

- A. Cloud Storage
- B. Cloud Spanner
- C. BigQuery
- D. Cloud SQL

Quiz

Answer

Which would be the best SQL-based storage option for a transactional workload that requires global scalability?

A. Cloud Storage

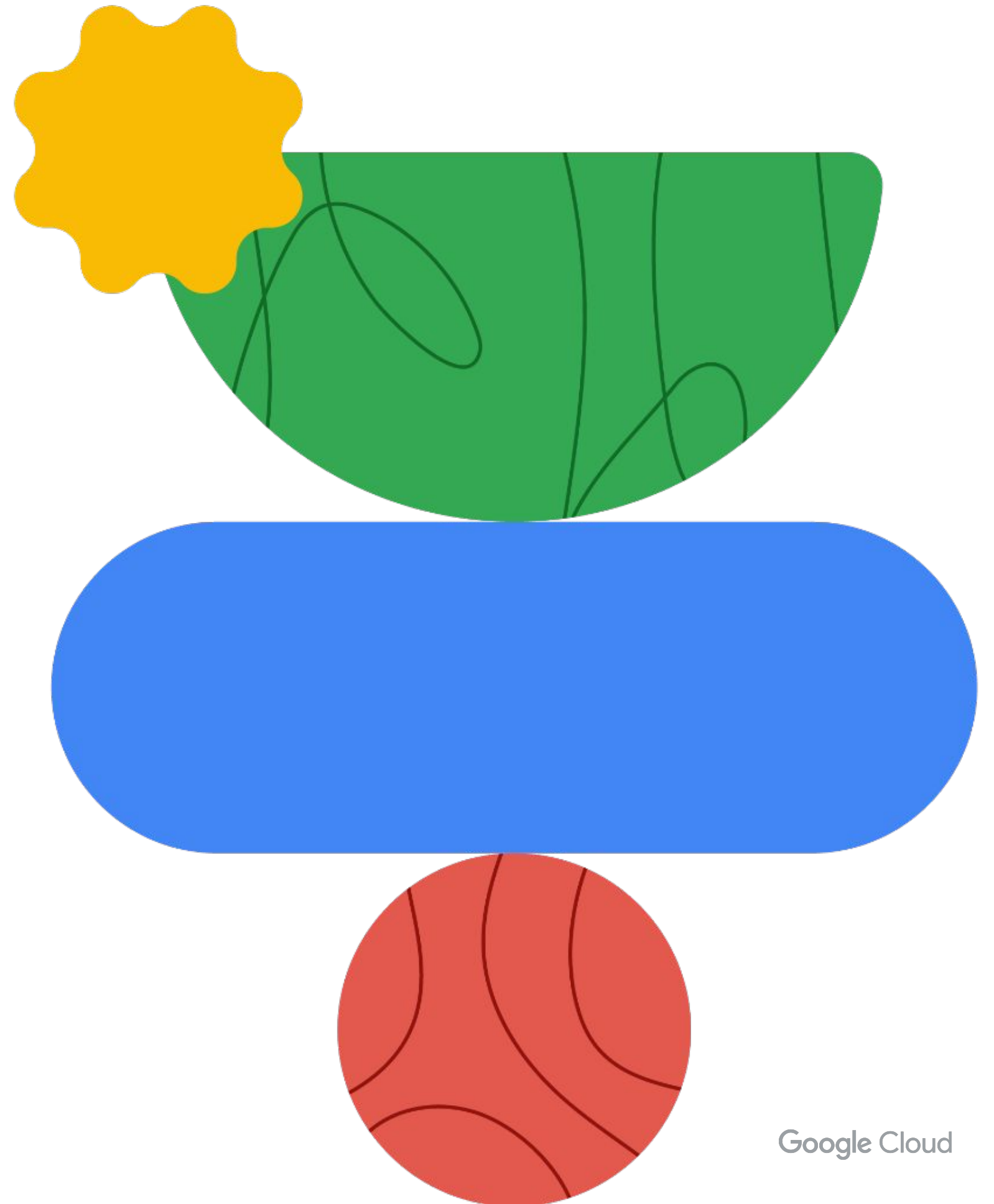
B. Cloud Spanner

C. BigQuery

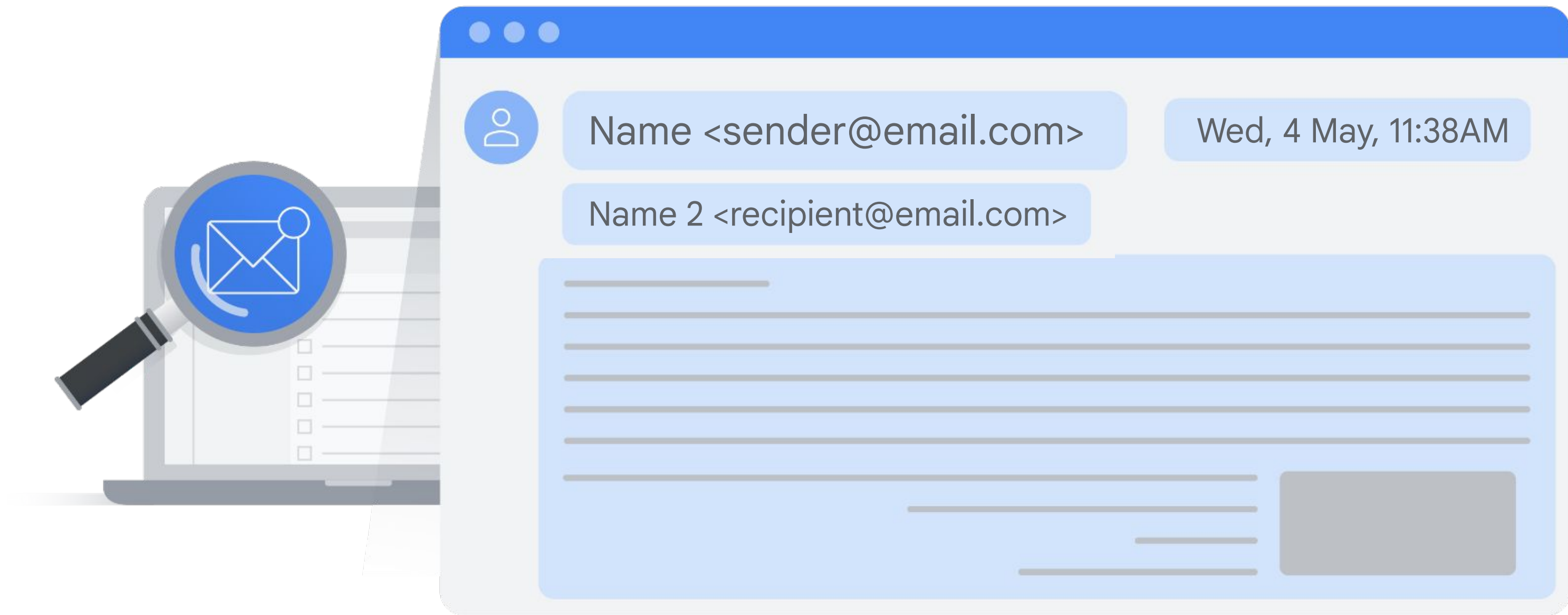
D. Cloud SQL



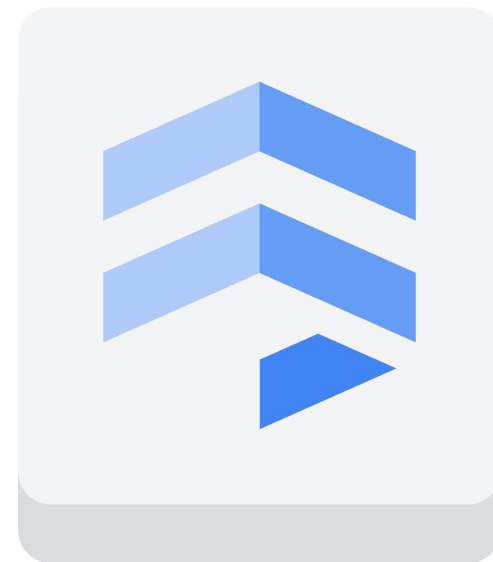
Semi-structured data storage



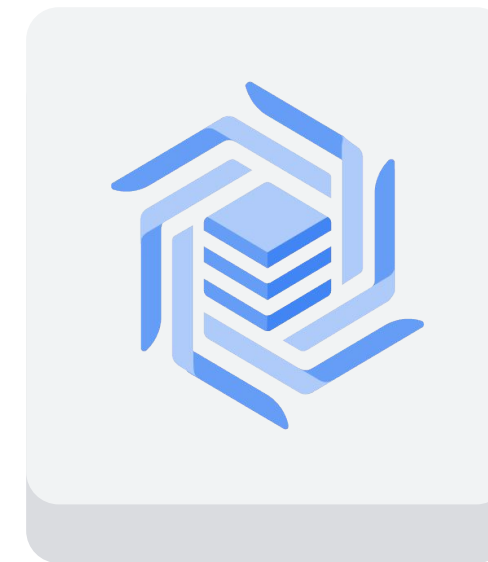
An email has unstructured data and structured data



Google Cloud semi-structured data storage products

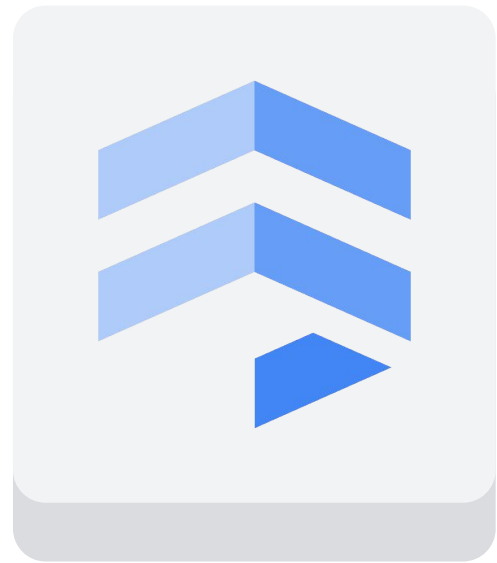


Firestore



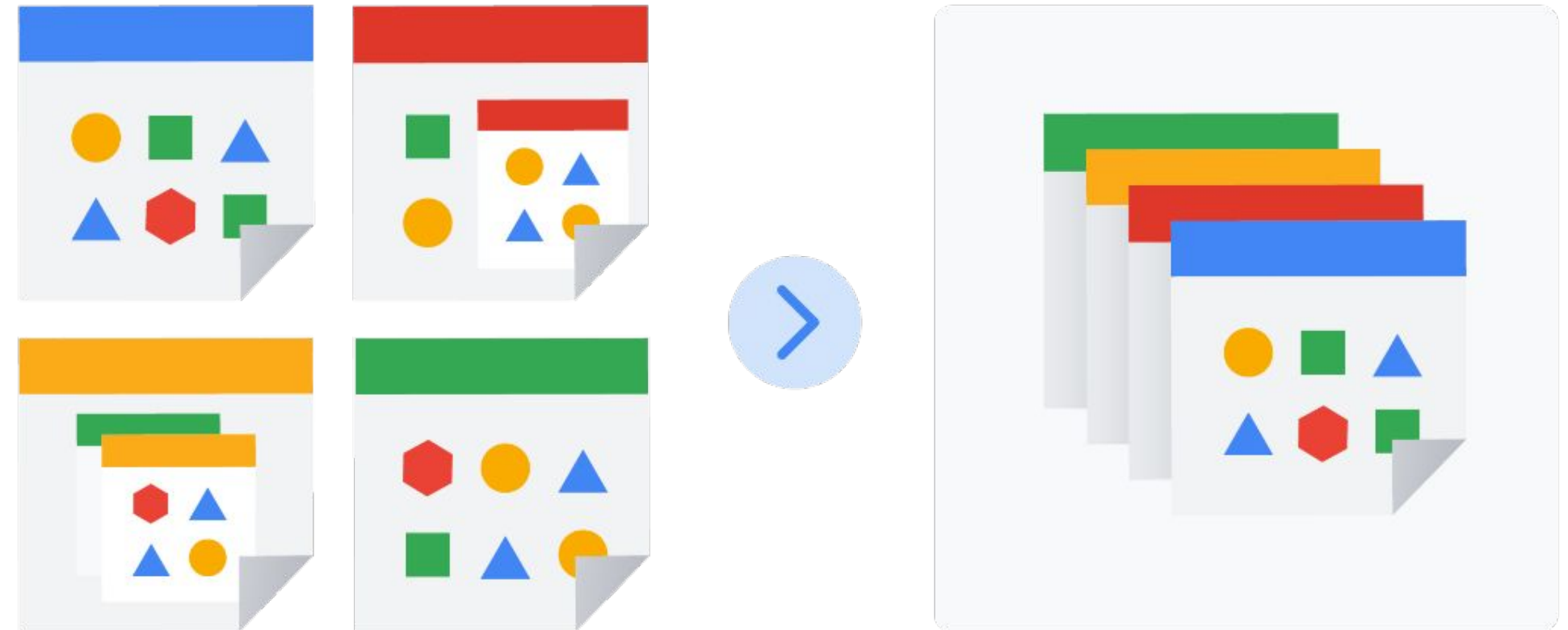
Cloud Bigtable

Firestore



Firestore

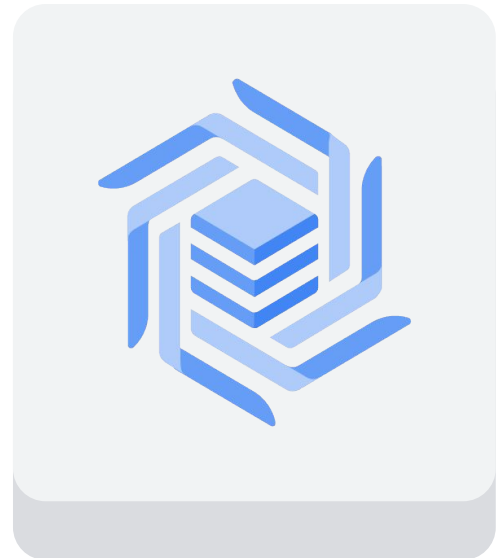
Firestore is a flexible, horizontally scalable, NoSQL cloud database for storing and syncing data in real-time..



It performs data storage in the form of **documents**.

Documents are stored in **collections**.

Cloud Bigtable

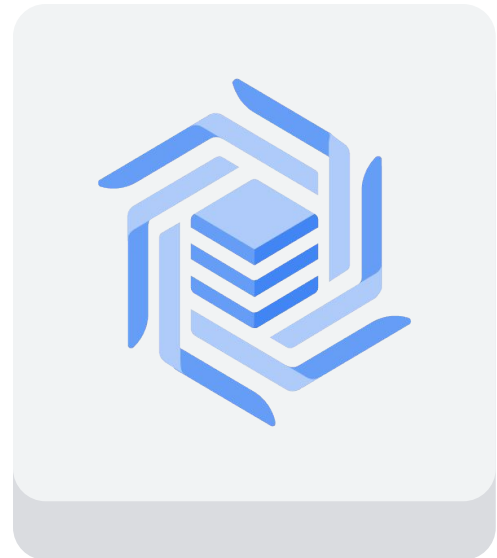


Cloud Bigtable

Google's NoSQL big data database service



Cloud Bigtable



Cloud Bigtable

Google's NoSQL big data database service

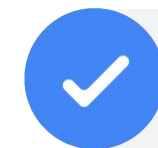


Designed to handle large workloads at consistent low latency

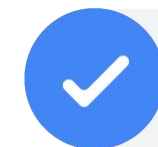


Can send and receive large amounts of data

Bigtable is a great choice for:

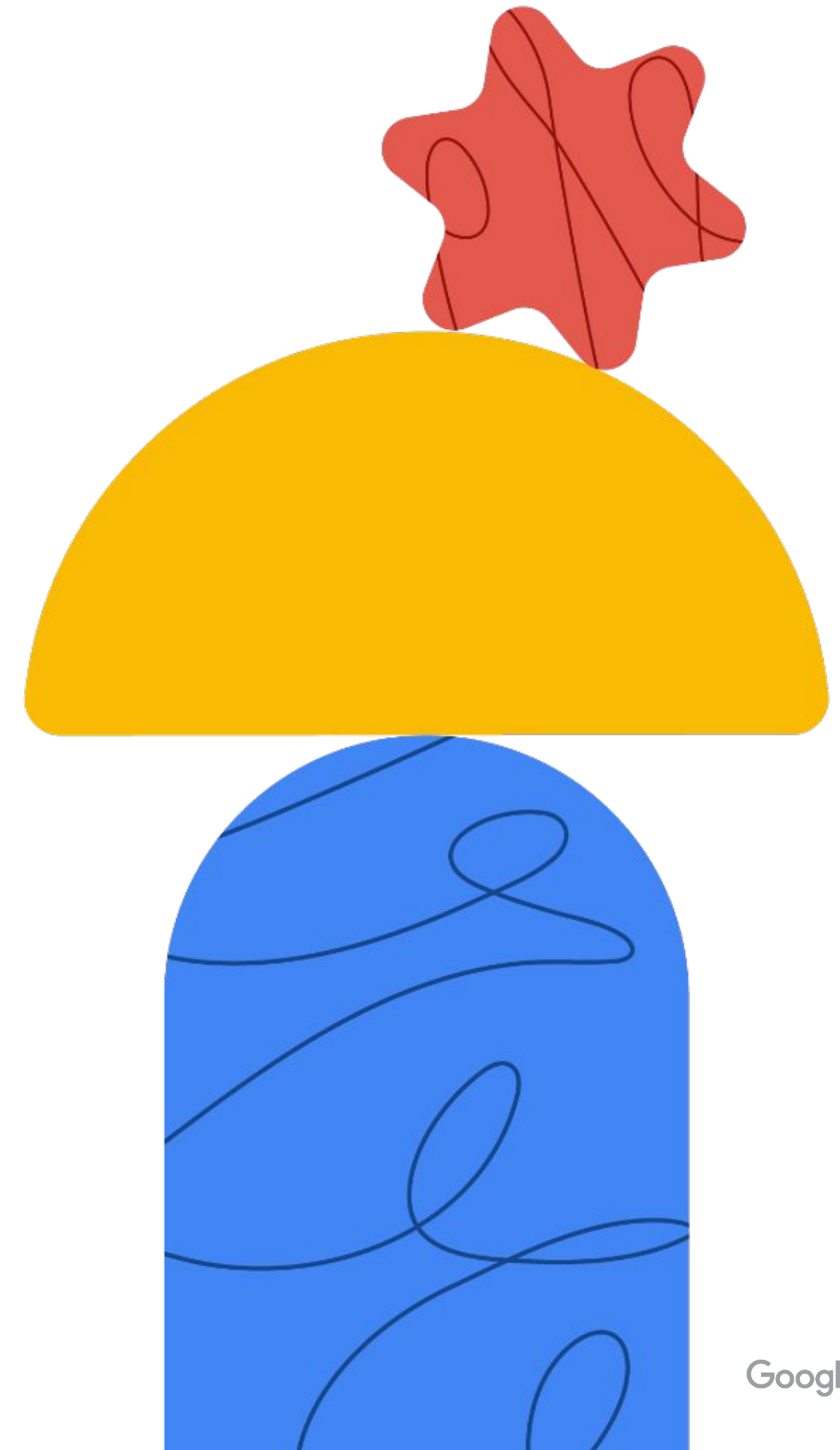


Operational applications

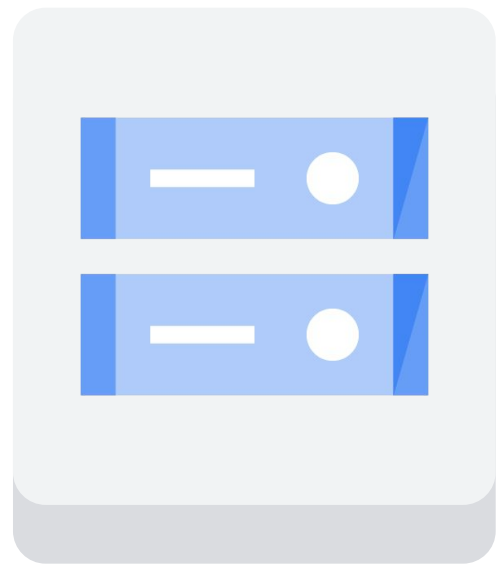


Analytical applications

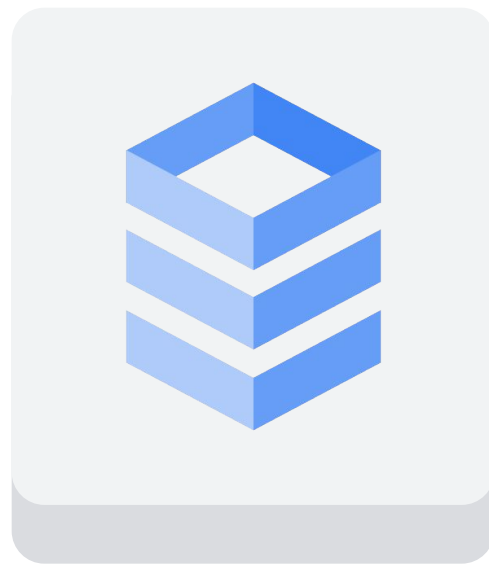
Choosing the right storage product



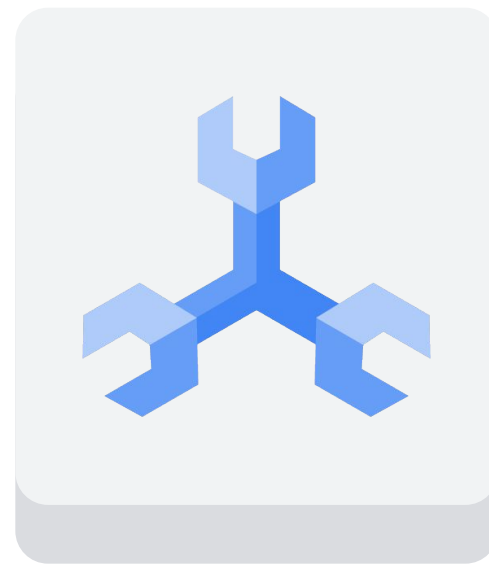
What's the right scenario for each storage option?



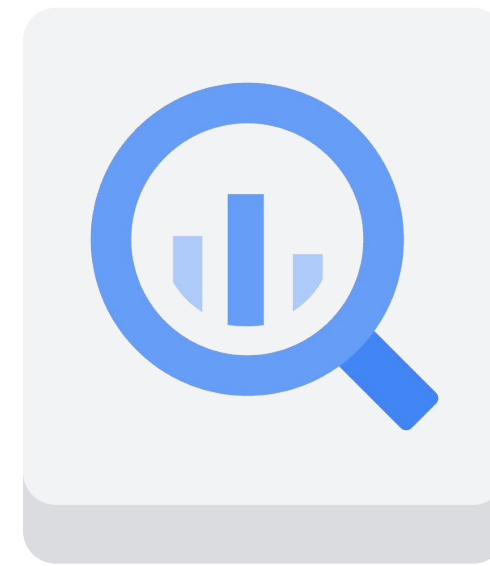
Cloud Storage



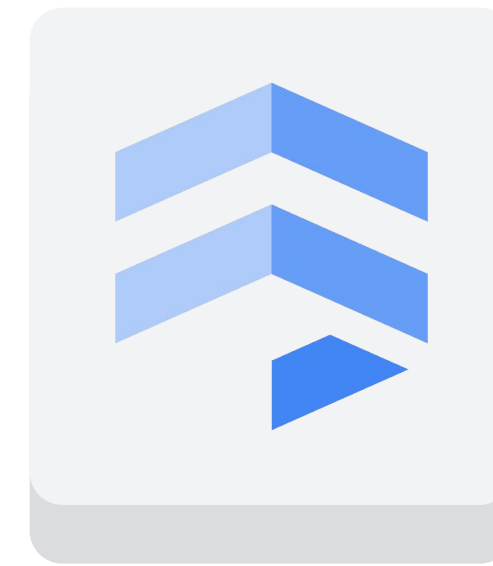
Cloud SQL



Cloud Spanner



BigQuery



Firestore



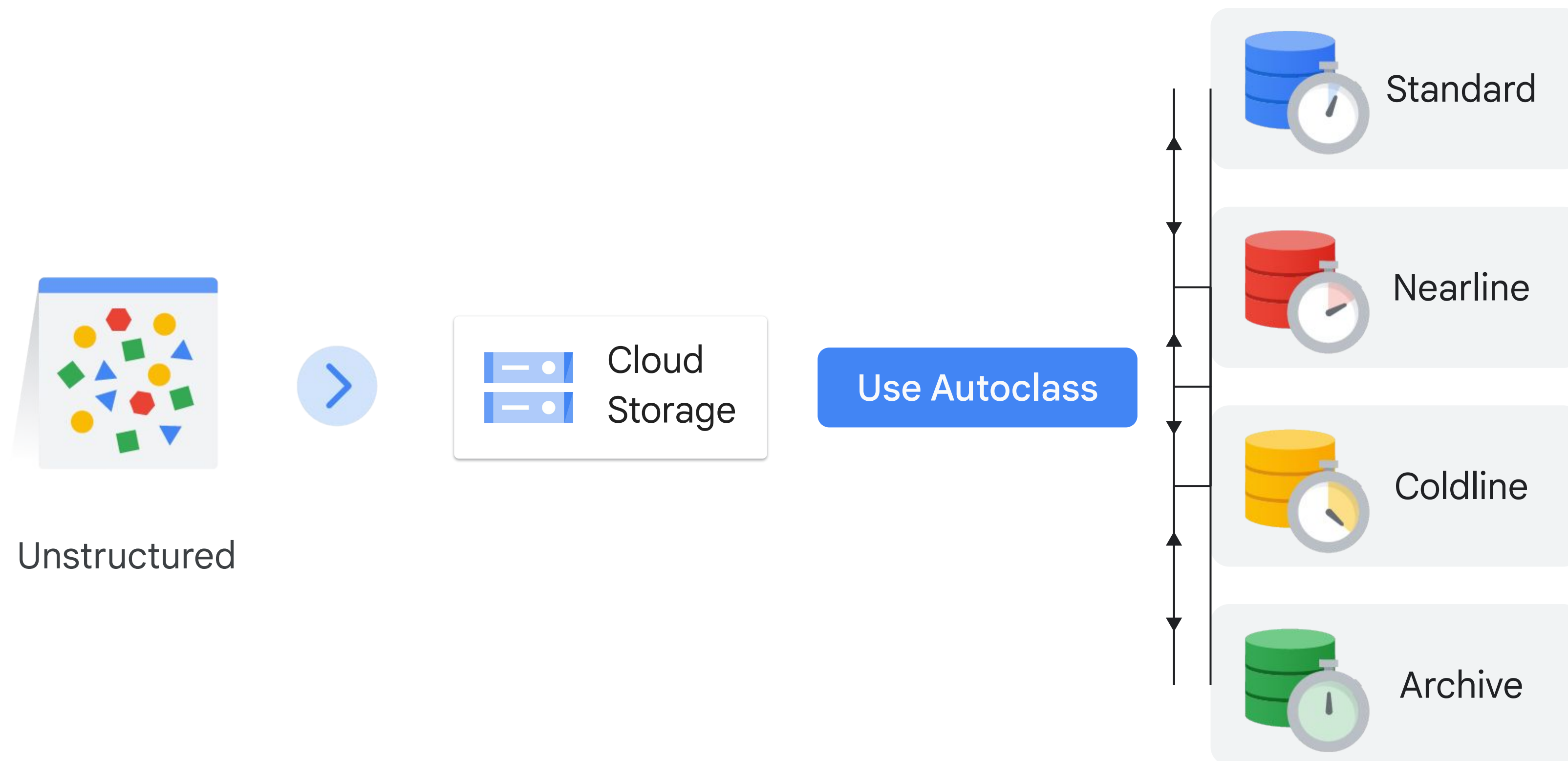
Cloud Bigtable

Data type

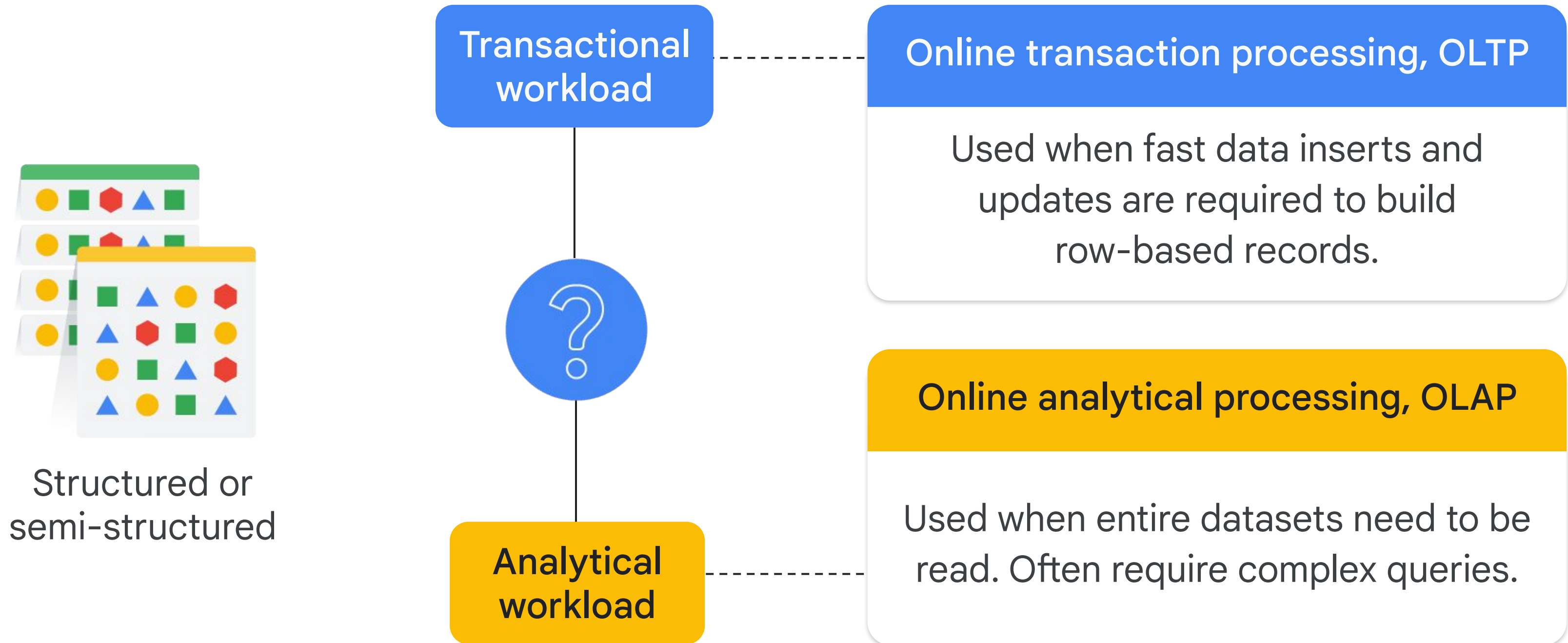


Business need

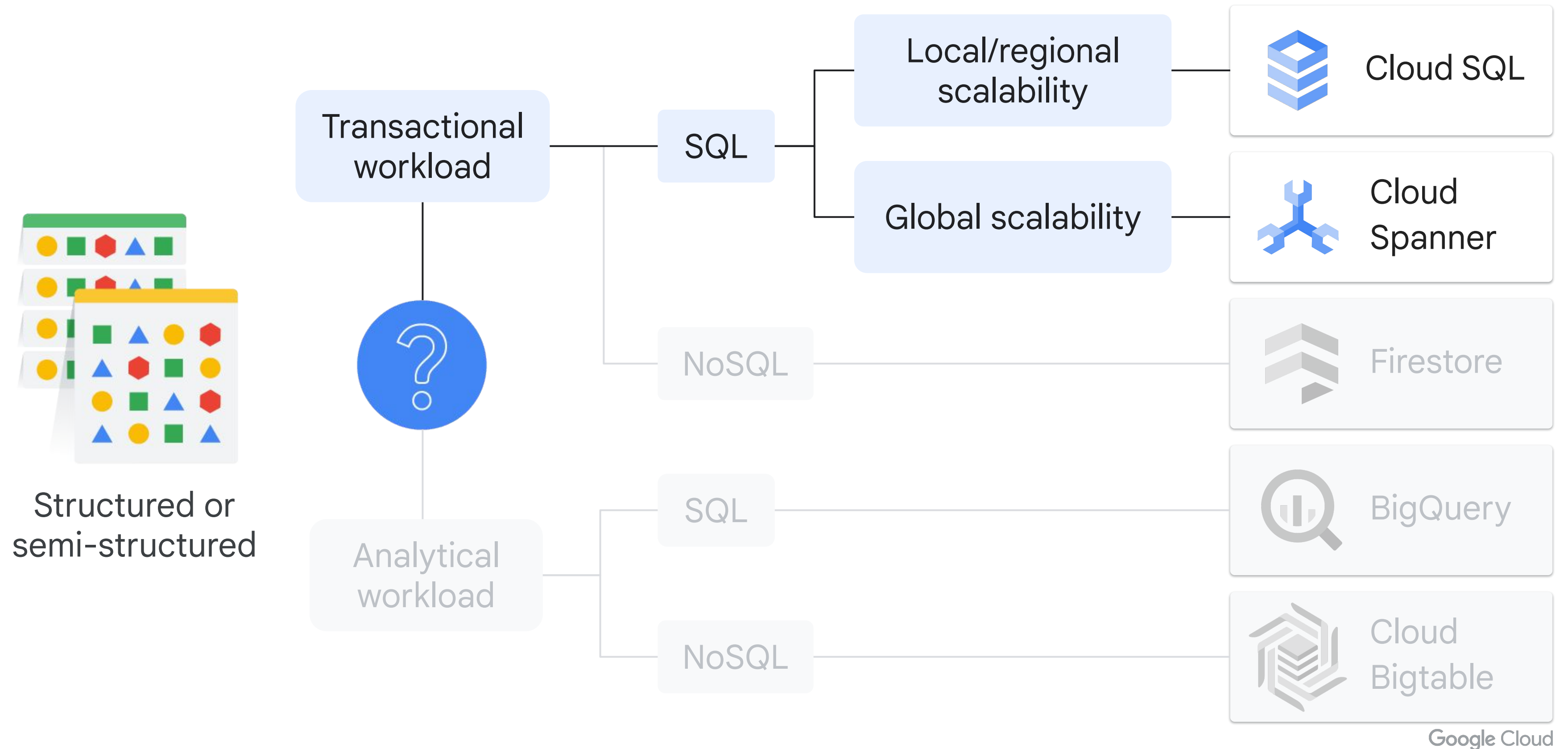
Use Cloud Storage for unstructured data



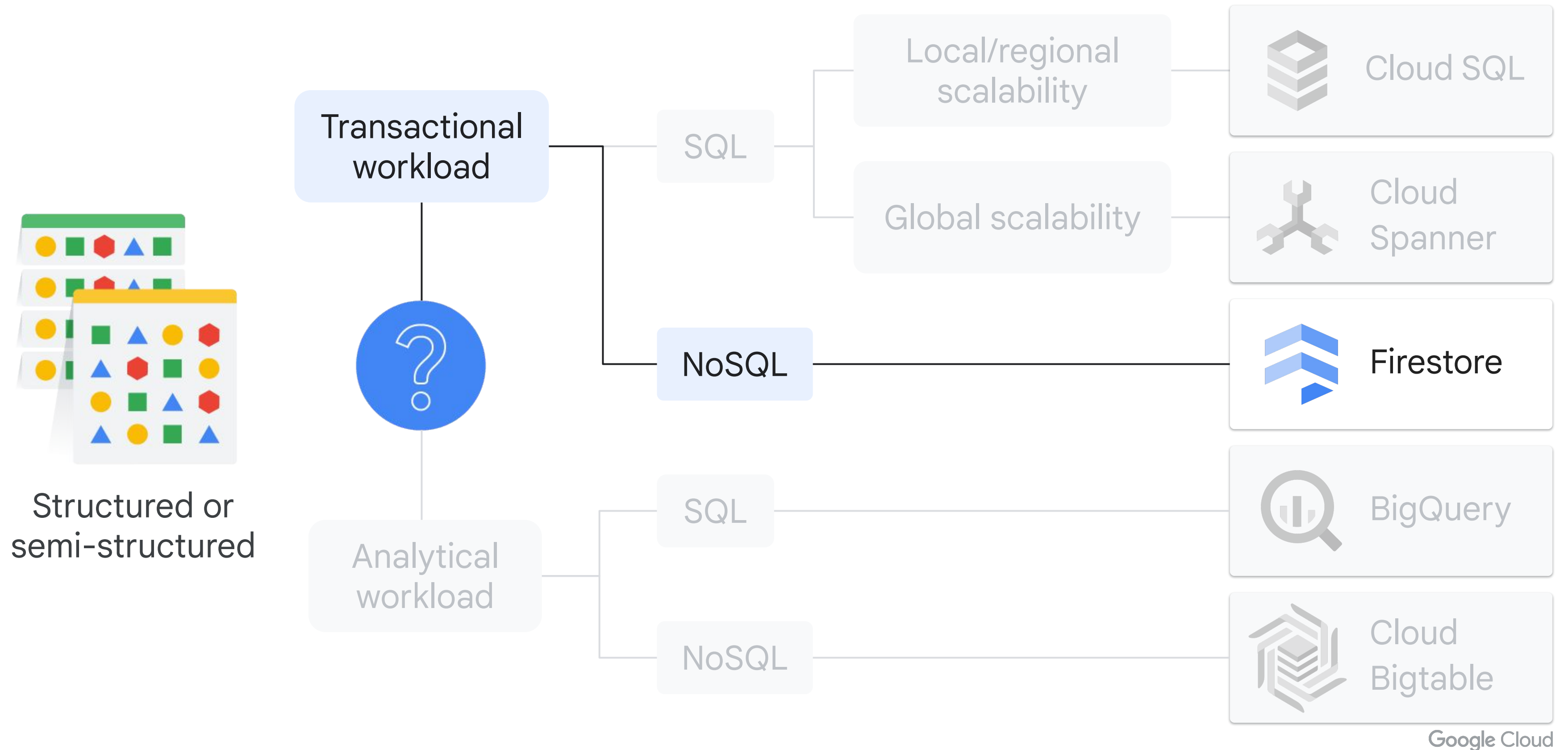
Structured or semi-structured data



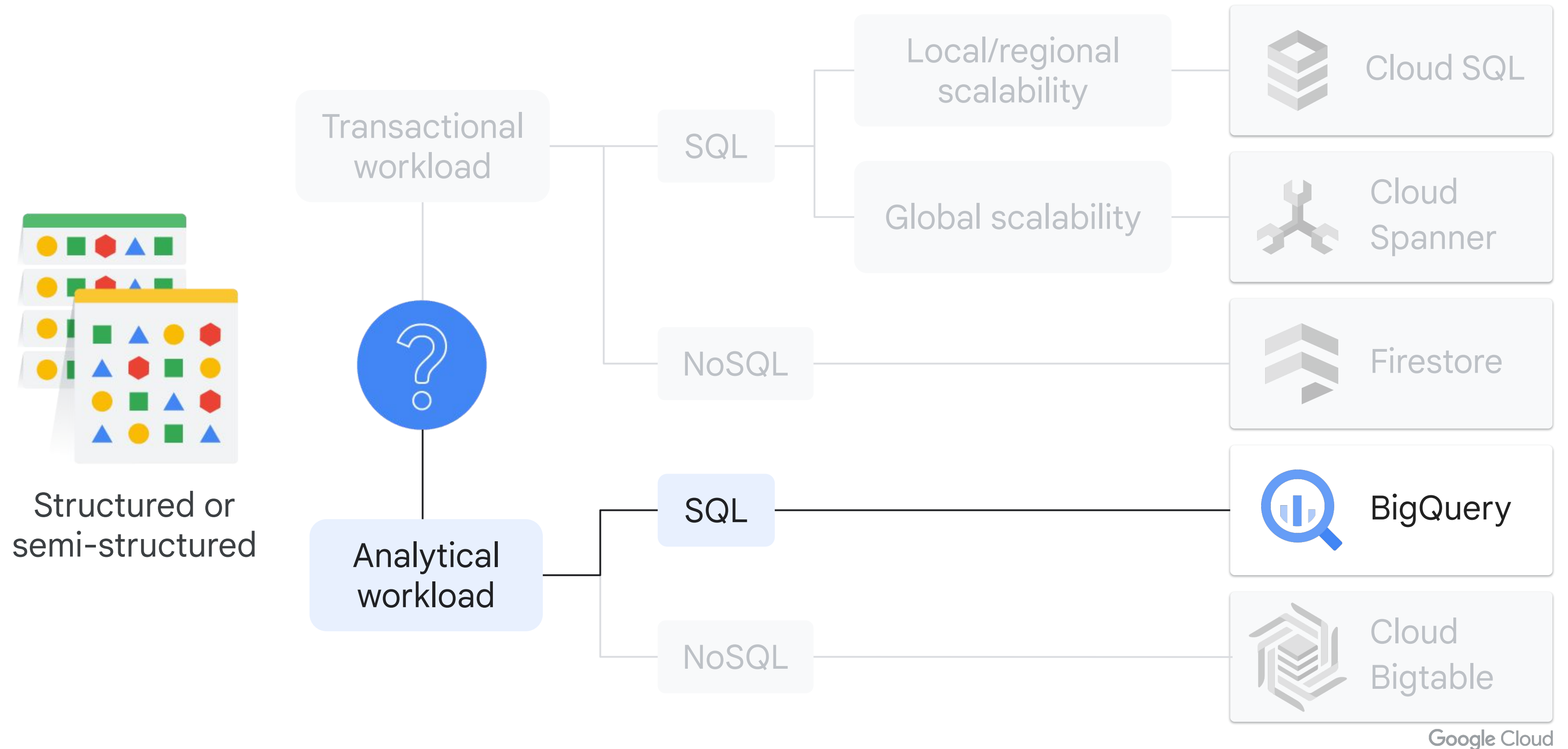
Transactional workloads accessed with SQL



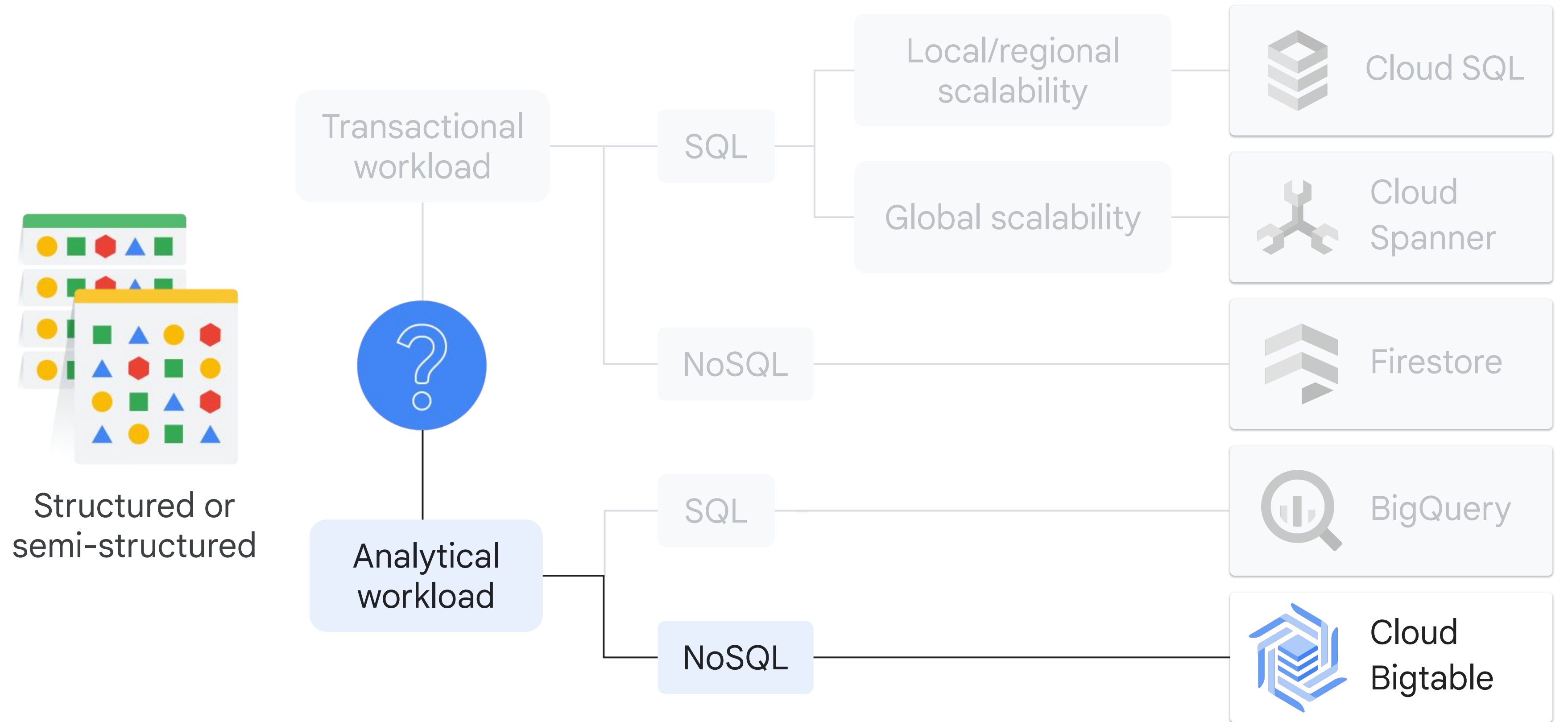
Transactional workloads accessed without SQL



Analytical workloads that require SQL commands



Analytical workloads that don't require SQL commands



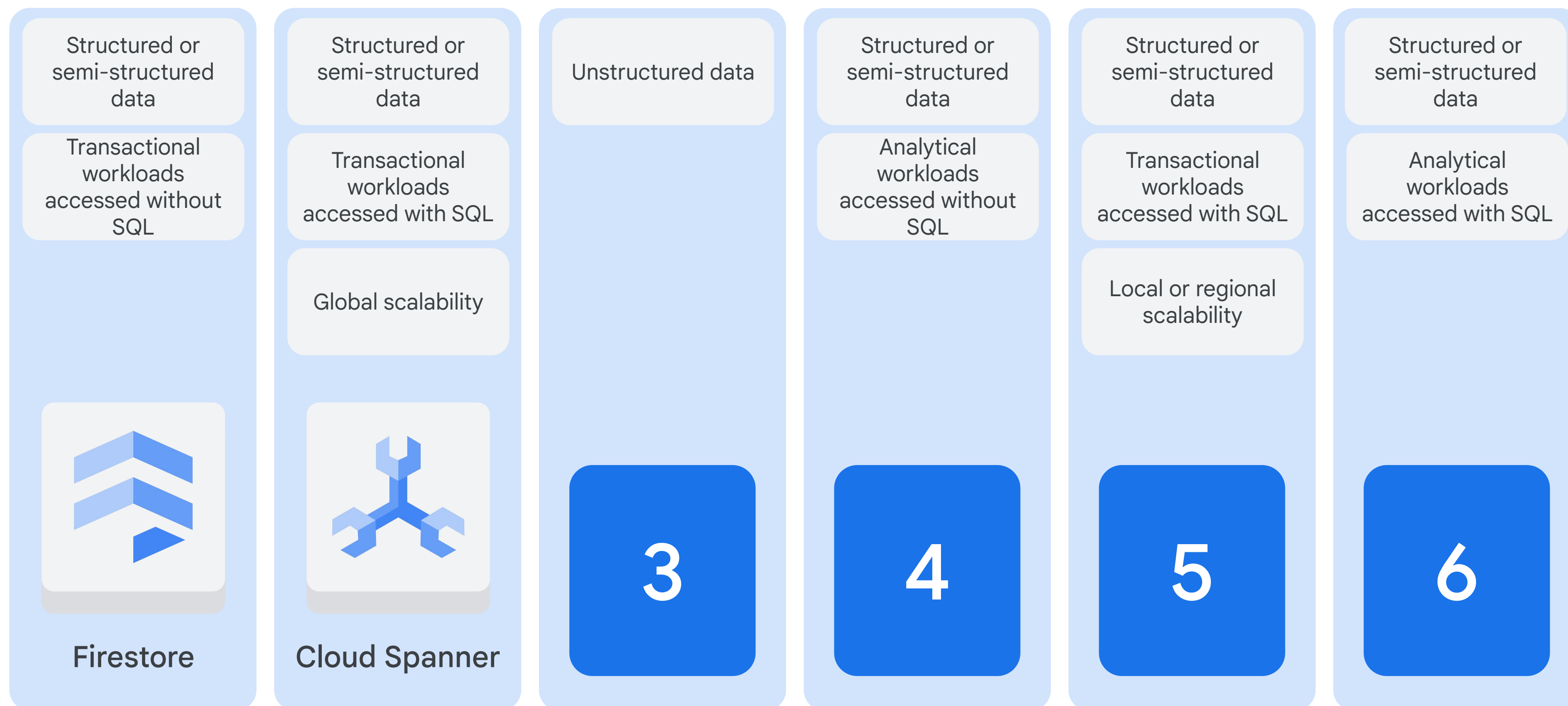
Which storage product...?



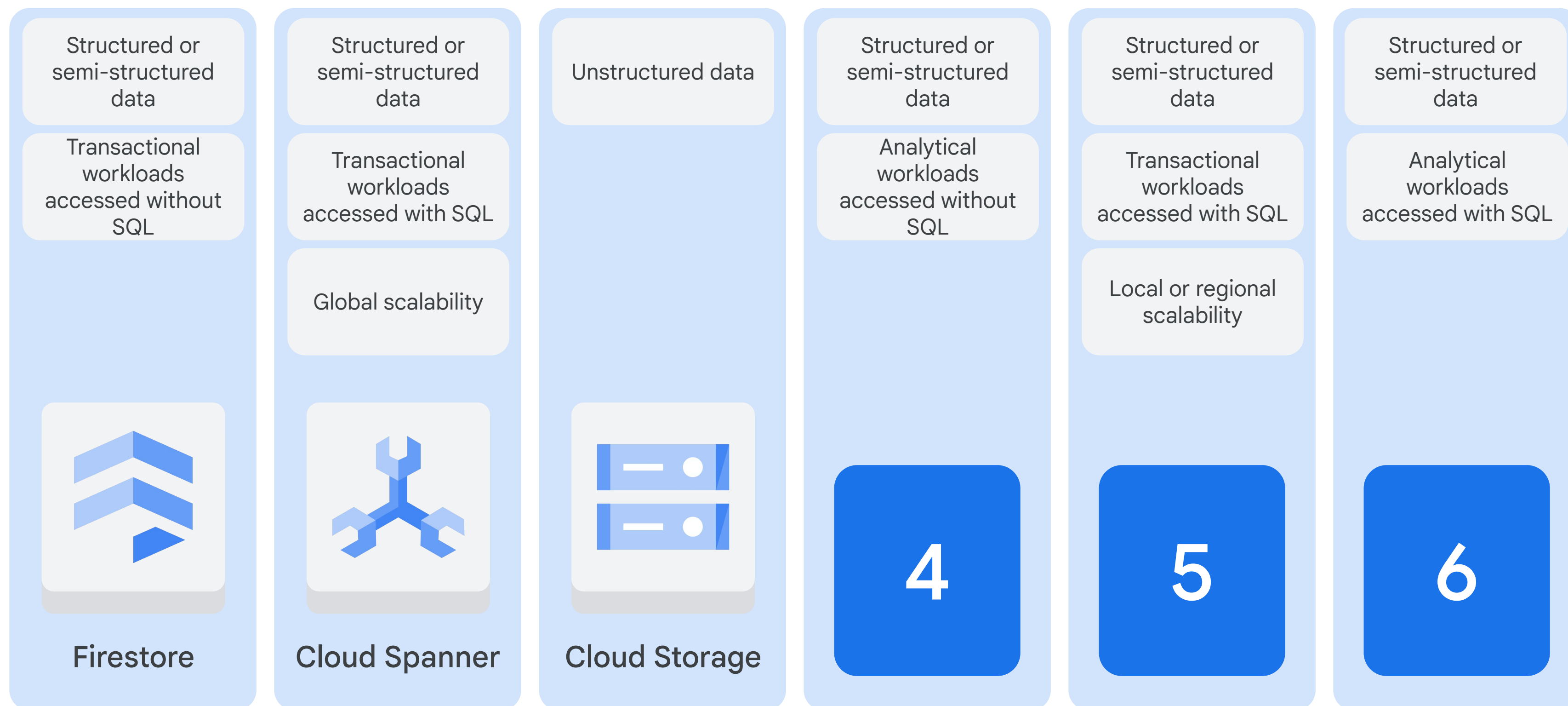
Which storage product...?



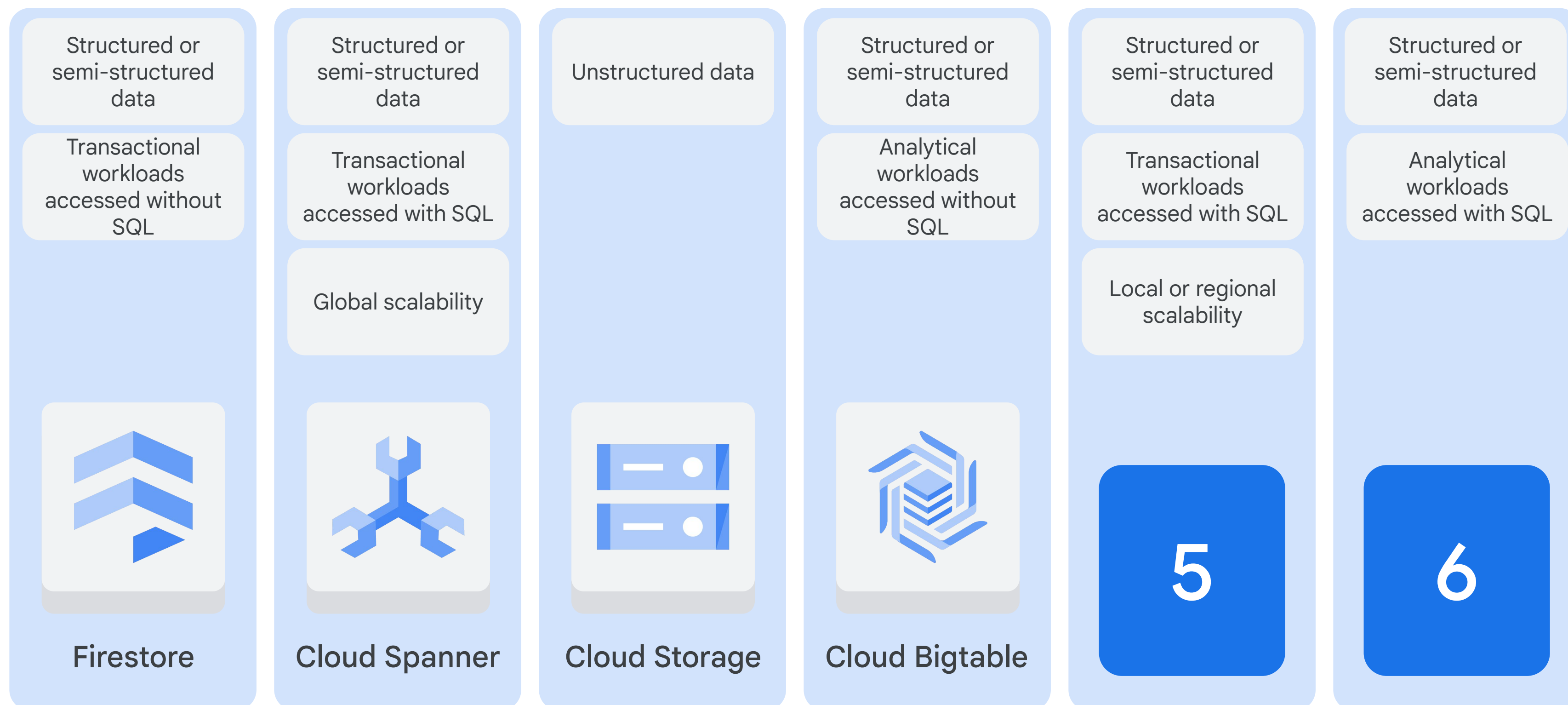
Which storage product...?



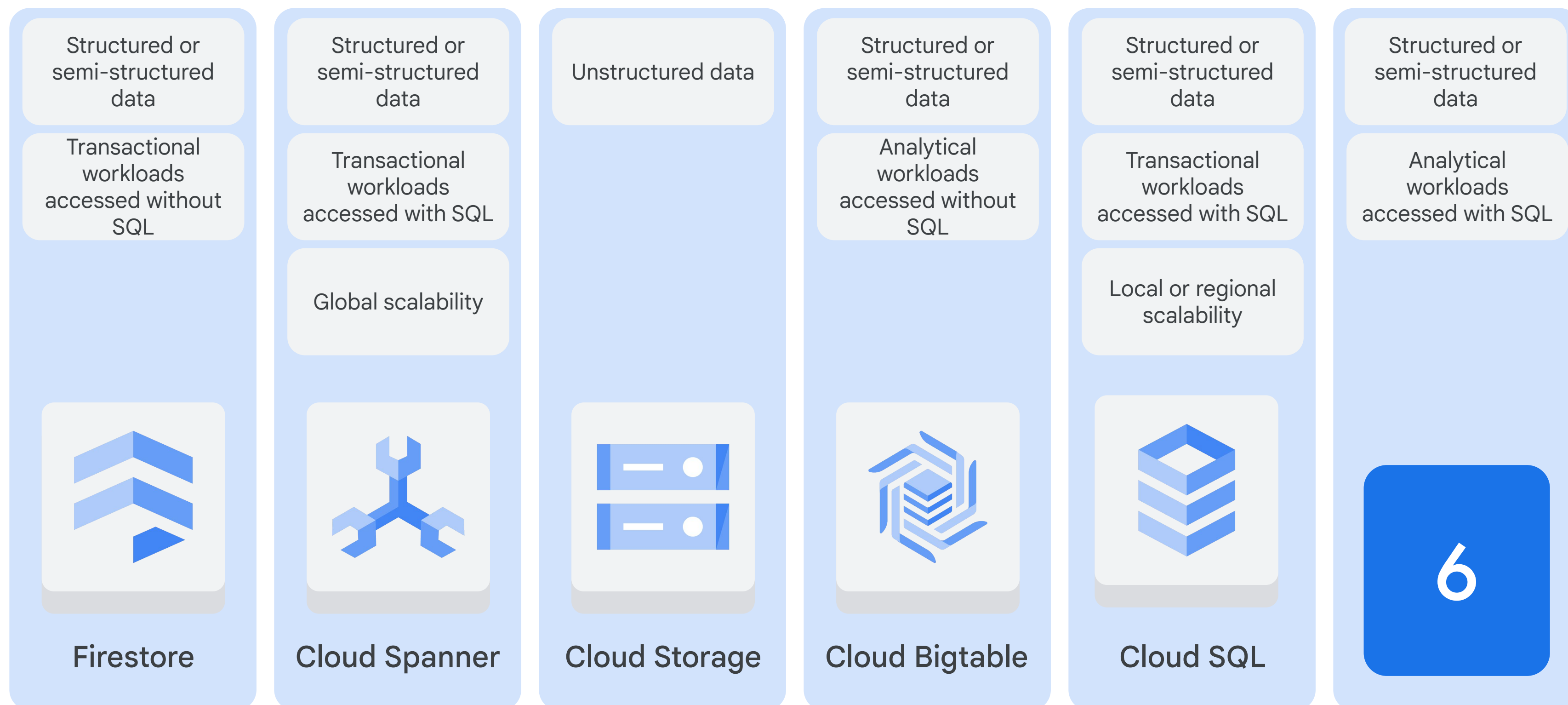
Which storage product...?



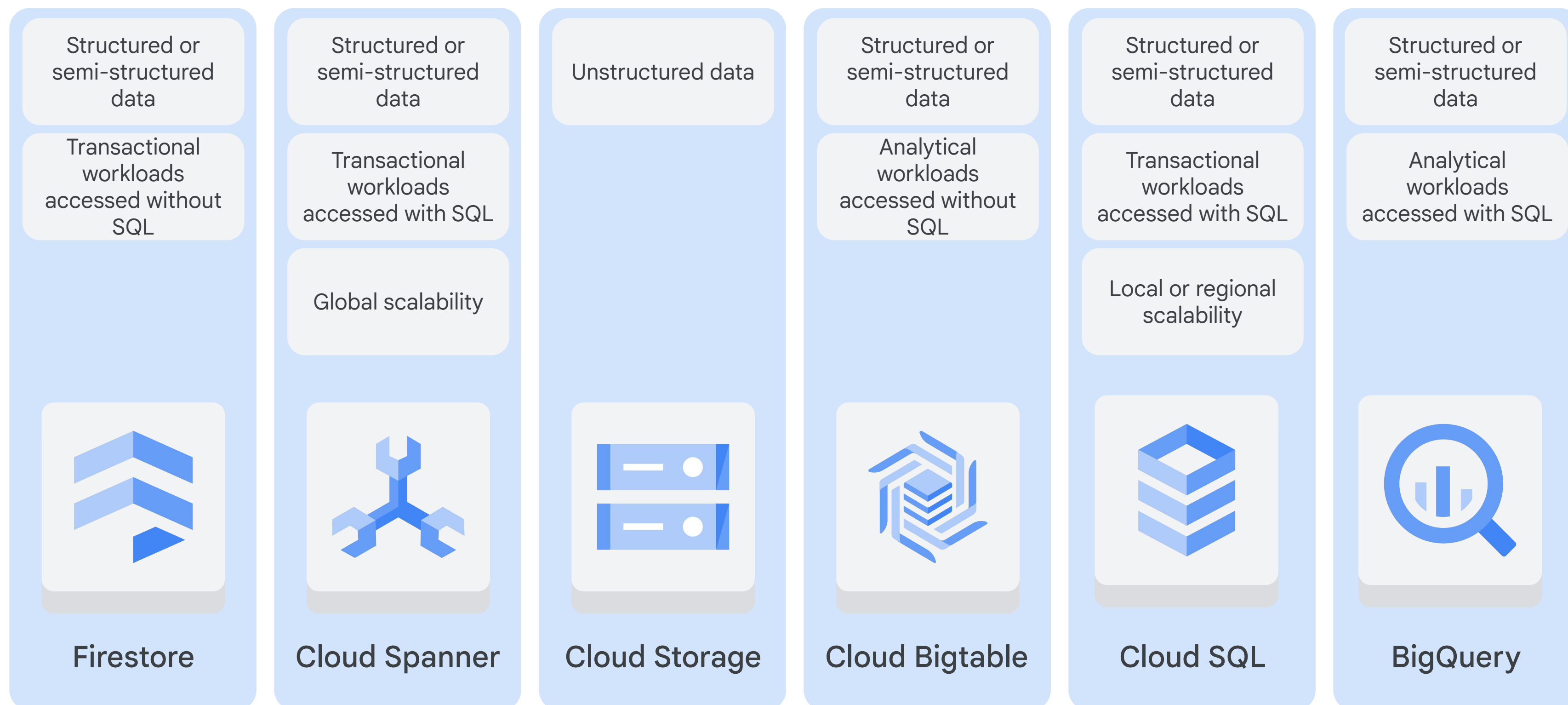
Which storage product...?



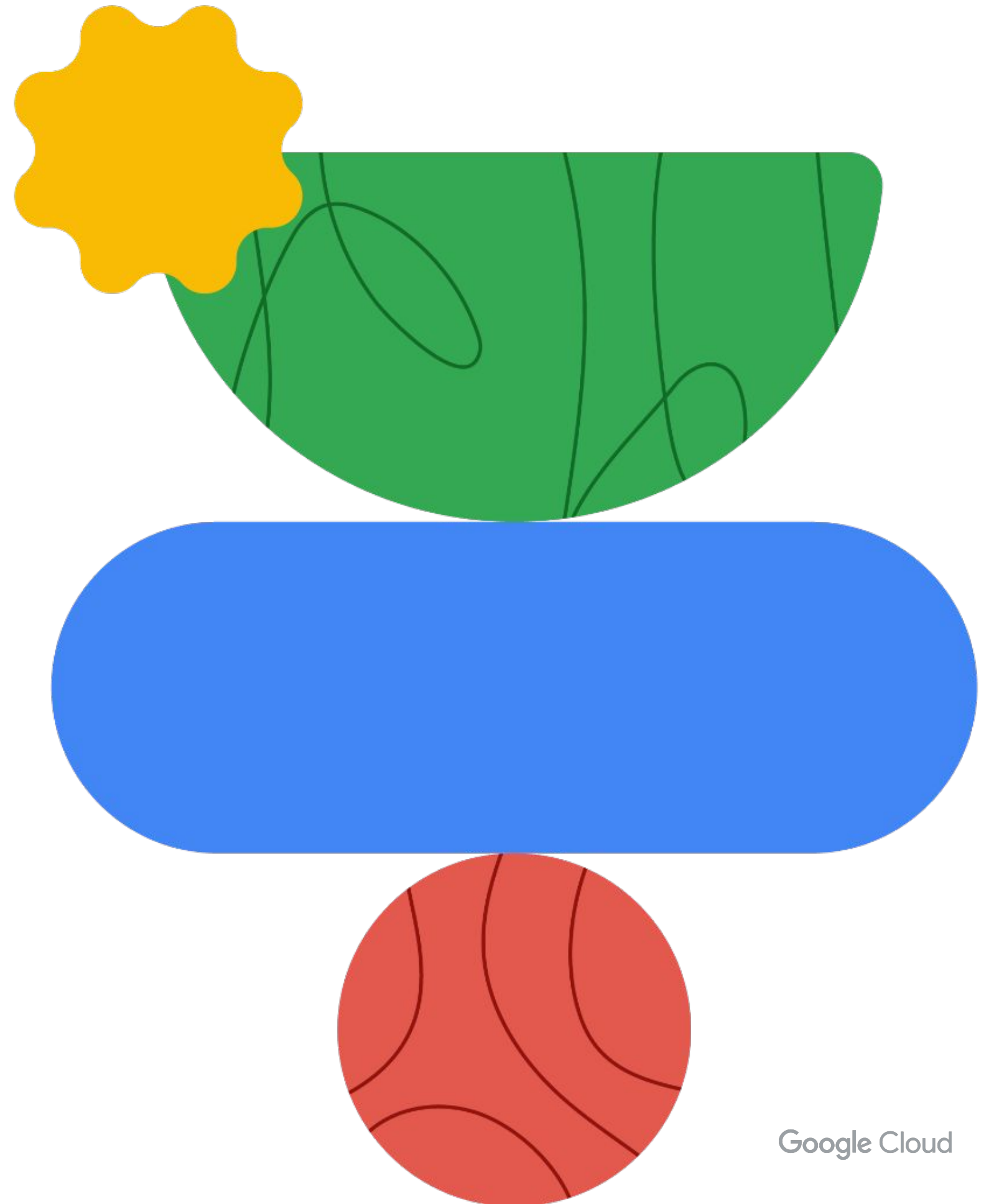
Which storage product...?



Which storage product...?



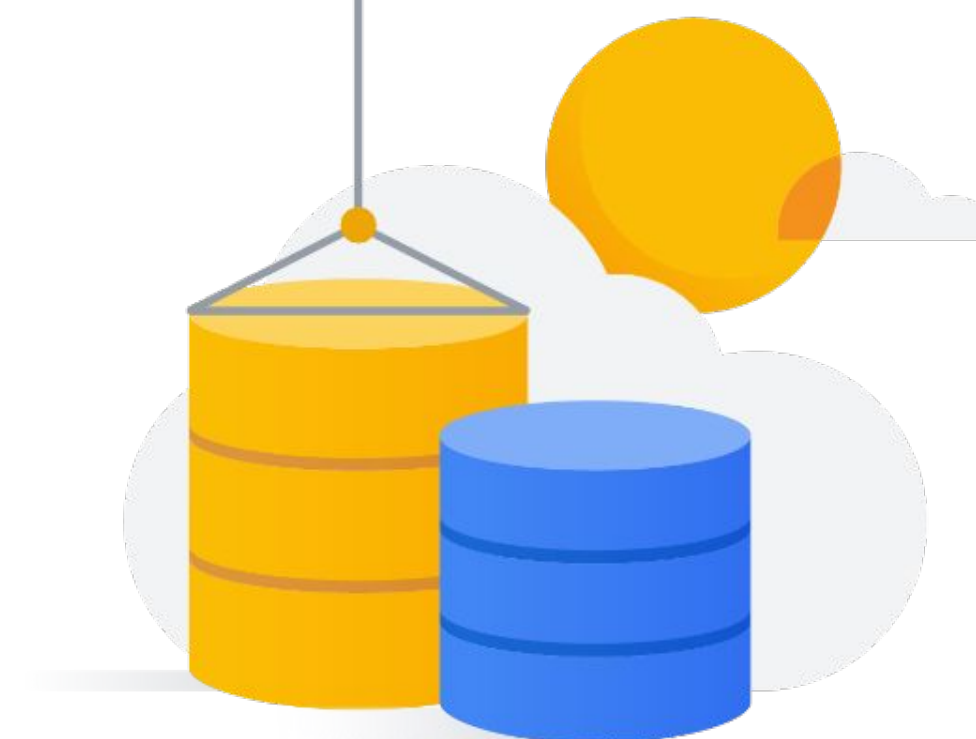
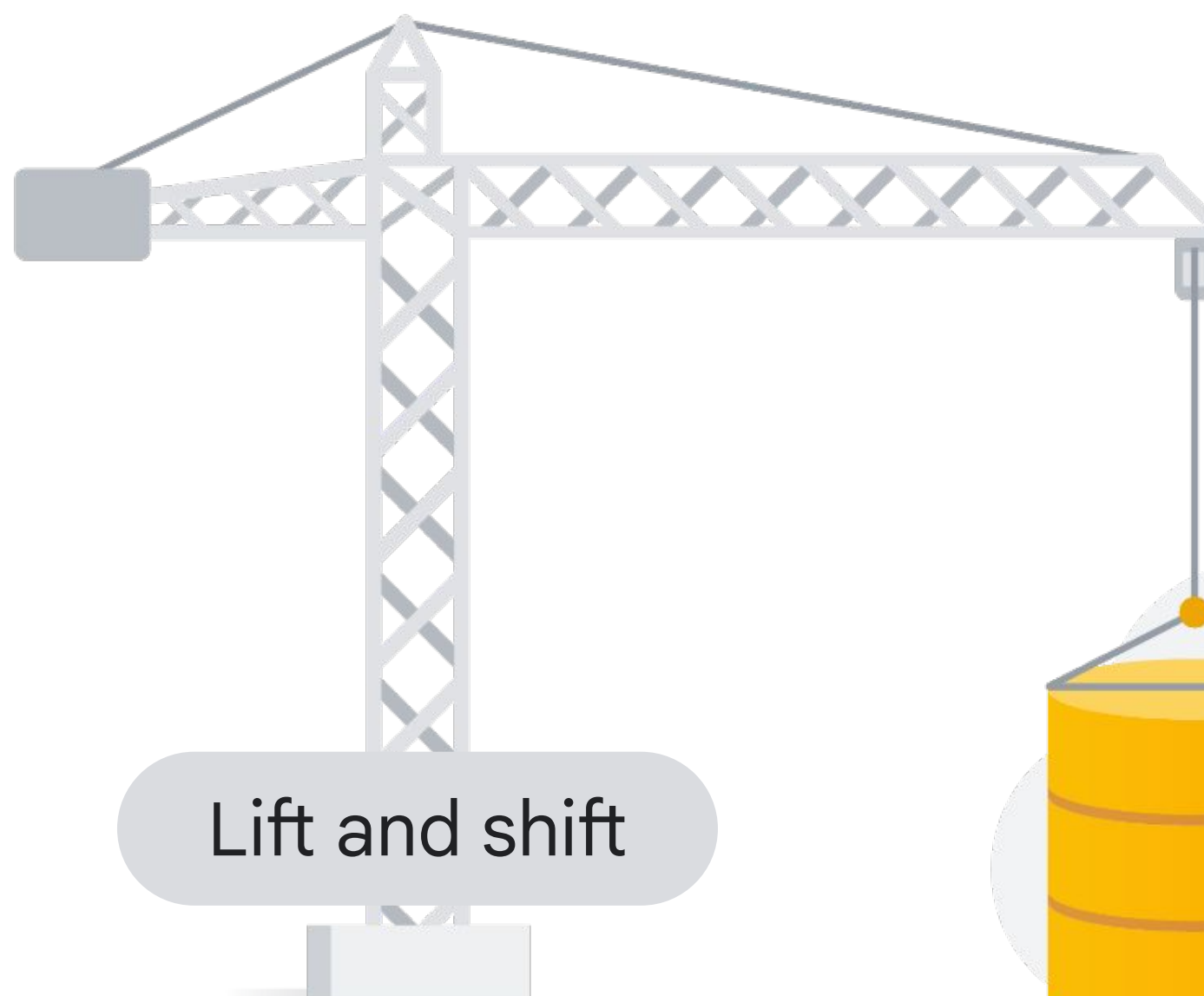
Database migration and modernization



Lift and shift databases

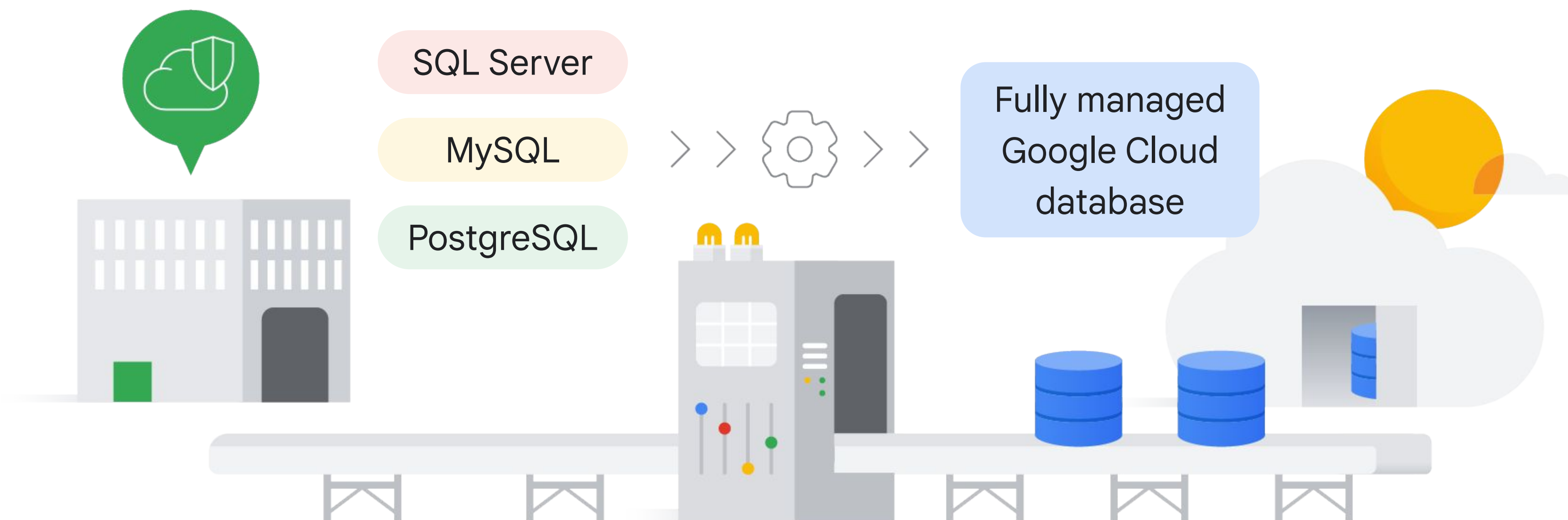


On-premises or private
cloud environment

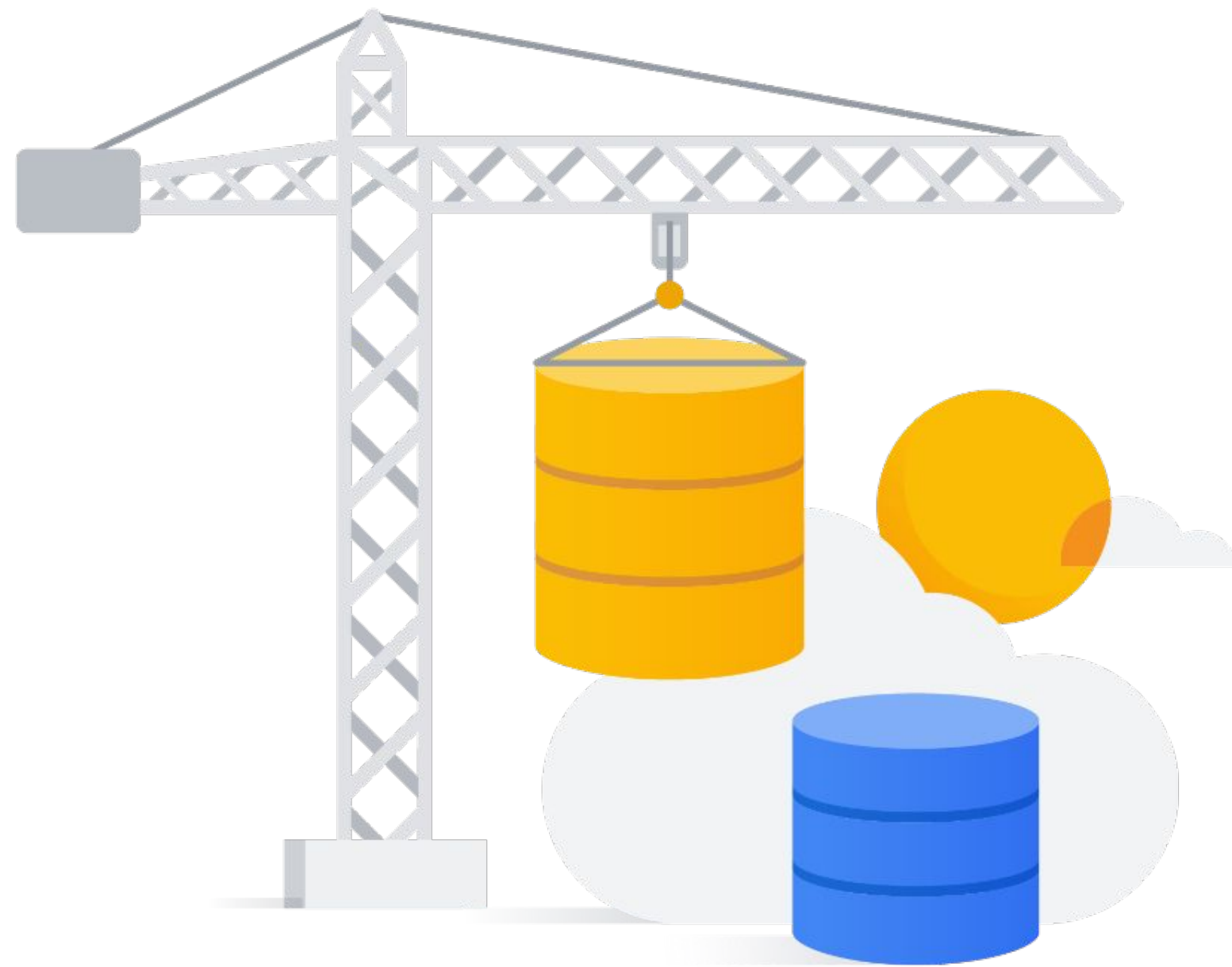


Public cloud provider

Managed database migration

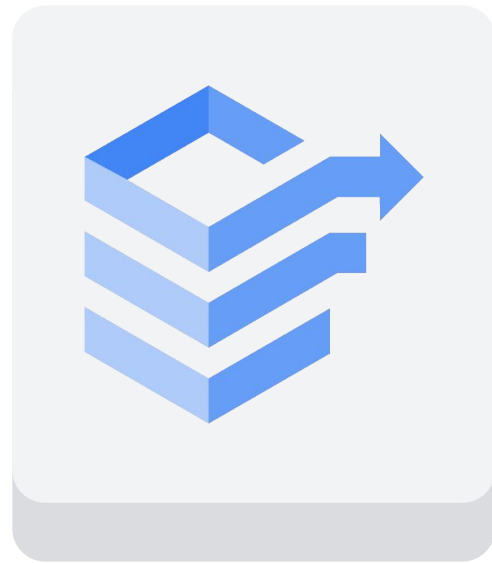


Lift and shift databases



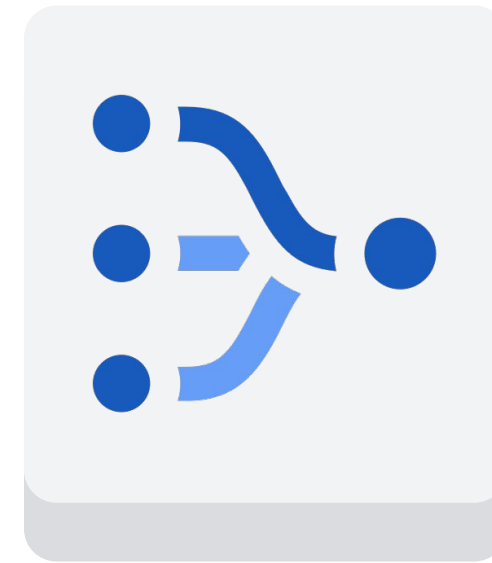
- ! More difficult to modernize databases
- ✓ Minimal upheaval
- ✓ Managed by the cloud provider

Managed database migration services



Database Migration Service (DMS)

Easily migrate your databases to Google Cloud.



Datastream

It's used to synchronize data across databases, storage systems, and applications.

Module 2

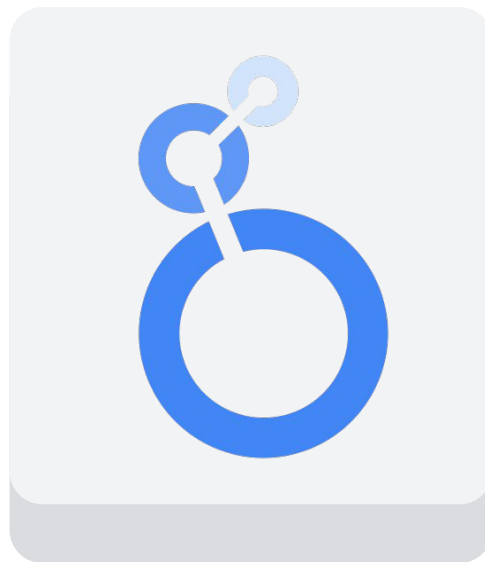
Exploring Data Transformation with Google Cloud



Lessons

- 01 The value of data
- 02 Google Cloud data management solutions
- 03 Making data useful and accessible

A platform to analyze, visualize, and share data



Looker

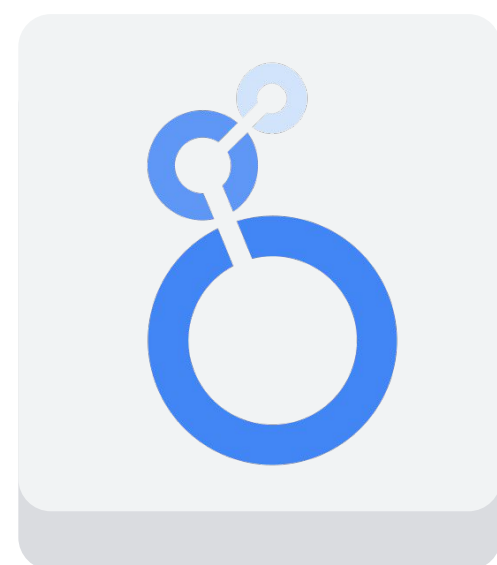
✓ Interactive dashboards

✓ Interactive reports

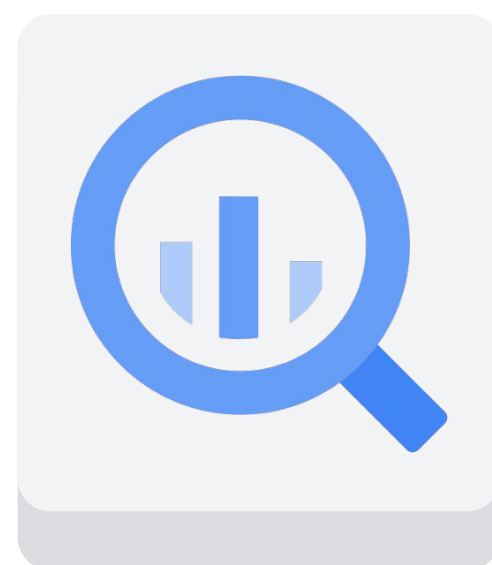
✓ Easy to understand

✓ Easy to share

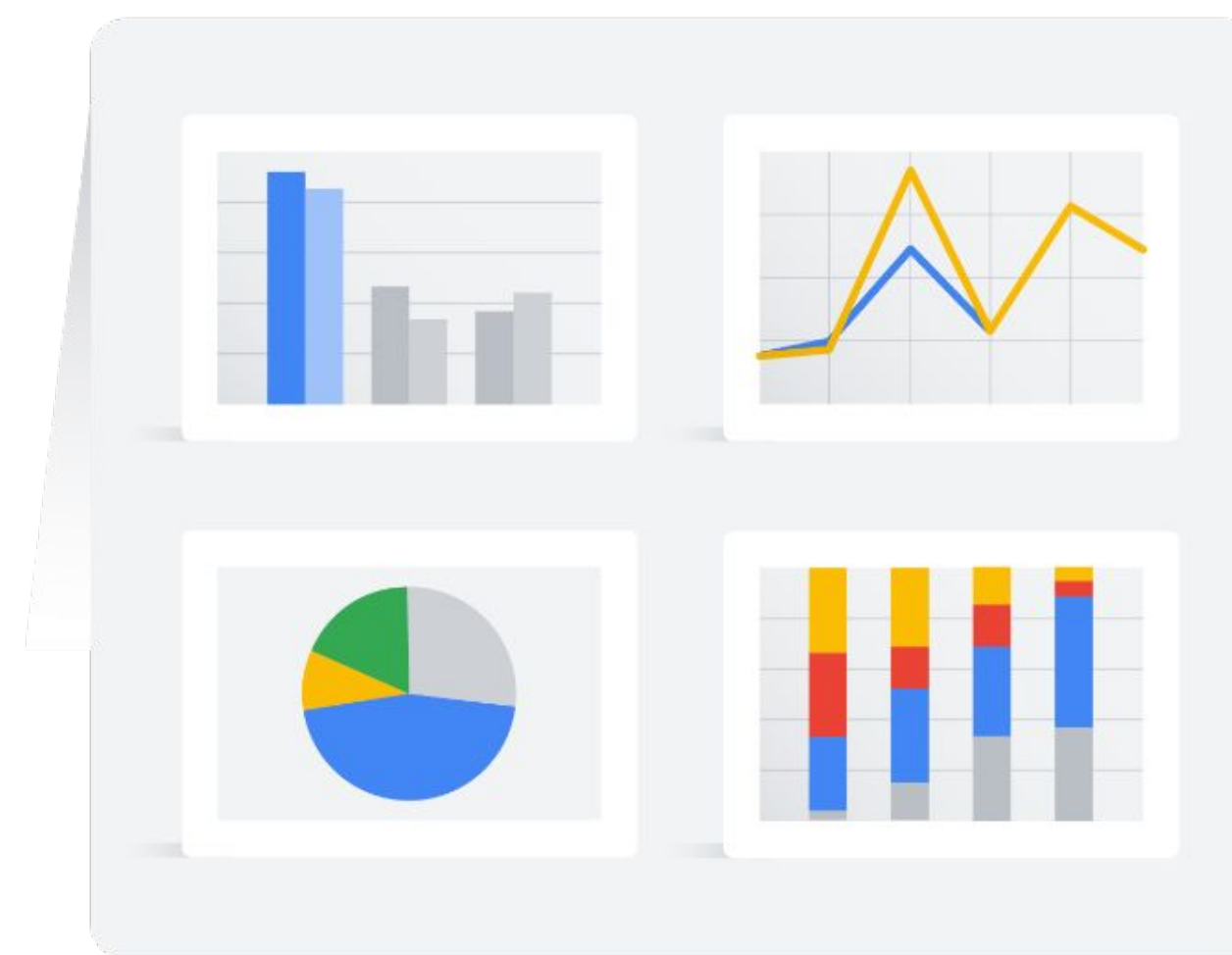
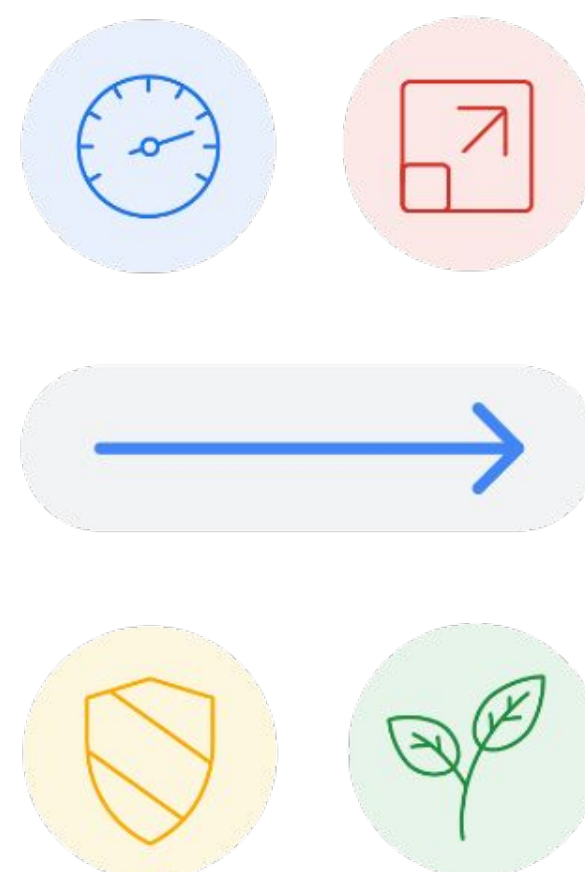
Looker supports BigQuery and 60+ SQL databases



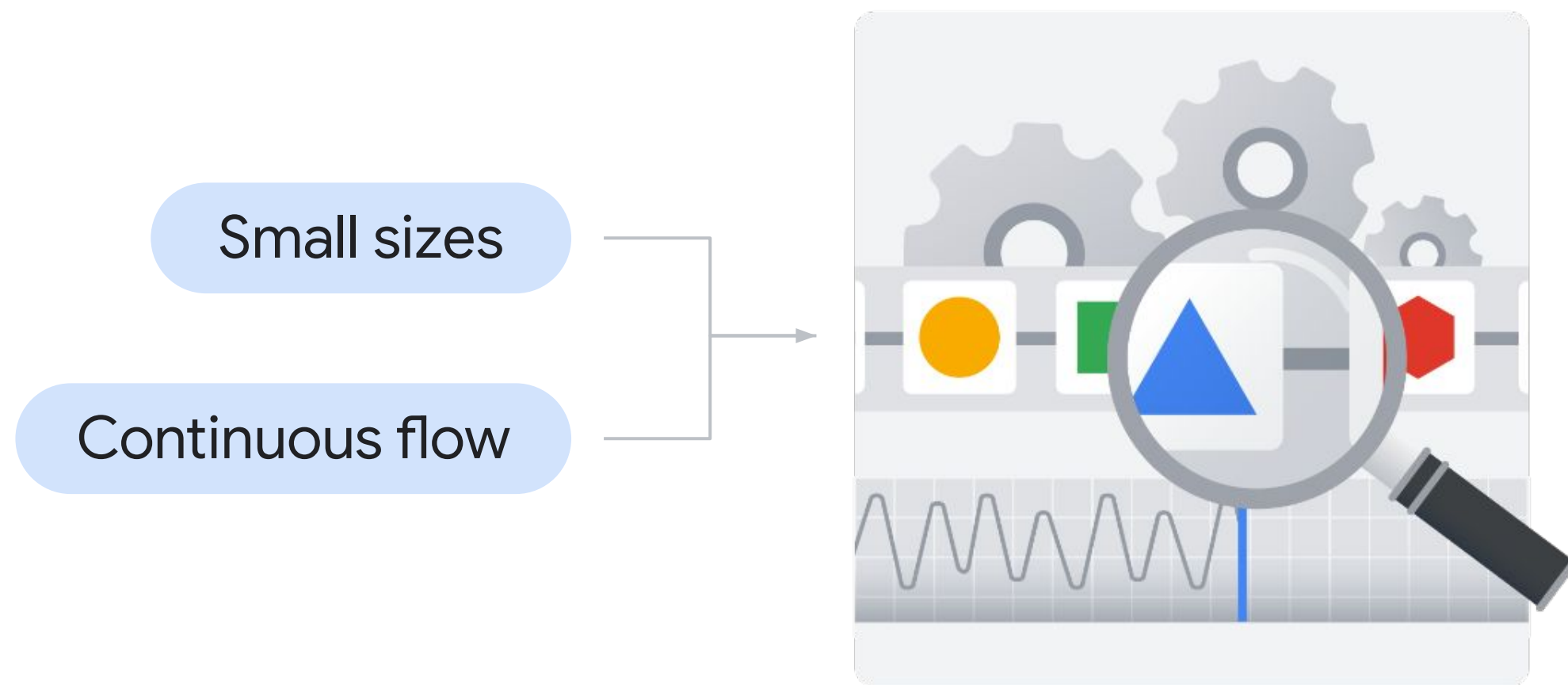
Looker



BigQuery

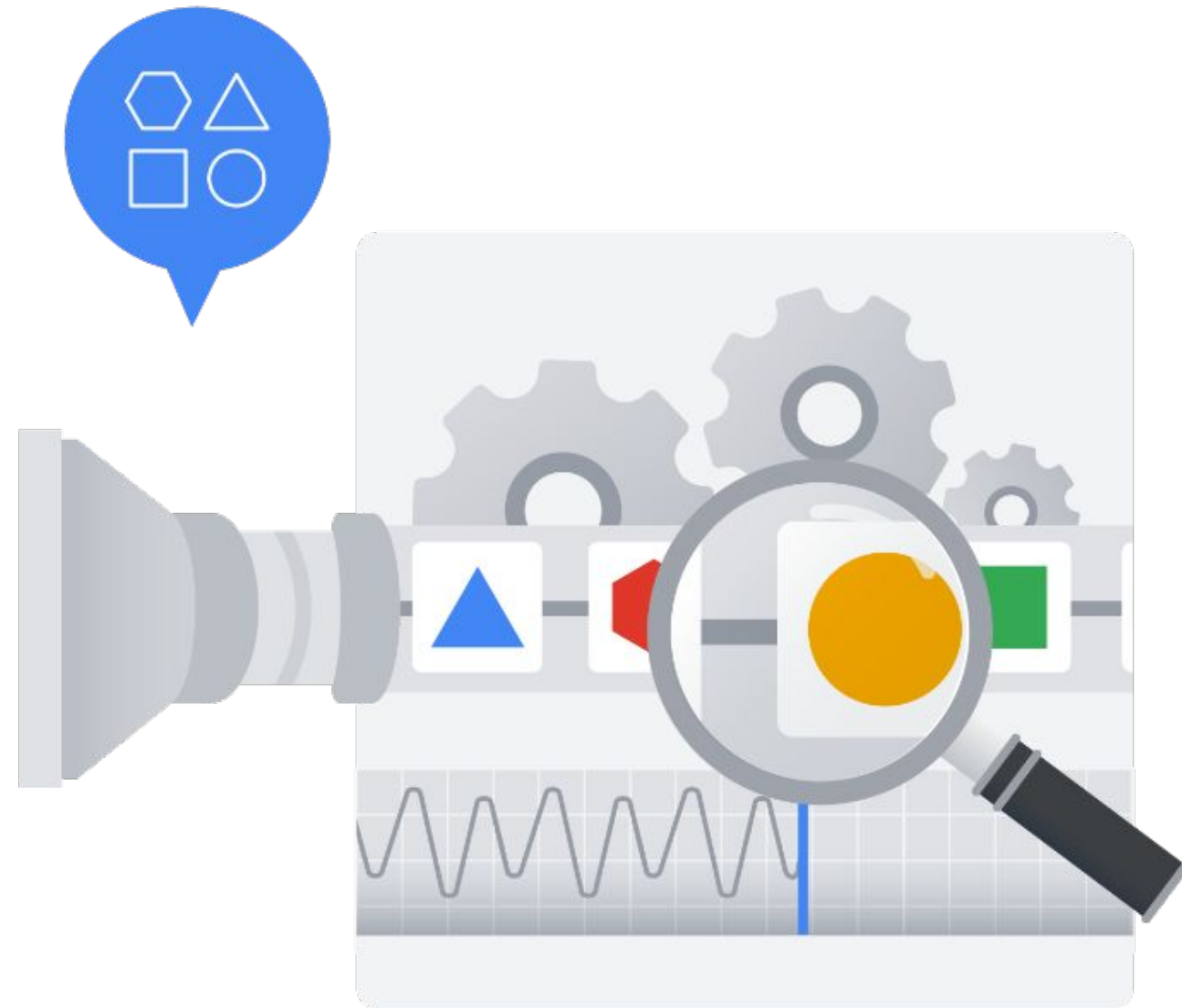


Streaming analytics



Streaming analytics is the processing and analyzing of data records **continuously** instead of in batches.

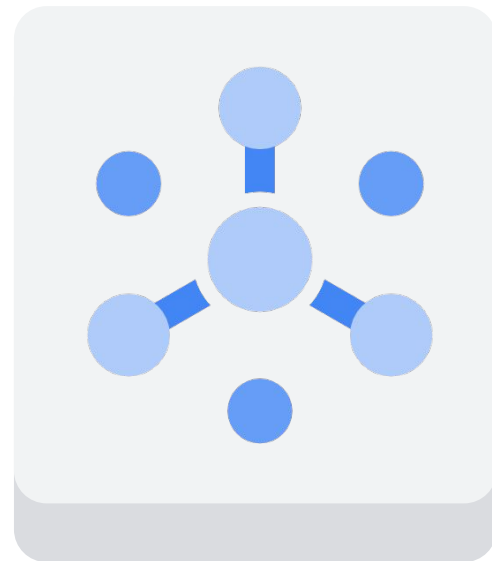
Sources of streaming data



- ✓ Equipment sensors
- ✓ Clickstreams
- ✓ Social media feeds
- ✓ Stock market quotes
- ✓ App activity

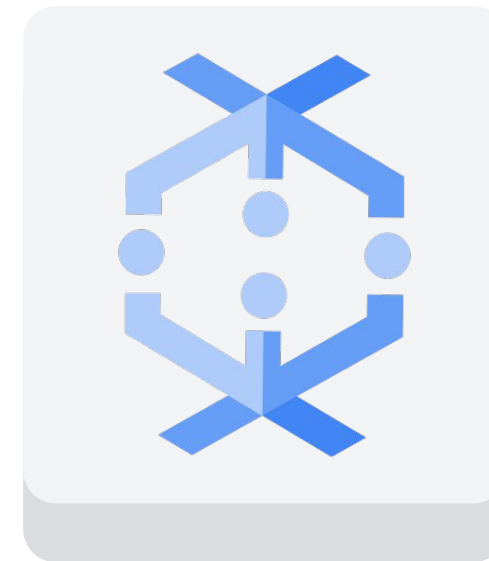
Google Cloud's streaming analytics products

Pub/Sub



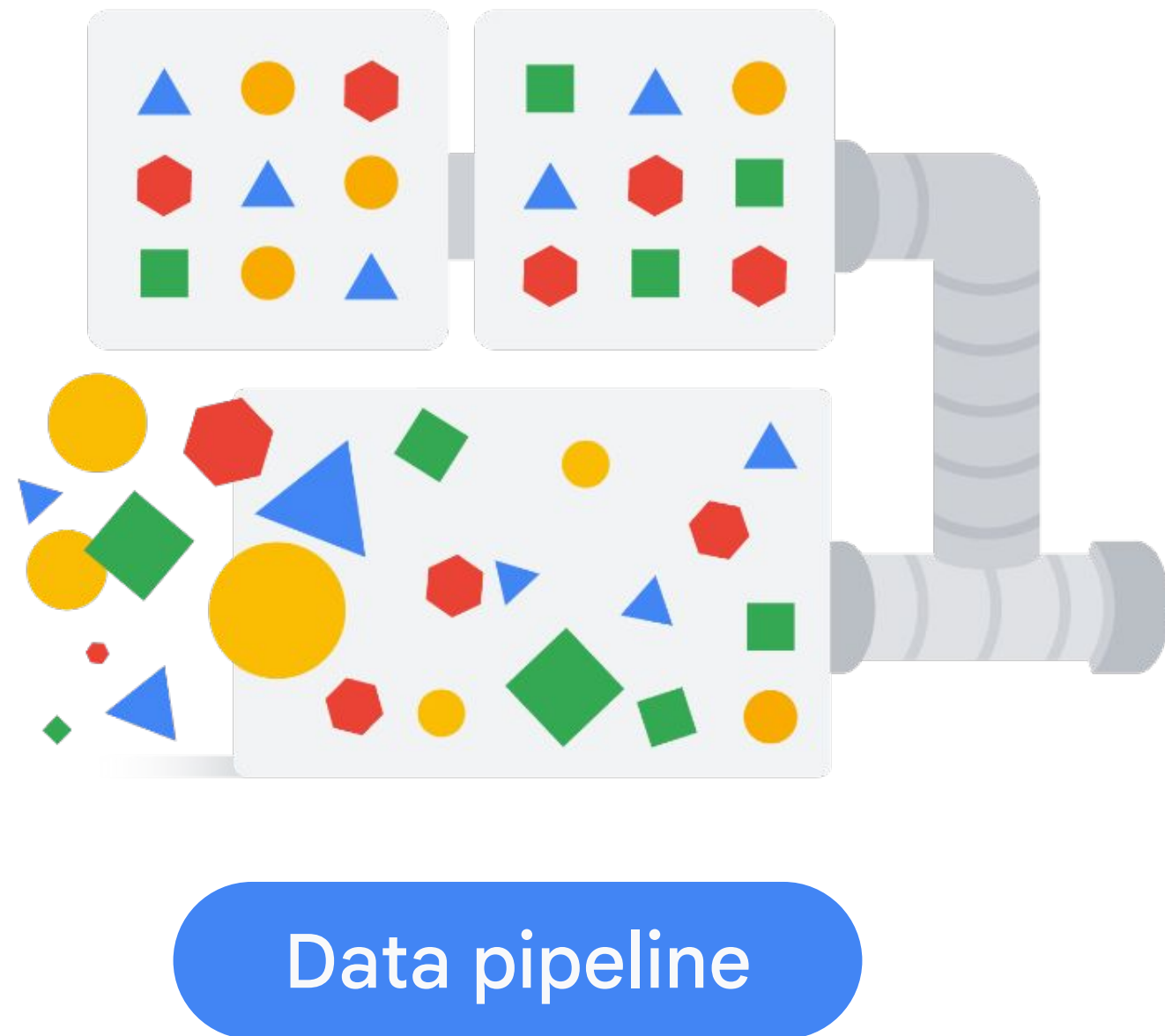
Ingests hundreds of millions of events per second.

Dataflow



Unifies streaming and batch data analysis and builds cohesive **data pipelines**.

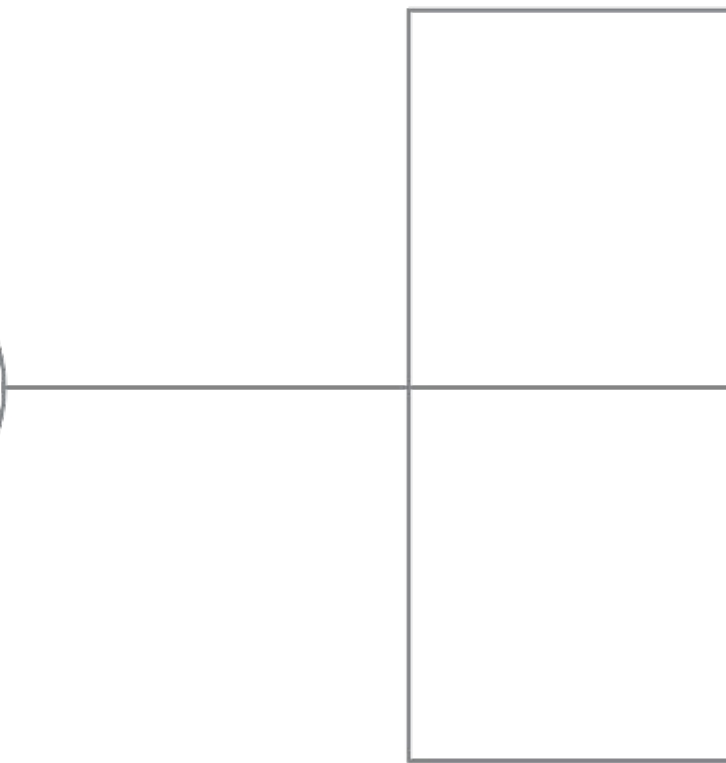
What is a data pipeline?



A series of actions, or stages, that ingest raw data from different sources and then move that data to a destination for storage and analysis.

Pub/Sub

Publisher/Subscriber,
or *publish messages
to subscribers*



Gaming
events



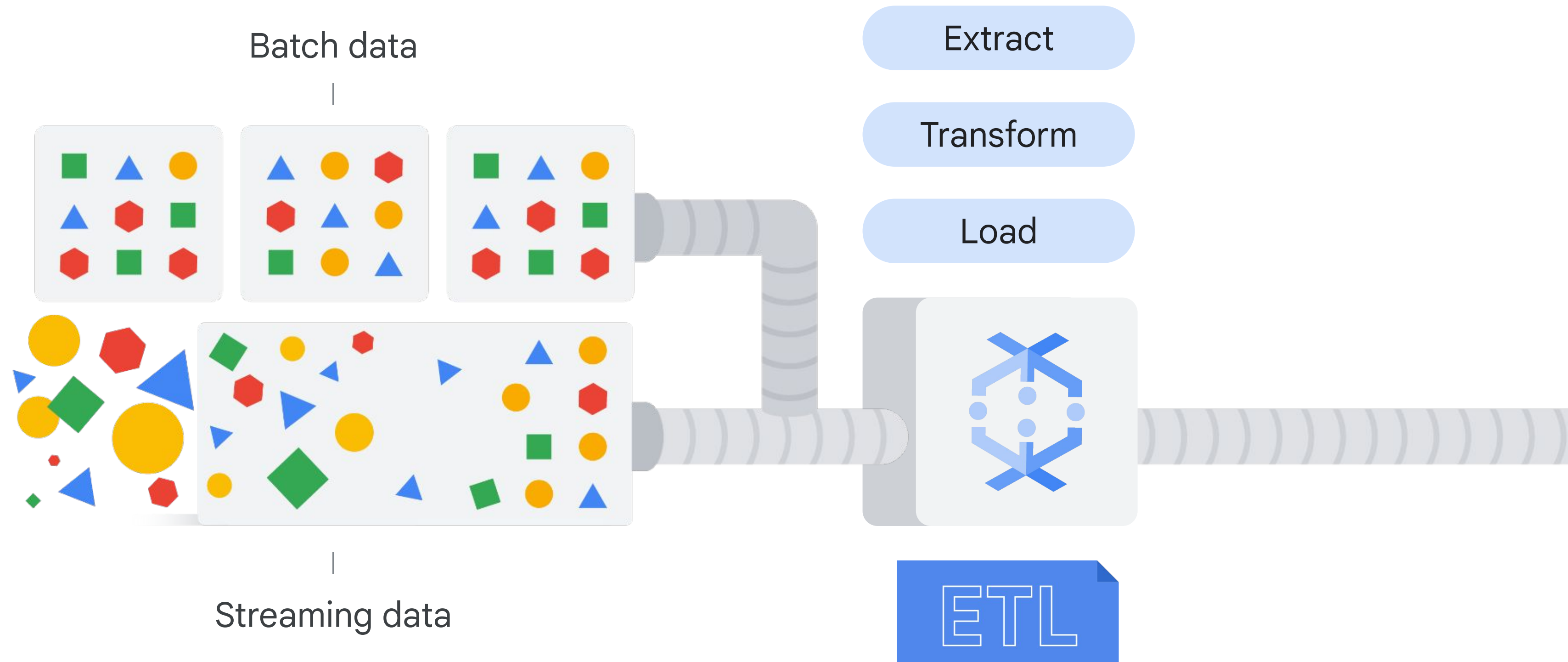
IoT
devices



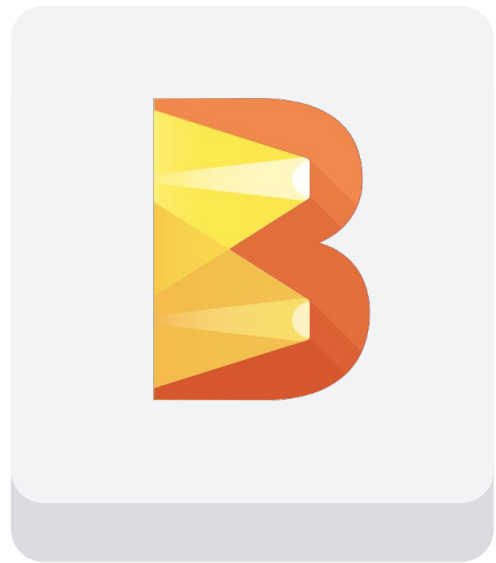
Application
streams

Distributed messaging service that can
receive messages from various device streams

Dataflow pipes data captured by Pub/Sub into a data warehouse



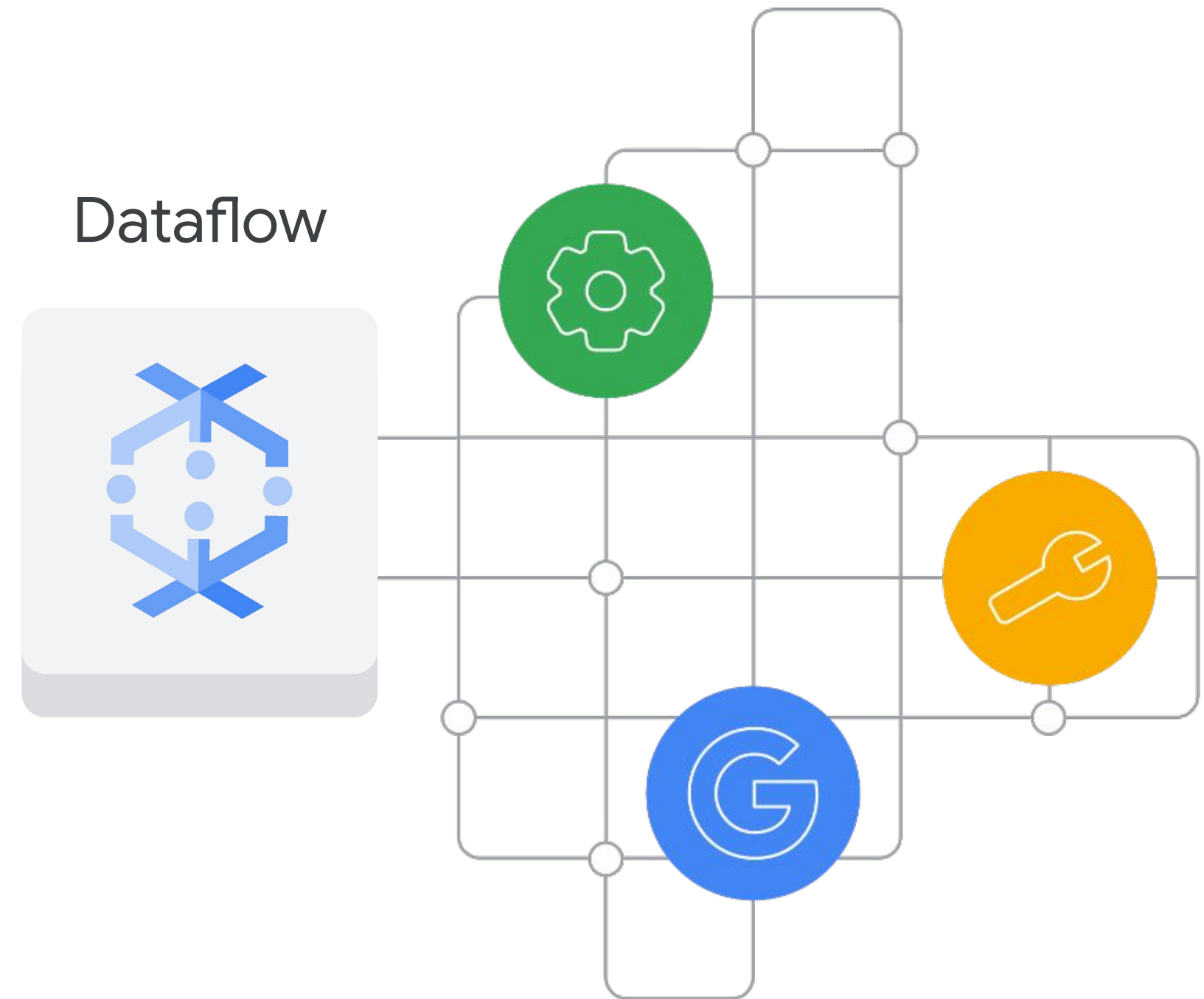
Apache Beam is a popular solution for pipeline design



An open source, unified programming model to **define** and **execute** data processing pipelines, including ETL, batch, and stream processing

Dataflow

- ✓ Handles infrastructure setup
- ✓ Handles maintenance
- ✓ Is built on Google infrastructure
- ✓ Allows for reliable auto scaling
- ✓ Meets data pipeline demands



Quiz

Question

New cloud tools make it possible to harness the potential of unstructured data. Which of these use cases best demonstrates this?

- A. Creating visualizations from seasonal weather data
- B. Analyzing historical sales figures to predict future trends
- C. Analyzing social media posts to identify sentiment toward a brand
- D. Using GPS coordinates to power a ride-sharing app

Quiz

Answer

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Quiz

Question

Data in the form of video, pictures, and audio recordings is well suited to object storage. Which product is best for storing this kind of data?

- A. Firestore
- B. Cloud Storage
- C. Cloud SQL
- D. BigQuery

Quiz

Answer

Data in the form of video, pictures, and audio recordings is well suited to object storage. Which product is best for storing this kind of data?

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Google Cloud